

Lecture Notes

Linear Algebra

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2021-2022

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Chapter 1

Linear equations

1.1 Introduction

While solving problems in many technical fields we obtain several equations with many unknowns. A major problem, in this case, is then to obtain all possible solutions to this set of equations. In other words, we start with equations of the form:

$$\left\{ \begin{array}{l} f_1(x_1, \dots, x_n) = b_1 \\ f_2(x_1, \dots, x_n) = b_2 \\ f_3(x_1, \dots, x_n) = b_3 \\ \vdots \\ f_m(x_1, \dots, x_n) = b_m, \end{array} \right. \quad (1.1)$$

where $f_1, \dots, f_m$ are given functions and $b_1, \dots, b_m$ are known constants. The objective is here to determine all values for the unknowns $x_1, \dots, x_n$ for which the above set of equations is satisfied. We see that we have m equations with n unknowns in (1.1).

Example 1.1 Assume we have obtained the distance to three GPS satellites. The GPS satellites are at location $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, respectively, with corresponding distances $\sqrt{5}$ and $\sqrt{3}$ and $\sqrt{5}$ (where we obviously simplified the numbers) then our location (x_1, x_2, x_3) satisfies the equations:

$$\left\{ \begin{array}{l} (x_1 - 1)^2 + x_2^2 + x_3^2 = 5 \\ x_1^2 + (x_2 - 1)^2 + x_3^2 = 3 \\ x_1^2 + x_2^2 + (x_3 - 1)^2 = 5. \end{array} \right. \quad (1.2)$$

This is of the form (1.1) with $m = n = 3$. It can be verified that $x_1 = 1$, $x_2 = 2$ and $x_3 = 1$ is a solution but are there other solutions? $\square$

Finding the solution of such sets of equations is in general a difficult problem. However, important special cases are so-called linear sets of equations. Investigating such sets of linear equations is an important problem in many areas and, for this special case, we have a rather complete solution which we will present in this chapter.

1.2 Linear Systems

Let us first clarify what we mean by a linear equation.

Definition 1.2 *An equation*

$$f(x_1, \dots, x_n) = b \quad (1.3)$$

*is called a **linear equation** if the function f is linear. In other words for any $x_1, \dots, x_n$ and $y_1, \dots, y_n$ and constant α we have:*

$$f(x_1 + \alpha y_1, x_2 + \alpha y_2, \dots, x_n + \alpha y_n) = f(x_1, \dots, x_n) + \alpha f(y_1, \dots, y_n)$$

This is clearly a quite abstract definition which, however, can be quite useful in checking whether a given f is linear. To make the definition more explicit, an equation (1.3) is linear if it can be written as:

$$a_1 x_1 + a_2 x_2 + \dots + a_n x_n = b \quad (1.4)$$

for certain $a_1, \dots, a_n$. Note that the unknowns are here $x_1, \dots, x_n$. On the other hand, $a_1, \dots, a_n$ as well as b are known constants. Note that b is not part of the definition of $f(x_1, \dots, x_n)$.

Example 1.3 The equation

$$5x_1 - 3x_2 + 2x_3 = 4$$

is a linear equation. It is of the form (1.4) for $a_1 = 5, a_2 = -3, a_3 = 2$ and $b = 4$. We have $n = 3$ since we have three unknowns. Note that the first equation from (1.2)

$$(x_1 - 1)^2 + x_2^2 + x_3^2 = 5$$

is a nonlinear equation of the form (1.3) but it can clearly not be written in the form (1.4). $\square$

If we have a set of equations (1.1) where each individual equation can be written in the form (1.4) then we can rewrite the entire set of equations as:

$$\left\{ \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ a_{31}x_1 + a_{32}x_2 + \cdots + a_{3n}x_n = b_3 \\ \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{array} \right. \quad (1.5)$$

Note that in the above, $a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn}$ as well as $b_1, \dots, b_m$ are all known constants. We will refer to (1.5) as a **linear system** consisting of m linear equations with n unknowns.

Since the linear system (1.5) will occur frequently, it makes sense to introduce some notation to write the above more efficiently. First we introduce vectors.

Definition 1.4 *We define:*

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad (1.6)$$

These are called **column vectors**. In these lecture notes, vectors will be denoted by boldface letters. The space $\mathbb{R}^k$ consists of all column vectors consisting of k elements and will be studied in more detail in Section 2.1. In the above $\mathbf{x}$ is in $\mathbb{R}^n$ while $\mathbf{b}$ is in $\mathbb{R}^m$. Since our objective is to solve this equation to obtain $\mathbf{x}$, the vector $\mathbf{x}$ is sometimes referred to as the **solution vector** which needs to be determined. Next, we denote:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

This is called a **matrix**. In relationship to the set of linear equations we sometimes will refer to this as the **coefficient matrix**. The space $\mathbb{R}^{m \times n}$ consists of all matrices which have m **rows** and n **columns**. Here rows run horizontally and are associated with one equation. The i 'th row is

given by:

$$(a_{i1} \ a_{i2} \ \cdots \ a_{in})$$

On the other hand columns run vertically and are associated with one unknown. The j 'th column is given by:

$$\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$$

Given A , $\mathbf{x}$ and $\mathbf{b}$ we can rewrite our linear system as

$$A\mathbf{x} = \mathbf{b}$$

How to interpret the above product of a matrix A and a column vector $\mathbf{x}$ will be clarified in Section 2.2.

Let us first consider some examples of linear systems of the form (1.5).

Example 1.5 Consider the linear system:

$$\begin{cases} 3x_1 + x_2 = 1 \\ 2x_1 + x_2 = 0 \end{cases}$$

If we subtract the above two equations, we get:

$$(3x_1 + x_2) - (2x_1 + x_2) = 1 - 0$$

which yields $x_1 = 1$. On the other hand, 2 times the first equation minus 3 times the second equation yields:

$$2(3x_1 + x_2) - 3(2x_1 + x_2) = 2 - 0$$

which yields $-x_2 = 2$ or $x_2 = -2$. In other words, we see that there is a **unique** solution $x_1 = 1$ and $x_2 = -2$. $\square$

Example 1.6 Consider the linear system:

$$\begin{cases} 3x_1 + x_2 = 1 \\ 6x_1 + 2x_2 = 2 \end{cases}$$

We note that the second equation is twice the first equation and hence all (x_1, x_2) which satisfy the first equation also satisfy the second equation.

Therefore, we only need to consider the first equation. We note that the first equation can be rewritten as:

$$x_2 = 1 - 3x_1$$

and hence for any arbitrary value for x_1 the linear system is solved provided the second unknown x_2 is given by $1 - 3x_1$. We call x_1 a free variable in this context and we note that the linear system has an infinite number of solutions. $x_1 = 1$ and $x_2 = -2$ works just as well as $x_1 = 3$ and $x_2 = -8$ or $x_1 = 2$ and $x_2 = -5$. $\square$

Example 1.7 Consider the linear system:

$$\begin{cases} 3x_1 + x_2 = 1 \\ 6x_1 + 2x_2 = 3 \end{cases}$$

Subtracting 2 times the first equation from the second equation we obtain:

$$(6x_1 + 2x_2) - 2(3x_1 + x_2) = 3 - 2$$

or

$$0 = 1$$

which clearly yields a contradiction. Therefore, we note that no matter how we choose the variables x_1 and x_2 we can never satisfy both of these linear equations at the same time. $\square$

In the above, we have seen three examples. One in which the linear system has one unique solution, one with an infinite number of solutions and one with no solutions at all. It turns out that these are the only three possibilities.

Definition 1.8 We call the linear system (1.5) **consistent** if the set of linear equations has at least one solution.

Definition 1.9 The **solution set** of a linear system (1.5) is the set of all possible solutions $(x_1, \dots, x_n)$ for this set of linear equations.

Example 1.10 If we have one linear equation in two variables then the solution set can be viewed as a **line** in $\mathbb{R}^2$. If we have one linear equation in three variables then the solution set can be viewed as a **plane** in $\mathbb{R}^3$.

We have the following result which shows that the solution set contains either 0, 1 or an infinite number of solutions.

Theorem 1.11 *A linear system (1.5) whose solution set contains at least two distinct solutions, has a solution set which contains an infinite number of distinct solutions.*

Proof : Assume $(u_1, \dots, u_n)$ and $(v_1, \dots, v_n)$ are both solutions of the linear system. In other words,

$$\begin{aligned} a_{i1}u_1 + a_{i2}u_2 + \dots + a_{in}u_n &= b_i \\ a_{i1}v_1 + a_{i2}v_2 + \dots + a_{in}v_n &= b_i \end{aligned}$$

for $i = 1, \dots, m$. But then $(x_1, \dots, x_n)$ given by:

$$x_j = u_j + \lambda(v_j - u_j)$$

for $j = 1, \dots, n$ is also a solution of the linear equations for all λ . This implies that we can generate an infinite number of solutions by varying λ provided $\mathbf{u}$ and $\mathbf{v}$ are not equal. ■

The above clarifies that we always have 0, 1 or an infinite number of solutions for a given linear system. In the next section, we will present a structured approach to obtain all solutions of a set of linear equations. It will be based on the ideas exploited in the above examples but it is not based on ad-hoc manipulations but has a structured approach which works for 2 equations with 2 unknowns as well as for large systems with say 1000 equations with 2000 unknowns.

1.3 Gauss-Jordan elimination

Given a linear system (1.5) we can define the coefficient matrix A and the column vector $\mathbf{b}$. However, it is sometimes also convenient to work with the so-called **augmented matrix**:

$$A^a = \left(A \mid \mathbf{b} \right) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ a_{31} & a_{32} & \cdots & a_{3n} & b_3 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right) \quad (1.7)$$

There are three fundamental operations which do not affect the solution set of a linear system:

Definition 1.12 An **elementary operation** for a linear system (1.5) is any one of the following three operations:

- Adding a multiple of one equation to another equation.
- Multiplying one equation with a number unequal to zero.
- Interchanging two equations.

Definition 1.13 We call two linear systems **equivalent** if we can transform one into the other by a sequence of elementary operations. Similarly, we call two augmented matrices **row equivalent** if we can transform one into the other by elementary operations.

Remark. Since the elementary operations can be reversed we should note that if we can obtain linear system 1 from linear system 2, then we can also obtain linear system 2 from linear system 1 by elementary operations. The same applies to augmented matrices and the associated elementary row operations.

The power of the elementary operations is expressed by the following theorem which we will not formally prove:

Theorem 1.14 Consider two linear systems which have the same number of equations and which have the same solution set. Then the two linear systems are equivalent.

On the other hand, if two linear systems are equivalent then they have the same number of equations and the same solution set.

Example 1.15 Note that the following linear system:

$$\begin{cases} 3x_1 - 2x_2 = 5 \\ 6x_1 - 4x_2 = 10 \end{cases}$$

has the same solution set as the linear system:

$$\begin{cases} 3x_1 - 2x_2 = 5 \end{cases}$$

even though they do not have the same number of equations.

By an elementary operation (subtracting two times the first equation from the second equation), we can transform the first system to:

$$\begin{cases} 3x_1 - 2x_2 = 5 \\ 0 = 0 \end{cases}$$

which is equal to the second linear system after we delete the trivial equation $0 = 0$. The above theorem is still valid without the condition about the number of equations provided we add an elementary operation which is adding or deleting the trivial equation $0 = 0$. $\square$

Example 1.16 Finding the solution set of a linear system can be done by using a sequence of elementary operations to simplify the system. Consider the system:

$$\begin{cases} 2x_1 - x_2 + x_3 - x_4 = -6 \\ -x_1 + x_2 + 5x_3 + 2x_4 = 2 \\ 2x_1 - x_2 + 2x_3 - 3x_4 = 3 \end{cases} \quad (1.8)$$

Interchanging the second and first equation, we obtain

$$\begin{cases} -x_1 + x_2 + 5x_3 + 2x_4 = 2 \\ 2x_1 - x_2 + x_3 - x_4 = -6 \\ 2x_1 - x_2 + 2x_3 - 3x_4 = 3 \end{cases}$$

and multiplying the first equation by -1 yields:

$$\begin{cases} x_1 - x_2 - 5x_3 - 2x_4 = -2 \\ 2x_1 - x_2 + x_3 - x_4 = -6 \\ 2x_1 - x_2 + 2x_3 - 3x_4 = 3 \end{cases}$$

then subtracting twice the first equation from the second equation and subtracting twice the first equation from the third equation yields:

$$\begin{cases} x_1 - x_2 - 5x_3 - 2x_4 = -2 \\ x_2 + 11x_3 + 3x_4 = -2 \\ x_2 + 12x_3 + x_4 = 7 \end{cases}$$

Next subtracting the second equation from the third equation yields:

$$\begin{cases} x_1 - x_2 - 5x_3 - 2x_4 = -2 \\ x_2 + 11x_3 + 3x_4 = -2 \\ x_3 - 2x_4 = 9 \end{cases} \quad (1.9)$$

Note that this already gives us a method to determine the solution set. The third equation can be used to determine x_3 in terms of x_4 . Substituting this result in the second equation, we obtain x_2 in terms of x_4 . Finally substituting the expressions for x_2 and x_3 in the first equation, we obtain x_1 in

terms of x_4 . However, this substitution can be avoided by using some extra elementary row operations. We subtract eleven times the third equation from the second equation and get:

$$\begin{cases} x_1 - x_2 - 5x_3 - 2x_4 = -2 \\ \quad x_2 \quad \quad + 25x_4 = -101 \\ \quad \quad x_3 - 2x_4 = 9 \end{cases}$$

Next, we add the second equation to the first equation

$$\begin{cases} x_1 - 5x_3 + 23x_4 = -103 \\ \quad x_2 \quad \quad + 25x_4 = -101 \\ \quad \quad x_3 - 2x_4 = 9 \end{cases} \quad (1.10)$$

and finally we add five times the third equation to the first equation:

$$\begin{cases} x_1 \quad \quad + 13x_4 = -58 \\ \quad x_2 \quad \quad + 25x_4 = -101 \\ \quad \quad x_3 - 2x_4 = 9 \end{cases}$$

We have applied eight elementary operations which do not change the solution set. But looking at the last linear system we immediately see that for any arbitrary choice for x_4 we can directly obtain x_1 , x_2 and x_3 from the three respective equations. A **parametric description** of the solution set can be given as:

$$\begin{cases} x_1 = -13x_4 - 58 \\ x_2 = -25x_4 - 101 \\ x_3 = 2x_4 + 9 \\ x_4 \text{ is free} \end{cases} \quad \square$$

Since the linear system (1.5) is completely described by the augmented matrix (1.7) we can apply these elementary operations directly to the augmented matrix. Instead of adding, multiplying or switching equations we then add, multiply or switch rows and therefore we then refer to these elementary operations as **elementary row operations**:

- Adding a multiple of one row to another row.
- Multiplying one row with a number unequal to zero.
- Interchanging two rows.

Example 1.17 Consider the linear system (1.8) from Example 1.16. The associated augmented matrix is given by:

$$\left(\begin{array}{cccc|c} 2 & -1 & 1 & -1 & -6 \\ -1 & 1 & 5 & 2 & 2 \\ 2 & -1 & 2 & -3 & 3 \end{array} \right)$$

Using the same row operations as in Example 1.16 but directly on the augmented matrix we get:

$$\begin{aligned} \left(\begin{array}{cccc|c} 2 & -1 & 1 & -1 & -6 \\ -1 & 1 & 5 & 2 & 2 \\ 2 & -1 & 2 & -3 & 3 \end{array} \right) &\sim \left(\begin{array}{cccc|c} -1 & 1 & 5 & 2 & 2 \\ 2 & -1 & 1 & -1 & -6 \\ 2 & -1 & 2 & -3 & 3 \end{array} \right) \\ &\sim \left(\begin{array}{cccc|c} 1 & -1 & -5 & -2 & -2 \\ 2 & -1 & 1 & -1 & -6 \\ 2 & -1 & 2 & -3 & 3 \end{array} \right) \\ &\sim \left(\begin{array}{cccc|c} 1 & -1 & -5 & -2 & -2 \\ 0 & 1 & 11 & 3 & -2 \\ 0 & 1 & 12 & 1 & 7 \end{array} \right) \\ &\sim \left(\begin{array}{cccc|c} 1 & -1 & -5 & -2 & -2 \\ 0 & 1 & 11 & 3 & -2 \\ 0 & 0 & 1 & -2 & 9 \end{array} \right) \end{aligned} \quad (1.11)$$

Here $\sim$ indicates that we can obtain one matrix from the other with the help of elementary row operations. We note that the last matrix we obtain is precisely the augmented matrix of the linear system (1.9). The final elementary row operations of Example 1.16 yield:

$$\begin{aligned} \left(\begin{array}{cccc|c} 1 & -1 & -5 & -2 & -2 \\ 0 & 1 & 11 & 3 & -2 \\ 0 & 0 & 1 & -2 & 9 \end{array} \right) &\sim \left(\begin{array}{cccc|c} 1 & -1 & -5 & -2 & -2 \\ 0 & 1 & 0 & 25 & -101 \\ 0 & 0 & 1 & -2 & 9 \end{array} \right) \\ &\sim \left(\begin{array}{cccc|c} 1 & 0 & -5 & 23 & -103 \\ 0 & 1 & 0 & 25 & -101 \\ 0 & 0 & 1 & -2 & 9 \end{array} \right) \\ &\sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & 13 & -58 \\ 0 & 1 & 0 & 25 & -101 \\ 0 & 0 & 1 & -2 & 9 \end{array} \right) \end{aligned} \quad (1.12)$$

which yields the augmented matrix of the linear system (1.10). Performing the row operations directly on the augmented matrix is equivalent to the

same operations applied to the linear system. However, using the augmented matrix is notationally easier. $\square$

In the above, we have shown an approach to simplify linear systems. We have seen that it is notationally easier to work with the augmented matrix introduced before. Second, we will use these elementary row operations to reduce the augmented matrix to either the echelon form or the reduced echelon form:

Definition 1.18 A matrix is in **echelon form** if

- Nonzero rows are situated above zero rows.
- The first nonzero entry (the **pivot entry**) of a row is always in a column to the right of the pivots of the rows above it.

The matrix is in **reduced echelon form** (sometimes also called **row-reduced echelon form**) if, additionally

- Each pivot equals one and is the only nonzero entry in its column.

Example 1.19 Note that the augmented matrix we obtained in (1.11) is in echelon form while the augmented matrix we ultimately obtained in (1.12) is in reduced echelon form. $\square$

Theorem 1.20 A matrix can always be transformed into a reduced echelon form by elementary row operations. Moreover, this reduced echelon form is unique. The echelon form is not unique.

We have the following algorithm to transform any matrix into either the echelon form or the reduced echelon form:

- (i) Determine the leftmost column which does not consist of only zeros. Interchange the top row with another row, if needed, such that the top element is nonzero (this will be called a pivot element). If possible, choose it equal to 1 since that makes the consequent steps easier by avoiding fractions.
- (ii) Add a suitable multiple of the top row to the rows below such that the term in the column with the pivot element becomes equal to zero.
- (iii) Form a virtual matrix by deleting – in your mind – the top row and applying the previous steps to the rows below. Continue until the matrix is in echelon form.

To get the matrix into reduced echelon form, we need two additional steps:

- (iv) Make all pivot elements equal to 1 by dividing the corresponding rows by a suitable number.
- (v) Add a suitable multiple of the row with a pivot element to all rows above to ensure that the pivot element is the only nonzero element in its column.

Example 1.21 Consider the matrix

$$A = \begin{pmatrix} 0 & 0 & 0 & 2 & -2 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 1 & 3 & 2 & -2 & 3 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix}$$

Using elementary row operations, we get:

$$\begin{pmatrix} 0 & 0 & 0 & 2 & -2 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 1 & 3 & 2 & -2 & 3 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 0 & 0 & 0 & 2 & -2 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 2 & 4 & -2 & -2 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix}$$

Here, we first interchange the first and fourth row. Next, we subtract 2 times the first row from the second row, subtract 2 times the first row from the third row and subtract 2 times the first row from the fifth row. This corresponds to steps (i) and (ii) of the algorithm. We then form a virtual matrix by ignoring the first row and we continue with the matrix under the line. We get:

$$\begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 2 & 4 & -2 & -2 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 6 & -6 \end{pmatrix}$$

Here, we first divide the second row by a factor 2. Next, we add 3 times the second row to the third row and add the second row to the fifth row. Note that we first applied step (iv) of the algorithm to the second row. By first creating this 1 as the first nonzero element of the second row the following steps are a bit easier. Next, we continued with (ii) of the algorithm. We then form a virtual matrix by ignoring the top two rows and we continue with the

matrix under the line.

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 6 & -6 \end{array} \right) \sim \left(\begin{array}{ccccc} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 6 & -6 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccccc} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Here we first interchange the third and fifth row and then divide the third row by the factor 6. We need one final step (subtracting 2 times the third row from the fourth row) and the matrix is in echelon form:

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccccc} \mathbf{1} & 3 & 2 & -2 & 3 \\ \hline 0 & \mathbf{1} & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Note that the bold digits indicate the pivot elements and the lines in the final matrix indicate the echelon structure. However, this is not in **reduced** echelon form since we then need two more steps:

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccccc} 1 & 0 & -4 & 1 & 6 \\ 0 & 1 & 2 & -1 & -1 \\ \hline 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \sim \left(\begin{array}{ccccc} \mathbf{1} & 0 & -4 & 0 & 7 \\ \hline 0 & \mathbf{1} & 2 & 0 & -2 \\ \hline 0 & 0 & 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Here we first subtracted 3 times the second row from the first row. Next, we added the third row to the second row and subtracted the third row from the first row. These steps corresponded to step (v) of the algorithm. $\square$

From the (reduced) echelon form of an augmented matrix, we can immediately deduce whether a linear system is consistent.

Theorem 1.22 A linear system $A\mathbf{x} = \mathbf{b}$ with associated augmented matrix:

$$\left(A \mid \mathbf{b} \right)$$

is consistent if and only if the (reduced) echelon form of the augmented matrix has no pivot element in the last column.

Example 1.23 Consider the linear system:

$$\begin{cases} 2x_4 - 2x_5 = 1 \\ 2x_1 + 8x_2 + 8x_3 - 6x_4 + 4x_5 = 2 \\ 2x_1 + 3x_2 - 2x_3 - x_4 + 9x_5 = 1 \\ x_1 + 3x_2 + 2x_3 - 2x_4 + 3x_5 = -1 \\ 2x_1 + 5x_2 + 2x_3 + 3x_4 + x_5 = 2 \end{cases}$$

This linear system has the same coefficient matrix A as in Example 1.21. The associated augmented matrix is equal to:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & -2 & 1 \\ 2 & 8 & 8 & -6 & 4 & 2 \\ 2 & 3 & -2 & -1 & 9 & 1 \\ 1 & 3 & 2 & -2 & 3 & -1 \\ 2 & 5 & 2 & 3 & 1 & 2 \end{array} \right)$$

Using the same elementary row operations as in Example 1.21 this augmented matrix is transformed into:

$$\left(\begin{array}{ccccc|c} \mathbf{1} & 0 & -4 & 0 & 7 & -8 \\ 0 & \mathbf{1} & 2 & 0 & -2 & 3 \\ 0 & 0 & 0 & \mathbf{1} & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 9 \end{array} \right) \quad (1.13)$$

If we convert this back to a linear system we get:

$$\begin{cases} x_1 - 4x_3 + 7x_5 = -8 \\ x_2 + 2x_3 + 2x_5 = 3 \\ x_4 - x_5 = 1 \\ 0 = -1 \\ 0 = 9 \end{cases}$$

From the last two equations we immediately see that this linear system is not consistent since it has no solutions. Note that our elementary row operations from Example 1.21 did not yield an augmented matrix (1.13) which is in

reduced echelon form. For that we need two more steps:

$$\begin{pmatrix} \overline{1} & 0 & -4 & 0 & 7 & -8 \\ 0 & \overline{1} & 2 & 0 & -2 & 3 \\ 0 & 0 & 0 & \overline{1} & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 9 \end{pmatrix} \sim \begin{pmatrix} \overline{1} & 0 & -4 & 0 & 7 & -8 \\ 0 & \overline{1} & 2 & 0 & -2 & 3 \\ 0 & 0 & 0 & \overline{1} & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 9 \end{pmatrix}$$

$$\sim \begin{pmatrix} \overline{1} & 0 & -4 & 0 & 7 & 0 \\ 0 & \overline{1} & 2 & 0 & -2 & 0 \\ 0 & 0 & 0 & \overline{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \overline{1} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

which are obtained by multiplying the fourth row by -1 and then, as a final step, we subtract 9 times the fourth row from the fifth row, we add 8 times the fourth row to the first row, we subtract 3 times the fourth row from the second row and we subtract the fourth row from the third row.

We see that the last column of the reduced echelon form contains a pivot element which, by Theorem 1.22, also yields that this linear system is not consistent. $\square$

Example 1.24 Consider the linear system:

$$\begin{cases} 2x_4 - 2x_5 = 0 \\ 2x_1 + 8x_2 + 8x_3 - 6x_4 + 4x_5 = 24 \\ 2x_1 + 3x_2 - 2x_3 - x_4 + 9x_5 = 14 \\ x_1 + 3x_2 + 2x_3 - 2x_4 + 3x_5 = 10 \\ 2x_1 + 5x_2 + 2x_3 + 3x_4 + x_5 = 18 \end{cases}$$

This linear system again has the same coefficient matrix A as in Example 1.21. The associated augmented matrix is equal to:

$$\left(\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & -2 & 0 \\ 2 & 8 & 8 & -6 & 4 & 24 \\ 2 & 3 & -2 & -1 & 9 & 14 \\ 1 & 3 & 2 & -2 & 3 & 10 \\ 2 & 5 & 2 & 3 & 1 & 18 \end{array} \right)$$

Using the same elementary row operations as in Example 1.21 this aug-

mented matrix is transformed into:

$$\left(\begin{array}{ccccc|c} \mathbf{1} & 0 & -4 & 0 & 7 & 4 \\ 0 & \mathbf{1} & 2 & 0 & -2 & 2 \\ 0 & 0 & 0 & \mathbf{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

This matrix is in reduced echelon form and we notice that the last column does not contain a pivot element. Hence by Theorem 1.22 we know that the linear system is consistent. We notice that the first, second and fourth columns contain a pivot element and we call the associated variables x_1 , x_2 and x_4 **basic variables**. The remaining variables x_3 and x_5 are called **free variables**.

If we convert this back to a linear system we get:

$$\left\{ \begin{array}{rcl} x_1 & -4x_3 & +7x_5 = 4 \\ & x_2 + 2x_3 & -2x_5 = 2 \\ & & x_4 - x_5 = 0 \\ & & 0 = 0 \\ & & 0 = 0 \end{array} \right. \quad (1.14)$$

The last two equations are clearly immediately satisfied. In order to find a parametric description of the solution set of this linear system we first use that x_3 and x_5 were free variables:

$$\left\{ \begin{array}{l} x_3 \text{ is free} \\ x_5 \text{ is free} \end{array} \right.$$

Next we express the basis variables in terms of the free variables using the (simplified) linear system (1.14):

$$\left\{ \begin{array}{l} x_1 = 4x_3 - 7x_5 + 4 \\ x_2 = -2x_3 + 2x_5 + 2 \\ x_4 = x_5 \end{array} \right.$$

In conclusion the solution set is given by:

$$\left\{ \begin{array}{l} x_1 = 4x_3 - 7x_5 + 4 \\ x_2 = -2x_3 + 2x_5 + 2 \\ x_3 \text{ is free} \\ x_4 = x_5 \\ x_5 \text{ is free} \end{array} \right.$$

□

Example 1.25 Consider the following system of linear equations:

$$\begin{cases} \alpha x_1 + (\alpha + 1)x_2 + \alpha x_3 = \beta + 2 \\ 2x_1 + 4x_2 + \alpha x_3 = \beta + 5 \\ x_1 + 2x_2 + x_3 = 2 \end{cases}$$

where $\alpha, \beta \in \mathbb{R}$. The question is for which values of α and β this system is consistent.

The associated augmented matrix is equal to:

$$\left(\begin{array}{ccc|c} \alpha & \alpha + 1 & \alpha & \beta + 2 \\ 2 & 4 & \alpha & \beta + 5 \\ 1 & 2 & 1 & 2 \end{array} \right)$$

We want to use elementary row operations but subtracting $2/\alpha$ times the first row from the second row, to create a second row which starts with a zero, is not the best way to proceed. First of all, the expressions become quite complicated but, more importantly, this does not work if α just happens to be zero. Instead we interchange the first and third row:

$$\left(\begin{array}{ccc|c} \alpha & \alpha + 1 & \alpha & \beta + 2 \\ 2 & 4 & \alpha & \beta + 5 \\ 1 & 2 & 1 & 2 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 4 & \alpha & \beta + 5 \\ \alpha & \alpha + 1 & \alpha & \beta + 2 \end{array} \right)$$

Now, we are going to use the 1 in the first element of the first row to make the other elements in the first column equal to zero:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & 4 & \alpha & \beta + 5 \\ \alpha & \alpha + 1 & \alpha & \beta + 2 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & \alpha - 2 & \beta + 1 \\ 0 & 1 - \alpha & 0 & \beta + 2 - 2\alpha \end{array} \right)$$

where we subtracted 2 times the first row from the second row and we subtracted α times the first row from the third row. Interchanging the second and third rows yields an echelon form:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 - \alpha & 0 & \beta + 2 - 2\alpha \\ 0 & 0 & \alpha - 2 & \beta + 1 \end{array} \right)$$

We see that if $\alpha \neq 1$ and $\alpha \neq 2$ then the augmented matrix is in echelon form and since the first three columns contain a pivot element and the last column does not contain a pivot element we conclude that the system is consistent.

If $\alpha = 1$, we obtain:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & 0 & \beta \\ 0 & 0 & -1 & \beta + 1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & -1 & \beta + 1 \\ 0 & 0 & 0 & \beta \end{array} \right)$$

where we interchanged the second and third row. We see that the system has a pivot element in the final column if $\beta \neq 0$. In that case the system is inconsistent. However, if $\beta = 0$ then the system is consistent.

If $\alpha = 2$, we obtain:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -1 & 0 & \beta - 2 \\ 0 & 0 & 0 & \beta + 1 \end{array} \right)$$

We see that the system has a pivot element in the final column if $\beta \neq -1$. In that case the system is inconsistent. However, if $\beta = -1$ then the system is consistent.

In conclusion, we see that the system is consistent except when $\alpha = 1, \beta \neq 0$ or $\alpha = 2, \beta \neq -1$. $\square$

1.4 Exercises

1.4.1 Self study exercises

1.1 Consider the following system of linear equations:

$$\begin{cases} x_1 - 2x_2 - 3x_3 & = 1 \\ -2x_1 + 4x_2 + 5x_3 + x_4 & = -1 \\ -x_1 + 2x_2 + x_3 + 2x_4 & = -2 \end{cases}$$

(a) Show that the reduced echelon form of the augmented matrix of this system is

$$\left(\begin{array}{cccc|c} 1 & -2 & 0 & -3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

(b) Is this system consistent? Use Theorem 1.22 to formulate your answer!

1.2 Consider the following system of linear equations:

$$\begin{cases} x_1 - x_2 + 2x_3 + x_4 & = 1 \\ 2x_1 + 3x_2 + 4x_3 - 3x_4 & = 2 \\ 5x_1 + 4x_2 + 10x_3 - 2x_4 & = 3 \end{cases}$$

(a) Examine if

$$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

is a solution of this system.

(b) Determine the augmented matrix

(c) Determine the reduced echelon form of the augmented matrix

(d) Determine the set of solutions of this system of linear equations

1.3 Give a system of 3 linear equations, each with nonzero coefficients for the 3 unknowns x_1 , x_2 and x_3 , such that the system has the unique solution

$$\begin{cases} x_1 = 1 \\ x_2 = -7 \\ x_3 = 3 \end{cases}$$

In other words, the coefficient matrix A should have no zero elements.

1.4 Study once again carefully Theorem 1.22 and use it to formulate your answers. Consider a matrix $A \in \mathbb{R}^{m \times n}$.

(a) How can one deduce from the (reduced) echelon form of A that the linear system $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^m$?

(b) Is it possible that the system $A\mathbf{x} = \mathbf{b}$ is inconsistent for all $\mathbf{b} \in \mathbb{R}^m$?

(c) How can one deduce from the (reduced) echelon form of A that the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^m$?

1.4.2 Tutorial exercises

1.5 Bring the following two matrices into the reduced echelon form by elementary row operations:

$$\begin{pmatrix} 1 & 2 & 4 & -3 \\ -1 & -1 & 5 & 5 \\ 2 & 4 & 8 & -6 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 0 & -2 \\ 2 & 3 & 1 & 0 \\ 4 & 7 & 1 & -4 \\ 3 & 4 & 2 & 2 \end{pmatrix}$$

- 1.6 Determine the set of solutions of the following system of linear equations:

$$\begin{cases} x_1 - 2x_2 + x_3 = 2 \\ 2x_1 - 3x_2 + x_3 = 1 \\ 7x_1 - 2x_2 + x_3 = 5 \end{cases}$$

- 1.7 Verify whether the following system of linear equations is consistent:

$$\begin{cases} 2x_1 - 3x_2 = 3 \\ 3x_1 + 2x_2 = 4 \\ x_1 - 4x_2 = 1 \end{cases}$$

- 1.8 Examine if there exists a point in $\mathbb{R}^3$ that lies on each of the three planes

$$p_1 : x_1 + x_2 - 2x_3 = 1, \quad p_2 : 2x_1 + x_2 - x_3 = 3 \text{ and } p_3 : x_1 - x_2 + 4x_3 = 1.$$

- 1.9 Does there exist a quadratic function $f(x) = ax^2 + bx + c$ whose graph passes through the four points $(1, 3)$, $(2, 2)$, $(3, 5)$ and $(4, 12)$?

- 1.10 (a) An *underdetermined* system is a linear system with fewer equations than unknowns. Can an underdetermined system have a unique solution? If Yes, give an example; if no, explain why not.

- (b) An *overdetermined* system is a linear system with more equations than unknowns. Can an overdetermined system have a unique solution? If yes, give an example; if no, explain why not.

- 1.11 Consider Example 1.1. Show that the equations in (1.2) are not linear equations by using Definition 1.2.

- 1.12 Study the proof of Theorem 1.11. Show that the vector $(x_1, \dots, x_n)$ given by

$$x_i = u_i + \lambda(v_i - u_i)$$

for $i = 1, \dots, n$, is indeed a solution of the linear equations for all $\lambda \in \mathbb{R}$.

- 1.13 Determine for each of the following augmented matrices for which value(s) of α and β the corresponding system of linear equations has no solutions; has exactly one solution and has infinitely many solutions. For those values of α and β for which the system has exactly one solution determine this solution.

$$(a) \begin{pmatrix} 1 & \alpha & -2 \\ 4 & 2 & -8 \end{pmatrix}$$

$$(b) \begin{pmatrix} 1 & \alpha & -2 \\ 4 & 2 & -7 \end{pmatrix}$$

$$(c) \begin{pmatrix} 1 & -2 & \alpha \\ 4 & -7 & 2 \end{pmatrix}$$

$$(d) \begin{pmatrix} \alpha & \alpha + \beta & -12 \\ 1 & 4 & -\beta \end{pmatrix}$$

1.14 Let

$$A = \begin{pmatrix} -2 & 5 & 4 \\ 1 & -2 & -3 \\ -1 & 3 & 1 \end{pmatrix}.$$

(a) Show that the system $A\mathbf{x} = \mathbf{b}$ is not consistent for all $\mathbf{b} \in \mathbb{R}^3$ (i.e. there exist $\mathbf{b} \in \mathbb{R}^3$ for which the system is not consistent).

(b) For which

$$\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

is the system $A\mathbf{x} = \mathbf{b}$ consistent?

Describe your answer with an equation in b_1 , b_2 and b_3 .

1.15 Consider the following system of linear equations:

$$\begin{cases} x_1 + 5x_2 + x_3 = 1 \\ x_1 + 6x_2 - x_3 = 1 \\ 2x_1 + \alpha x_2 - 6x_3 = \beta \end{cases}$$

where $\alpha, \beta \in \mathbb{R}$.

(a) Determine all values of α and β for which this system is consistent.

(b) Take $\alpha = 14$ and $\beta = 2$. Determine the solution set of this system.

(c) Let A be the coefficient matrix of this system. Determine the value(s) for α for which the system $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^3$.

1.16 Consider the following system of linear equations:

$$\begin{cases} x_1 - 2x_2 - 3x_3 + x_4 = 3 \\ 3x_1 - 6x_2 - 8x_3 + (\alpha + 3)x_4 = 7 \\ -2x_1 + 4x_2 + 4x_3 - (\alpha + 1)x_4 = \beta - 6 \end{cases}$$

where $\alpha, \beta \in \mathbb{R}$.

- (a) Determine an echelon form for the augmented matrix of this system.
- (b) Determine all values of α and β for which this system is consistent.
- (c) Take $\alpha = -1$ and $\beta = 4$. Determine the solution set of this system.

1.17 Consider two lines in $\mathbb{R}^2$. The first line is described as the solution set of the linear equation $3x_1 - 2x_2 = 1$. The second line is described as the solution set of the linear equation $x_1 + x_2 = 1$.

- (a) Determine the intersection of the two lines.
- (b) Sketch the two lines in $\mathbb{R}^2$ and verify whether your intersection is consistent with your computation in part a.

1.18 Consider two planes in $\mathbb{R}^3$. The first plane is described as the solution set of the linear equation $x_1 - 2x_2 + x_3 = 1$. The second plane is described as the solution set of the linear equation $x_1 + x_2 + 2x_3 = 2$. Determine the intersection of the two planes.

1.19 Consider three planes in $\mathbb{R}^3$. The first plane is described as the solution set of the linear equation $x_1 - 2x_2 + x_3 = 1$. The second plane is described as the solution set of the linear equation $x_1 + x_2 + x_3 = 1$ and the third plane is described as the solution set of the linear equation $2x_1 + x_2 + x_3 = 3$. Determine the intersection of the three planes.

Chapter 2

Vector spaces and matrix algebra

2.1 The vector space $\mathbb{R}^n$

In Section 1.1 we already mentioned column vectors. The space $\mathbb{R}^n$ consists of column vectors with exactly n elements. In other words, the following two vectors are in $\mathbb{R}^2$:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 0 \end{pmatrix},$$

while the following vectors are in $\mathbb{R}^3$:

$$\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \quad \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}.$$

The space $\mathbb{R}^n$ is the most common example of a vector space. A vector space consists of elements for which two operations are defined: addition and scalar multiplication. These operations must satisfy several properties:

Definition 2.1 A **vector space** V is a collection of elements, called **vectors** for which there are two operations: addition and scalar multiplication which satisfy the following ten axioms:

- $\mathbf{u}, \mathbf{v} \in V$ implies $\mathbf{u} + \mathbf{v} \in V$.
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.
- There exists a vector $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$.
- For every $\mathbf{u} \in V$ there exists a vector $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.

- The scalar multiplication of $\mathbf{u} \in V$ with $c \in \mathbb{R}$, denoted by $c\mathbf{u}$, is in V .
- $c(d\mathbf{u}) = (cd)\mathbf{u}$ for all $\mathbf{u} \in V$ and $c, d \in \mathbb{R}$.
- $1\mathbf{u} = \mathbf{u}$ for all $\mathbf{u} \in V$.
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{R}$.
- $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ for all $\mathbf{u} \in V$ and $c, d \in \mathbb{R}$.

It is not hard to verify that the column vectors in $\mathbb{R}^n$ satisfy the above conditions of a vector space if we define addition and scalar multiplication componentwise:

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}, \quad c \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} cu_1 \\ cu_2 \\ \vdots \\ cu_n \end{pmatrix}$$

In this context, the zero vector $\mathbf{0}$ mentioned in the above definition equals:

$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Example 2.2 Consider the vector space $\mathbb{R}^2$ which contains the vectors:

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}.$$

In that case, we have:

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad 3\mathbf{v} = \begin{pmatrix} 6 \\ -9 \end{pmatrix}.$$

□

2.2 Matrices and linear equations

In Section 1.1 we also introduced the space $\mathbb{R}^{m \times n}$ consisting of matrices with m rows and n columns. We first note that these also form a vector space as

defined in Definition 2.1 provided we define addition as:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ b_{31} & b_{32} & \cdots & b_{3n} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ a_{31} + b_{31} & a_{32} + b_{32} & \cdots & a_{3n} + b_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

and scalar multiplication as:

$$c \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ ca_{31} & ca_{32} & \cdots & ca_{3n} \\ \vdots & \vdots & & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{pmatrix}$$

Example 2.3 Consider the space $\mathbb{R}^{3 \times 2}$. The following matrices are in this space:

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

We have:

$$A + B = \begin{pmatrix} 3 & 3 \\ 3 & 0 \\ 1 & 2 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & 6 \\ 6 & 3 \\ 3 & 3 \end{pmatrix}. \quad \square$$

Note that the space $\mathbb{R}^k$ of column vectors and the space $\mathbb{R}^{k \times 1}$ of matrices with exactly one column are basically the same.

Let us first define a number of special matrices which are often used:

- A is called **square** if it has the same number of columns and rows.

- A is called the $k \times n$ **zero matrix**, notation 0 , if all a_{ij} are equal to zero.
- A is called a **diagonal matrix** of size n if A is square and all a_{ij} with $i \neq j$ are equal to zero.
- A is called the **identity matrix** of size n , notation I_n , if A is a diagonal matrix with all diagonal elements $a_{ii} = 1$.
- A is called an **upper triangular matrix** if $a_{ij} = 0$ if $i > j$.
- A is called a **lower triangular matrix** if $a_{ij} = 0$ if $i < j$.

Often, we will use for the identity matrix simply I instead of I_n where we deduce the size of the matrix from the context. For instance, if we write $I - A$ where A is a square 4×4 matrix then we conclude that the identity matrix must also have size 4×4 . Clearly, we can then also write I_4 instead of I to make this explicit.

We noted earlier that the set $\mathbb{R}^{k \times n}$ of matrices with k rows and n columns form a vector space. In a vector space we always have a zero element. In this case that is precisely the zero matrix.

In the case of matrices, we can however also define a product if the size of the matrices are appropriate. We first define the product of a matrix and a vector.

Definition 2.4 Given a matrix A in $\mathbb{R}^{m \times n}$ and a vector $\mathbf{x}$ in $\mathbb{R}^n$, we define:

$$\mathbf{Ax} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n \end{pmatrix}$$

provided

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix}$$

We note that $\mathbf{Ax}$ is in $\mathbb{R}^m$.

Using the above definition, we note that a linear system (1.5) can be written as:

$$\mathbf{Ax} = \mathbf{b} \tag{2.1}$$

where A is the coefficient matrix and $\mathbf{x}$ and $\mathbf{b}$ as defined in (1.6). Note that this matrix-vector product has some important properties:

Lemma 2.5 Consider an $m \times n$ matrix A , vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and a constant $\alpha \in \mathbb{R}$. We have:

- $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$
- $A(\alpha\mathbf{u}) = (\alpha A)\mathbf{u} = \alpha(A\mathbf{u})$

Using the above matrix-vector product, we will now define the product of two matrices:

Definition 2.6 Given a matrix A in $\mathbb{R}^{m \times n}$ and a matrix $B \in \mathbb{R}^{k \times \ell}$ where

$$B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1\ell} \\ b_{21} & b_{22} & \cdots & b_{2\ell} \\ b_{31} & b_{32} & \cdots & b_{3\ell} \\ \vdots & \vdots & & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{k\ell} \end{pmatrix} = (\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_\ell)$$

Note that B can be seen as a collection of columns where

$$\mathbf{b}_i = \begin{pmatrix} b_{1i} \\ b_{2i} \\ b_{3i} \\ \vdots \\ b_{ki} \end{pmatrix}$$

Now the product $AB \in \mathbb{R}^{m \times \ell}$ with $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{k \times \ell}$ is defined as:

$$AB = (\mathbf{Ab}_1 \quad \mathbf{Ab}_2 \quad \cdots \quad \mathbf{Ab}_\ell)$$

provided $n = k$. So the matrix product AB is only defined if the number of columns of A is equal to the number of rows of B .

We should also note that AB and BA are **completely** different objects. It is even possible that AB is defined while BA is not defined.

Example 2.7 Consider the matrices A and B of Example 2.3. Note that AB is not defined since the number of columns of A (which is 2) is not equal to the number of rows of B (which is 3). Similarly we can note that BA is not defined either.

Next consider

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}.$$

In that case AB is well-defined:

$$AB = (\mathbf{A}\mathbf{b}_1 \quad \mathbf{A}\mathbf{b}_2 \quad \mathbf{A}\mathbf{b}_3)$$

where

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

We find:

$$AB = \begin{pmatrix} 0 & 1 & 3 \\ 3 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}.$$

Similar as above, we can also compute BA . We obtain:

$$BA = \begin{pmatrix} 5 & 6 \\ 0 & -1 \end{pmatrix}.$$

Note that BA and AB are indeed completely different objects which do not even have the same size. $\square$

The matrix multiplication has a number of useful properties:

Lemma 2.8 *We have:*

- $A(BC) = (AB)C$
- $A(B + C) = AB + AC$
- $(A + B)C = AC + BC$
- $A(\alpha B) = (\alpha A)B = \alpha(AB)$.

provided the matrices have the appropriate size such that all matrix products are well-defined.

Note that the second and fourth properties in the lemma above reduce to the result of Lemma 2.5 in case B and C contain only one column.

The identity matrix I has nice properties with respect to the multiplication. We note that $IA = A$ for any matrix A when the size of I is equal to the number of rows of A . On the other hand, $AI = A$ if the size of I is equal to the number of columns of A .

Using matrix multiplication, we can also define the power of a **square** matrix A . We note:

$$A^0 = I, \quad A^r A^s = A^{r+s}, \quad (A^r)^s = A^{rs} \quad (2.2)$$

where r and s are natural numbers. Here I is an identity matrix which has the same size as A .

It is easy to make mistakes with the multiplication of matrices since a number of standard properties are no longer true:

- In most cases we will have $AB \neq BA$.
- If we have $AB = 0$ where 0 refers to the zero matrix then we cannot conclude that $A = 0$ or $B = 0$.
- If we have $AC = AD$ with $A \neq 0$ then we cannot conclude that $C = D$.

The first issue we have seen before. The latter two issues are illustrated by the following two examples:

Example 2.9 Consider:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

then it is easily verified that we have $AB = 0$ but clearly neither A nor B equals the zero matrix. Define additionally:

$$C = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 4 & 1 \end{pmatrix}$$

then $AC = AD$ for a nonzero A but clearly we do not have that $C = D$. $\square$

An interesting property is that the elementary row operations can be seen as matrix multiplications. In other words, if we can obtain matrix A_2 from matrix A_1 by an elementary row operation then there exists a matrix T such that $TA_1 = A_2$. Note that the converse is clearly not the case. One key feature of an elementary row operation is that it is reversible in the sense that there also exists a matrix S such that $SA_2 = A_1$. The matrix T described above can be found by applying the given elementary row operation to an identity matrix. This is illustrated in the following example.

Example 2.10 Consider

$$\begin{aligned} A_1 &= \begin{pmatrix} 2 & -1 & 1 & -1 & -6 \\ -1 & 1 & 5 & 2 & 2 \\ 2 & -1 & 2 & -3 & 3 \end{pmatrix} & A_2 &= \begin{pmatrix} -1 & 1 & 5 & 2 & 2 \\ 2 & -1 & 1 & -1 & -6 \\ 2 & -1 & 2 & -3 & 3 \end{pmatrix} \\ A_3 &= \begin{pmatrix} 1 & -1 & -5 & -2 & -2 \\ 2 & -1 & 1 & -1 & -6 \\ 2 & -1 & 2 & -3 & 3 \end{pmatrix} & A_4 &= \begin{pmatrix} 1 & -1 & -5 & -2 & -2 \\ 0 & 1 & 11 & 3 & -2 \\ 2 & -1 & 2 & -3 & 3 \end{pmatrix} \end{aligned}$$

We obtain A_2 from A_1 by interchanging the first two rows. If we apply this elementary row operation to the identity matrix, we get:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and hence if we label the matrix we thus obtained as T_1

$$T_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

then we can easily verify that $A_2 = T_1 A_1$. Next, we obtain A_3 from A_2 by multiplying the first row by -1 . If we apply this elementary row operation to the identity matrix, we get:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and hence if we label the matrix we thus obtained as T_2

$$T_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

then we can easily verify that $A_3 = T_2 A_2$. Finally, we obtain A_4 from A_3 by subtracting two times the first row from the second row. If we apply this elementary row operation to the identity matrix, we get:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and hence if we label the matrix we thus obtained as T_3

$$T_3 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

then we can easily verify that $A_4 = T_3 A_3$.

All these steps are reversible. Interchanging two rows is of course trivially reversed by reversing the two rows again. In other words, applying the same elementary row operation. Hence for $S_1 = T_1$ we get $A_1 = S_1 A_2$. Multiplying a row by -1 is of course reversed by multiplying this row once more with -1 and hence we again obtain that for $S_2 = T_2$ we have $A_2 = S_2 A_3$. The final elementary row operation was to subtract two times the first row from the second row. This is of course reversed by **adding** two times the first row to the second row. So in this case we choose:

$$S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and we obtain that $A_3 = S_3 A_4$. □

2.3 Matrix inverse

Definition 2.11 A square matrix A is called **invertible** if there exists a matrix B of the same size such that

$$BA = AB = I$$

where I is the identity matrix of the same size as A . In this case, the matrix A is also called **regular** and the matrix B is referred to as the **inverse** of A . If such a matrix B does not exist then we call the matrix A **singular**.

It is clear that not all matrices are invertible. For instance, the zero matrix is clearly not invertible. We note some basic properties of the inverse:

Theorem 2.12 For an invertible matrix A , the inverse is always unique.

Proof : Assume B and C are both inverses of A and therefore: $AB = BA = I$ and $AC = CA = I$. In that case:

$$B = BI = B(AC) = (BA)C = IC = C$$

which shows that B and C are equal. ■

Since the inverse is unique, we can use the notation A^{-1} to denote the inverse. Another useful property is that we only need to check $AB = I$ or $BA = I$:

Lemma 2.13 *Given is a square matrix A . If there exists a matrix B such that:*

$$AB = I$$

then A is invertible and $A^{-1} = B$.

If there exists a matrix C such that:

$$CA = I$$

then A is invertible and $A^{-1} = C$.

We skip the rather technical proof. One reason to determine the inverse of a matrix A has to do with linear systems:

Theorem 2.14 *Consider a square, invertible matrix A . In that case the linear system:*

$$A\mathbf{x} = \mathbf{b}$$

has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

On the other hand for a square, singular matrix A there always exists at least one $\mathbf{b}$ for which

$$A\mathbf{x} = \mathbf{b} \tag{2.3}$$

has no solution. If for a singular matrix A and vector $\mathbf{b}$, the linear system (2.3) has a solution, then the linear system actually has an infinite number of solutions.

In other words, for a regular matrix A the associated linear system always has a unique solution. On the other hand for a singular matrix A we either have no solution or an infinite number of solutions.

We clearly still need a technique to determine whether a matrix A is invertible and also a technique to compute the inverse. Consider an $n \times n$ matrix A . In order to check whether this matrix is invertible, we need to check whether a matrix X exists such that:

$$AX = I$$

Let the columns of X be denoted by $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ respectively. The columns of I are denoted by: $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$. Note that $AX = I$ is therefore equivalent to:

$$A\mathbf{x}_1 = \mathbf{e}_1, \quad A\mathbf{x}_2 = \mathbf{e}_2, \quad \dots, \quad A\mathbf{x}_n = \mathbf{e}_n$$

where $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are unknown. In other words, we need to check whether

$$A\mathbf{x}_i = \mathbf{e}_i$$

is consistent for $i = 1, \dots, n$. That is, we need to solve n linear systems. The matrix A is invertible if all these linear systems have a solution. Moreover, in that case, these solutions then immediately yield all columns of the inverse A^{-1} . With the techniques presented in Chapter 1 we can check the existence of solutions to these linear equations as well as compute the solution. However, these linear systems all have the same coefficient matrix A and therefore we can actually compute the inverse more efficiently than a method based on solving n linear systems.

The idea is based on the fact that solving two linear systems

$$A\mathbf{x} = \mathbf{b}_1, \quad A\mathbf{x} = \mathbf{b}_2,$$

can be done simultaneously by using the reduction to the echelon form of the matrix:

$$\left(A \mid \mathbf{b}_1 \quad \mathbf{b}_2 \right)$$

If we obtain:

$$\left(A \mid \mathbf{b}_1 \quad \mathbf{b}_2 \right) \sim \left(\tilde{A} \mid \tilde{\mathbf{b}}_1 \quad \tilde{\mathbf{b}}_2 \right)$$

then $A\mathbf{x} = \mathbf{b}_1$ is equivalent to $\tilde{A}\mathbf{x} = \tilde{\mathbf{b}}_1$. Similarly, $A\mathbf{x} = \mathbf{b}_2$ is equivalent to $\tilde{A}\mathbf{x} = \tilde{\mathbf{b}}_2$.

For computation of the inverse, we consider the following expanded matrix:

$$\left(A \mid I \right) = \left(A \mid \mathbf{e}_1 \quad \mathbf{e}_2 \quad \cdots \quad \mathbf{e}_n \right) \quad (2.4)$$

which we can bring into the reduced echelon form:

$$\left(B \mid C \right) \quad (2.5)$$

We have the following result:

Theorem 2.15 *The matrix A is invertible if and only if the reduced echelon form (2.5) of (2.4) has the property that $B = I$. Moreover, in that case $A^{-1} = C$.*

Example 2.16 Consider the matrix A :

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 4 & 2 & 3 \\ 5 & 3 & 7 \end{pmatrix}$$

To check whether the inverse exists, we consider the following:

$$\begin{aligned}
 \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 4 & 2 & 3 & 0 & 1 & 0 \\ 5 & 3 & 7 & 0 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 2 & 7 & -4 & 1 & 0 \\ 0 & 3 & 12 & -5 & 0 & 1 \end{array} \right) \sim \\
 &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{7}{2} & -2 & \frac{1}{2} & 0 \\ 0 & 3 & 12 & -5 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{7}{2} & -2 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{3}{2} & 1 & -\frac{3}{2} & 1 \end{array} \right) \sim \\
 &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & \frac{7}{2} & -2 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{2}{3} & -1 & \frac{2}{3} \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{5}{3} & -1 & \frac{2}{3} \\ 0 & 1 & 0 & -\frac{13}{3} & 4 & -\frac{7}{3} \\ 0 & 0 & 1 & \frac{2}{3} & -1 & \frac{2}{3} \end{array} \right)
 \end{aligned}$$

We see that the first matrix in the row-reduced echelon form equals I and hence the matrix A is invertible. Moreover, the second matrix is equal to its inverse. In other words,

$$A^{-1} = \begin{pmatrix} \frac{5}{3} & -1 & \frac{2}{3} \\ -\frac{13}{3} & 4 & -\frac{7}{3} \\ \frac{2}{3} & -1 & \frac{2}{3} \end{pmatrix}$$

To check that this result is correct: verify that

$$AA^{-1} = I.$$

□

Example 2.17 Consider the matrix A :

$$A = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 2 & -1 & -1 \end{pmatrix}$$

To check whether the inverse exists, we consider the following:

$$\begin{aligned}
 \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 2 & -1 & -1 & 0 & 0 & 1 \end{array} \right) &\sim \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 3 & -3 & -2 & 0 & 1 \end{array} \right) \sim \\
 &\sim \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -2 & -3 & 1 \end{array} \right)
 \end{aligned}$$

This is clearly not yet equal to the reduced echelon form. But we can immediately see that if we continue with our elementary row operations to bring the matrix in reduced echelon form, the first matrix will not change any further and therefore will never become equal to the identity matrix. Hence the matrix A is singular (i.e. it is not invertible). □

2.4 Transpose of a matrix

We define the transpose of a matrix:

Definition 2.18 The **transpose** of A , denoted by A^T , is defined as the matrix obtained by interchanging the rows and columns. In other words, if a_{ij} and b_{ij} are the elements in the i th row and j th column of A and A^T respectively then we have $b_{ij} = a_{ji}$ for all i and j .

A matrix A is called **symmetric** if $A^T = A$. Note that a symmetric matrix is always square.

Example 2.19 Consider the matrices

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 6 & 5 \\ 7 & 9 & 8 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 3 & 2 \\ 3 & 6 & 5 \\ 2 & 5 & 8 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 3 \\ 2 & 6 \\ 7 & 9 \end{pmatrix}$$

then we have:

$$A^T = \begin{pmatrix} 1 & 2 & 7 \\ 3 & 6 & 9 \\ 2 & 5 & 8 \end{pmatrix}, \quad C^T = \begin{pmatrix} 1 & 2 & 7 \\ 3 & 6 & 9 \end{pmatrix}$$

while B is a symmetric matrix, i.e. $B^T = B$. □

We have the following useful properties for the transpose:

Lemma 2.20 We have:

- $(A^T)^T = A$ for any $k \times n$ matrix A
- $(A + B)^T = A^T + B^T$ for any $k \times n$ matrices A and B
- $(AB)^T = B^T A^T$ for any $k \times n$ matrix A and any $n \times m$ matrix B .

Proof : The first two properties are easily verified. The last property is a bit harder to check. Let a_{ij} and b_{ij} be the coefficients of A and B respectively. Then it can be checked that

$$(AB)_{ij} = \sum_{s=1}^n a_{is} b_{sj}$$

but then

$$(B^T A^T)_{ij} = \sum_{s=1}^n b_{si} a_{js}$$

It is then easily verified that we have:

$$(B^T A^T)_{ji} = \sum_{s=1}^n b_{sj} a_{is} = \sum_{s=1}^n a_{is} b_{sj} = (AB)_{ij}$$

which shows that $B^T A^T$ is the transpose of AB . ■

2.5 Exercises

2.5.1 Self study exercises

2.1 Let

$$\mathbf{u} = \begin{pmatrix} 2 \\ 0 \\ -1 \\ 3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 5 \\ 4 \\ 7 \\ -1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 6 \\ 2 \\ 0 \\ 9 \end{pmatrix}$$

- (a) Determine the vectors $3(\mathbf{u} - 7\mathbf{v})$ and $2\mathbf{v} - (\mathbf{u} + \mathbf{w})$.
- (b) Determine the vector $\mathbf{x}$ such that

$$2\mathbf{u} - \mathbf{v} + \mathbf{x} = 7\mathbf{x} + \mathbf{w}.$$

2.2 Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 2 \\ 3 & -1 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & -2 & 2 \\ 1 & 0 & -1 \\ 2 & 3 & 1 \end{pmatrix}$$

- (a) Determine the matrix $3A + 2B$.
- (b) Determine the matrices AB and BA .
- (c) Determine the matrix A^T .
- (d) Determine B^0 and B^2 Hint: recall equation (2.2).

2.3 Let

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}.$$

Show that $AI = A$ and $IA = A$ using Definition 2.6. Note that the size of I depends on whether A is multiplied on the left or on the right!

(conclusion: in matrix multiplication the identity matrix I has similar properties as the number 1 in multiplication with numbers).

2.4 Determine the inverse (if it exists) of the following matrices:

$$A = \begin{pmatrix} -2 & 5 \\ 1 & -3 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

2.5 Under what condition(s) are square upper and square lower triangular matrices invertible? Explain!

2.5.2 Tutorial exercises

2.6 The matrices A , B and C are given by:

$$A = \begin{pmatrix} -1 & 3 \\ 0 & 4 \\ -3 & 2 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 2 & -4 \end{pmatrix}; \quad C = \begin{pmatrix} 0 & 3 & -2 \\ 4 & -1 & 2 \end{pmatrix}.$$

Determine the following products if they exist: AB , BA , AC , CA , BC and CB .

2.7 Consider the problem in Exercise 1.9 and its solution $f(x) = 2x^2 - 7x + 8$. Write the corresponding linear system in the form $Ax = \mathbf{b}$ and verify, using Definition 2.4, that this equation holds.

2.8 Let A and B be matrices such that the product AB is defined.

- (a) If $B \in \mathbb{R}^{4 \times 7}$ and $AB \in \mathbb{R}^{5 \times 7}$, what is the size of A ?
- (b) If all elements in the last column of B are zero, what can you say about the matrix AB ? Explain!
- (c) Suppose

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \quad \text{and} \quad AB = \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}.$$

Determine B .

2.9 A is a 4×3 matrix. Construct (in each case) a matrix T such that TA is obtained from A by

- (a) adding 2 times the first row to the third row
- (b) multiplying the second row by 3.
- (c) interchanging the second and the fourth row.
- (d) First interchanging the first and the third row, and then subtracting three times the fourth row from the third row.

- (e) Matrices that correspond to elementary row operations are called *elementary matrices*. Explain why each elementary matrix is invertible.

2.10 Consider Example 2.16. Verify that $AA^{-1} = I$ and $A^{-1}A = I$.

2.11 Consider Theorem 2.14, where $A \in \mathbb{R}^{n \times n}$ is an invertible matrix.

- (a) Show that $\mathbf{x} = A^{-1}\mathbf{b}$ is indeed a solution of the system $A\mathbf{x} = \mathbf{b}$.
 (b) Show that if $\mathbf{v}$ is a solution of $A\mathbf{x} = \mathbf{b}$, then necessarily $\mathbf{v} = A^{-1}\mathbf{b}$.
 (so $A^{-1}\mathbf{b}$ is the *unique* solution of the system $A\mathbf{x} = \mathbf{b}$)

2.12 The matrices A and B and the vectors $\mathbf{b}_1$, $\mathbf{b}_2$ and $\mathbf{b}_3$ are given by:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 3 & 4 \end{pmatrix}; \quad B = \begin{pmatrix} 3 & -1 & -1 \\ -1 & -1 & 2 \\ 0 & 1 & -1 \end{pmatrix};$$

$$\mathbf{b}_1 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}; \quad \mathbf{b}_2 = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}; \quad \mathbf{b}_3 = \begin{pmatrix} 7 \\ -6 \\ 4 \end{pmatrix}$$

- (a) Show, without computing A^{-1} , that $B = A^{-1}$.
 (b) Use part (a) to determine the solution sets of $A\mathbf{x} = \mathbf{b}_1$, $A\mathbf{x} = \mathbf{b}_2$ and $A\mathbf{x} = \mathbf{b}_3$.

2.13 The matrix A is given by:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

- (a) Show that if $ad - bc \neq 0$ then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

- (b) Determine in two ways the inverse of

$$A = \begin{pmatrix} 2 & -8 \\ -1 & 3 \end{pmatrix};$$

by using part (a) and by using the procedure of Example 2.16.

2.14 Determine the inverse (if it exists) of the following matrices:

$$A = \begin{pmatrix} 2 & 1 & -5 \\ 1 & 2 & -7 \\ -2 & -2 & 8 \end{pmatrix}; \quad B = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix}.$$

2.15 Consider Theorem 2.15 and study carefully Examples 2.16 and 2.17. Let $A \in \mathbb{R}^{n \times n}$.

- (a) Suppose $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^n$.
Explain that A must be invertible.
- (b) Suppose there exists $\mathbf{b} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$ is inconsistent. What can you say about the number of solutions of $A\mathbf{x} = \mathbf{0}$?
- (c) Use Theorem 2.15 to prove the second part of Theorem 2.14.

2.16 Let $A, B \in \mathbb{R}^{n \times n}$ be invertible matrices. Show that:

- (a) $(A^{-1})^{-1} = A$.
- (b) $(A^{-1})^T = (A^T)^{-1}$.
- (c) $(AB)^{-1} = B^{-1}A^{-1}$.
- (d) Suppose that $A_1, \dots, A_k \in \mathbb{R}^{n \times n}$ are invertible matrices.
Give an expression for the inverse of the product $A_1 \cdots A_k$.

2.17 Let $A \in \mathbb{R}^{n \times n}$ be a square matrix.

- (a) Suppose $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^n$.
Is it possible that there exists $\mathbf{c} \in \mathbb{R}^n$ such that the system $A\mathbf{x} = \mathbf{c}$ has infinitely many solutions? Explain!
- (b) Suppose that the system $A\mathbf{x} = \mathbf{0}$ has a unique solution.
Explain that $A\mathbf{x} = \mathbf{b}$ has a unique solution for each $\mathbf{b} \in \mathbb{R}^n$.

2.18 (a) Explain that for $A \in \mathbb{R}^{m \times n}$ the products AA^T and $A^T A$ are always defined. Under what condition is the product AA defined?

- (b) Explain why in general $(AB)^T \neq A^T B^T$.
- (c) Show that $(ABC)^T = C^T B^T A^T$. (use Lemma 2.20)
- (d) Let $A \in \mathbb{R}^{n \times n}$ and $k \in \mathbb{N}$. Show that $(A^k)^T = (A^T)^k$.

2.19 (a) Let $A \in \mathbb{R}^{m \times m}$ and $B, C \in \mathbb{R}^{m \times n}$.

Show that if A is invertible and $AB = AC$, then $B = C$.

- (b) Let $A, B, P \in \mathbb{R}^{n \times n}$. Suppose that P is invertible and $A = PBP^{-1}$.
Express B in terms of A and P .
- (c) Let $A, B, X \in \mathbb{R}^{n \times n}$ be invertible matrices.
Solve the equation $(A^2 X)^{-1} = (AB^{-1})^{-1} (ABA^{-1})^2$ for X (use Exercise 2.16).
- (d) Let $A, B \in \mathbb{R}^{n \times n}$. Suppose that the product AB is invertible.
Show that both A and B must be invertible.

2.20 Consider the definition of a vector space V (Definition 2.1). Although in this definition the elements of V are called "vectors" (which is indeed the case if we take $V = \mathbb{R}^n$), vector spaces can also consist of other objects, like matrices or functions. For example, the set of all real valued functions on a fixed interval $[a, b]$ (with the usual addition and scalar multiplication for functions) is also a vector space.

Show that $\mathbb{R}^{m \times n}$ (the set of all matrices with m rows and n columns, with the addition and scalar multiplication defined in Section 2.2) is a vector space.

E.g: the second property of vector spaces: $A + B = B + A$ for all matrices $A, B \in \mathbb{R}^{m \times n}$ can be verified as follows:

For each row i and column j , the (i, j) -element of the matrix $A + B$ equals the (i, j) -element of the matrix $B + A$ since, by definition of matrix addition

$$(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}.$$

Therefore $A + B = B + A$.

Chapter 3

Subspaces and bases

The fundamental concept in linear algebra is the concept of vector space. A vector space is a collection (of vectors) on which two operations are defined: addition and scalar multiplication. The properties that we always impose are given in Definition 2.1. We have noted that $\mathbb{R}^n$ satisfies these properties. There are many other examples of vector spaces but in this chapter we will focus on $\mathbb{R}^n$. However, we will see that also certain subsets of $\mathbb{R}^n$ satisfy these properties and are therefore vector spaces themselves. We call these subsets linear subspaces.

3.1 Subspaces

Definition 3.1 A subset D of $\mathbb{R}^n$ is called a **(linear) subspace** of $\mathbb{R}^n$ if it satisfies the following three conditions:

- (i) $\mathbf{0} \in D$
- (ii) For all $\mathbf{u}$ and $\mathbf{v}$ in D we have that $\mathbf{u} + \mathbf{v} \in D$.
- (iii) For all $\alpha \in \mathbb{R}$ and $\mathbf{v} \in D$ we have that $\alpha\mathbf{v} \in D$.

Note that we will sometimes abbreviate "linear subspace" and simply call a set a "subspace". Note that the first condition is essentially superfluous since from the third property we obtain for $\alpha = 0$ that we have $\mathbf{0} \in D$. The only issue is that the first condition guarantees that the subset D cannot be the empty set.

We call a subset D closed under addition if the property (ii) is satisfied and closed under scalar multiplication if the property (iii) is satisfied. Note that if D is a linear subspace then also $-\mathbf{v} = (-1)\mathbf{v}$ is in D and for $\mathbf{u}$ and $\mathbf{v}$ in D we also have that $\mathbf{u} - \mathbf{v} \in D$. The smallest subspace in $\mathbb{R}^n$ is equal to $\{\mathbf{0}\}$ and the largest subspace in $\mathbb{R}^n$ is clearly $\mathbb{R}^n$ itself. Note that a linear subspace D is always a vector space and satisfies all conditions given in

Definition 2.1. We only need to verify the conditions in Definition 3.1 since the other conditions are automatically satisfied because we know that $\mathbb{R}^n$ is a vector space and satisfies all these properties in Definition 2.1.

Example 3.2 A line ℓ in $\mathbb{R}^n$ is a linear subspace of $\mathbb{R}^n$ if it goes through the origin. The fact that it needs to go through the origin is an immediate consequence of condition (i) of Definition 3.1. A line through the origin describes a set of points which can always be described as:

$$\ell = \{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \alpha \mathbf{q}, \alpha \in \mathbb{R} \}$$

where $\mathbf{q} \neq \mathbf{0}$ is an arbitrarily chosen vector on ℓ .

Assume $\mathbf{u} = \beta \mathbf{q}$ and $\mathbf{v} = \gamma \mathbf{q}$ in ℓ then:

$$\mathbf{u} + \mathbf{v} = \beta \mathbf{q} + \gamma \mathbf{q} = (\beta + \gamma) \mathbf{q}$$

and

$$\alpha \mathbf{v} = \alpha(\gamma \mathbf{q}) = (\alpha\gamma) \mathbf{q}$$

are both in ℓ . Hence ℓ is a linear subspace of $\mathbb{R}^n$.

We note that all vectors in this linear subspace are scalar multiples of one vector $\mathbf{q}$. We say that ℓ is generated or spanned by the vector $\mathbf{q}$ which we will denote by $\ell = \langle \mathbf{q} \rangle$. We will introduce this concept in a more general context later in this chapter. $\square$

3.2 Solution set of linear equations

We have seen subsets of $\mathbb{R}^n$ before. After all, we considered the solution set of a linear system of the form:

$$A\mathbf{x} = \mathbf{b} \tag{3.1}$$

with $A \in \mathbb{R}^{k \times n}$ and $\mathbf{b} \in \mathbb{R}^k$. The set of all $\mathbf{x}$ which satisfy this linear system is clearly a subset of $\mathbb{R}^n$. However, it is in general not a linear subspace. This is obvious since the first condition that a linear subspace needs to satisfy is that $\mathbf{0}$ must be in the set. For a linear system we have that $\mathbf{0}$ is in the solution set if and only if $\mathbf{b} = \mathbf{0}$. In that case we call the linear system **homogeneous**. For $\mathbf{b} \neq \mathbf{0}$ we refer to (3.1) as a **nonhomogeneous** linear system.

Lemma 3.3 *The solution set of a homogeneous linear system*

$$A\mathbf{x} = \mathbf{0}$$

with $A \in \mathbb{R}^{k \times n}$ is a linear subspace of $\mathbb{R}^n$.

Proof : Clearly, $\mathbf{0} \in D$ since $A\mathbf{0} = \mathbf{0}$. Now assume $\mathbf{u}$ and $\mathbf{v}$ are both in the solution set of the linear system. Hence we have:

$$A\mathbf{u} = \mathbf{0}, \quad A\mathbf{v} = \mathbf{0}$$

In that case we obtain:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0},$$

and

$$A(\alpha\mathbf{v}) = \alpha A\mathbf{v} = \alpha\mathbf{0} = \mathbf{0}.$$

This clearly implies that both $\mathbf{u} + \mathbf{v}$ and $\alpha\mathbf{v}$ are in the solution set. Hence the solution set is a linear subspace of $\mathbb{R}^n$. ■

The subspace defined here is called the null space of the matrix A :

Definition 3.4 The **null-space** of a $k \times n$ -matrix A is defined as the following subspace of $\mathbb{R}^n$:

$$\text{Null } A = \{ \mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0} \}$$

In other words, the null-space of the matrix A is nothing else than the solution set of the homogeneous linear system $A\mathbf{x} = \mathbf{0}$. Normally, to find the solution set of a linear system, we compute the reduced echelon form of the augmented matrix. For a homogeneous linear system, it is sufficient to compute the reduced echelon form $\tilde{A}$ of the coefficient matrix A because the last column of the augmented matrix contains merely zeros and will therefore never change when performing elementary row operations. We have:

$$\text{Null } A = \text{Null } \tilde{A}.$$

which is basically a direct consequence of Theorem 1.14.

3.3 Linear combinations and span

In Example 3.2, we have seen a subspace generated by a single vector. In this section we consider subspaces generated by multiple vectors:

Definition 3.5 A vector $\mathbf{w}$ is said to be a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ if $\mathbf{w}$ can be expressed in the following form:

$$\mathbf{w} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \dots + \alpha_p\mathbf{v}_p$$

for certain $\alpha_1, \alpha_2, \dots, \alpha_p \in \mathbb{R}$.

Example 3.6 Let

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix}$$

be two vectors in $\mathbb{R}^3$. Next consider two other vectors

$$\mathbf{w}_1 = \begin{pmatrix} 7 \\ 9 \\ 2 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 8 \\ 4 \\ -1 \end{pmatrix}.$$

In that case, $\mathbf{w}_1$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. On the other hand, $\mathbf{w}_2$ is **not** a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. This is clear since:

$$\mathbf{w}_1 = -3\mathbf{v}_1 + 2\mathbf{v}_2$$

while $\mathbf{w}_2 = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2$ would imply that:

$$\begin{cases} -\alpha_1 + 2\alpha_2 = 8 \\ \alpha_1 + 6\alpha_2 = 4 \\ 2\alpha_1 + 4\alpha_2 = -1 \end{cases}$$

We solve this linear system:

$$\left(\begin{array}{cc|c} -1 & 2 & 8 \\ 1 & 6 & 4 \\ 2 & 4 & -1 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & -2 & -8 \\ 0 & 8 & 12 \\ 0 & 8 & 15 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & -2 & -8 \\ 0 & 8 & 12 \\ 0 & 0 & 3 \end{array} \right)$$

We clearly see that this linear system has no solutions. Therefore $\mathbf{w}_2$ cannot be written as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. $\square$

Often a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is given and we are looking for the smallest linear subspace $D \subseteq \mathbb{R}^n$ which contains these vectors. Clearly every vector $\mathbf{w}$ which is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ should be in D (see Exercise 3.3). We now look at the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ which turns out to already be a linear subspace. After all, if

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \dots + \alpha_p\mathbf{v}_p$$

$$\mathbf{w} = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \dots + \beta_p\mathbf{v}_p$$

then:

$$\begin{aligned} \beta\mathbf{v} &= \beta(\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \dots + \alpha_p\mathbf{v}_p) \\ &= (\beta\alpha_1)\mathbf{v}_1 + (\beta\alpha_2)\mathbf{v}_2 + \dots + (\beta\alpha_p)\mathbf{v}_p \end{aligned}$$

and

$$\begin{aligned}\mathbf{v} + \mathbf{w} &= (\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_p \mathbf{v}_p) + (\beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \cdots + \beta_p \mathbf{v}_p) \\ &= (\alpha_1 + \beta_1) \mathbf{v}_1 + (\alpha_2 + \beta_2) \mathbf{v}_2 + \cdots + (\alpha_p + \beta_p) \mathbf{v}_p\end{aligned}$$

Obviously $\mathbf{0}$ is also a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ and hence we find that D equals the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$:

Definition 3.7 Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ be vectors in $\mathbb{R}^n$. We call:

$$\begin{aligned}\langle \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \rangle \\ = \{ \mathbf{w} \in \mathbb{R}^n \mid \mathbf{w} \text{ is a linear combination of } \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \}\end{aligned}$$

the linear subspace **spanned** by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. We call $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ the **generators** of this subspace.

We will often use the alternative notation:

$$\text{Span}\{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \}$$

for this linear subspace spanned by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$.

Using vectors, we can describe the solution set of a system of linear equations differently:

Example 3.8 In example 1.16, we described the solution set in a parametric form:

$$\begin{cases} x_1 = -13x_4 - 58 \\ x_2 = -25x_4 - 101 \\ x_3 = 2x_4 + 9 \\ x_4 \text{ is free} \end{cases}$$

We can, equivalently, describe the solution set using vectors:

$$\left\{ \mathbf{x} \in \mathbb{R}^4 \mid \mathbf{x} = \begin{pmatrix} -58 \\ -101 \\ 9 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -13 \\ -25 \\ 2 \\ 1 \end{pmatrix} \text{ for some } x_4 \in \mathbb{R} \right\}$$

Note that this set describes a line in $\mathbb{R}^4$. □

The solution set of a homogeneous linear system can always be described by:

$$\text{Span}\{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \}$$

For a consistent, nonhomogeneous linear system, we clearly cannot use the above description since the solution set is no longer a linear subspace. However, the solution set of a system of linear equations can then always be described by:

$$\mathbf{p} + \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\} \quad (3.2)$$

where $\mathbf{p}$ is one particular solution of the system. An element $\mathbf{x}$ of this set has the form

$$\mathbf{x} = \mathbf{p} + \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_p \mathbf{v}_p$$

where $\alpha_1, \alpha_2, \dots, \alpha_p \in \mathbb{R}$. The notation (3.2) is called the solution set in **parametric vector form**. We have $\mathbf{p} = \mathbf{0}$ in case of a homogeneous linear system.

Example 3.9 In Example 3.8 the solution set can be described in parametric vector form as follows:

$$\begin{pmatrix} -58 \\ -101 \\ 9 \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -13 \\ -25 \\ 2 \\ 1 \end{pmatrix} \right\} \quad (3.3)$$

Note that this description of the solution set is not unique. It can be verified (see Exercise 3.4) that the solution set can also be described by:

$$\begin{pmatrix} -6 \\ -1 \\ 1 \\ -4 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 13 \\ 25 \\ -2 \\ -1 \end{pmatrix} \right\} \quad (3.4)$$

The solution set is clearly unique; it is only in our description of this set that we have some freedom. $\square$

Since the solution set of a homogeneous linear system $A\mathbf{x} = \mathbf{0}$ is nothing else than the null-space of the matrix A , we obtain:

$$\text{Null } A = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

for suitably chosen $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. We find the following characterization:

Theorem 3.10 Consider the nonhomogeneous linear system

$$A\mathbf{x} = \mathbf{b}$$

Assume that $\mathbf{p}$ is one particular solution of the system.
In that case the solution set of the system is given by:

$$U = \mathbf{p} + \text{Null } A$$

Proof : See Exercise 3.10. ■

A similar concept is the set of all $\mathbf{b}$ for which the linear system $A\mathbf{x} = \mathbf{b}$ is consistent. In other words, we consider the set of all $\mathbf{b}$ for which there exists $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$.

Definition 3.11 Given is a $k \times n$ matrix A . We define the following set:

$$\{\mathbf{b} \in \mathbb{R}^k \mid \text{there exists } \mathbf{x} \in \mathbb{R}^n \text{ such that } A\mathbf{x} = \mathbf{b}\}$$

This set is called the **column-space** of the matrix A and will be denoted by $\text{Col } A$.

The column-space is a subspace. After all, if for $\mathbf{b}_1$ and $\mathbf{b}_2$ there exists $\mathbf{x}_1$ and $\mathbf{x}_2$ such that

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2$$

then we have:

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{b}_1 + \mathbf{b}_2$$

and

$$A(c\mathbf{x}_1) = cA\mathbf{x}_1 = c\mathbf{b}_1$$

Therefore $\mathbf{b}_1 + \mathbf{b}_2$ and $c\mathbf{b}_1$ are both in the column space as well. Also $\mathbf{0} \in \text{Col } A$, since the system $A\mathbf{x} = \mathbf{0}$ has the solution $\mathbf{x} = \mathbf{0}$. This proves the column space is a subspace.

Note that if $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of the $k \times n$ matrix A then we have:

$$\text{Col } A = \text{Span } \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$$

3.4 Linear independence and bases

In the previous section, we considered the span of a set of vectors. The parametric vector form expresses the solution set of a linear system using this concept. For a homogeneous linear system the solution set is equal to:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

while for nonhomogeneous linear systems we get:

$$\mathbf{p} + \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

As noted in Example 3.9, the parametric vector form is not unique. In particular, not even the number of elements in the span is fixed:

Example 3.12 We have:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$$

where:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

After all, we have: $\mathbf{v}_3 = 3\mathbf{v}_1 - \mathbf{v}_2$. But then:

$$\begin{aligned} \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 &= \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 (3\mathbf{v}_1 - \mathbf{v}_2) \\ &= (\alpha_1 + 3\alpha_3) \mathbf{v}_1 + (\alpha_2 - \alpha_3) \mathbf{v}_2 \end{aligned}$$

which shows that any linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ can also be represented as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$. $\square$

In the above, we have given an example where the span of three vectors can be represented equally well by two vectors. In such a case we call the set of vectors dependent:

Definition 3.13 A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is called **(linearly) dependent** if

- either $p = 1$ and $\mathbf{v}_1 = \mathbf{0}$
- or $p > 1$ and one of the vectors in S is a linear combination of the other vectors in S .

A set of vectors S is called **(linearly) independent** if it is not linearly dependent.

The following characterization is useful when we want to check if a set is linearly dependent or independent.

Theorem 3.14 A set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is (linearly) independent if and only if:

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$$

implies that $\alpha_1 = \alpha_2 = \dots = \alpha_p = 0$.

The proof of the above theorem is addressed in Exercise 3.16c.

Example 3.15 Consider the set of vectors:

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}$$

In order to check whether the set of vectors is linearly dependent or independent we consider:

$$\alpha_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \mathbf{0}.$$

This can be rewritten as:

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \mathbf{0}.$$

In order to solve this linear system, we consider the coefficient matrix:

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}.$$

Using elementary row operations we get:

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

where we first subtracted the first row from the second and third row. Next,

we multiply the second row by -1 . In the next step we subtract 2 times the second row from the first row. In the final step we add the third row to the second row and subtract 3 times the third row from the first row. It is then easily seen that our homogeneous linear system only has the trivial solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Therefore, our set S is linearly independent. $\square$

Each subspace V of $\mathbb{R}^n$ can be written as the span of a number of vectors:

$$V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

for suitable $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. We have seen in Example 3.12 that the number of vectors that span V is not fixed. Clearly, we would like to have a minimal number of vectors which span V . In this regard, the following definition is extremely useful:

Definition 3.16 If V is a subspace of $\mathbb{R}^n$ given by:

$$V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

with $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ linearly independent then we call the set:

$$\mathcal{A} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

a **basis** of the subspace V . In that case, the number p is called the **dimension** of the subspace V .

Theorem 3.17 Let V be a subspace of $\mathbb{R}^n$. The subspace V has a basis. Moreover, assume

$$\mathcal{A} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

is a basis for the subspace V . In that case we have:

- $p \leq n$. If $p = n$ then $V = \mathbb{R}^n$.
- If we have an alternative representation for the subspace V :

$$V = \text{Span}\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q\}$$

for certain $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q$ then we have $p \leq q$. Moreover, $p = q$ if the vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q$ are linearly independent.

The proof of the above theorem is addressed in Exercise 3.16de. This theorem implies that the dimension of the subspace V is well-defined since every basis for the subspace V has the same number of elements (see Exercise 3.19). If we have a number of vectors that span the subspace V then we can use this to obtain a basis for the subspace V .

Lemma 3.18 *Assume*

$$V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ are linearly dependent then there exists a vector $\mathbf{v}_i$ which can be expressed in terms of the other vectors $\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_p$. Moreover,

$$V = \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_p\} \quad (3.5)$$

Proof : Since the vectors are linearly dependent, we know from Definition 3.13 that there exists one vector, say $\mathbf{v}_i$, which can be expressed in terms of the other vectors. This implies (3.5) since if an arbitrary vector $\mathbf{x}$ is expressed as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ then we can use the expression for $\mathbf{v}_i$ in terms of the other vectors, to eliminate $\mathbf{v}_i$ which implies that $\mathbf{x}$ is expressed in terms of the remaining vectors of the set. ■

The above lemma enables us to trim a dependent set of vectors that spans the subspace V by leaving out one element. Afterwards, either the remaining set of vectors is a basis for the subspace V or they still are dependent in which case we can repeat this reduction step. After a finite number of steps we will have found a basis for the subspace V . Note that the above works but it is computationally very inefficient.

However, there are better ways to determine a basis for a subspace. We are given a set of vectors:

$$\mathcal{A} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}. \quad (3.6)$$

We then construct the matrix A whose columns are exactly $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$.

The first method we will present, is using the echelon form. Let the echelon form of the matrix A be given by the matrix B . In that case, we know that the solution set of $A\mathbf{x} = \mathbf{0}$ is the same as the solution set of $B\mathbf{x} = \mathbf{0}$. After all, elementary row operations do not affect the solution set. Next, it is easily seen that in the echelon form all the columns containing a pivot element are independent. On the other hand, columns without a pivot element can be expressed in terms of the columns that contain a pivot element. Therefore, a basis for the span of the columns of B is given by the columns containing a pivot element.

But this was of course not the question that we set out to solve. We wanted to find a basis for the span of the columns of A . But now we use that the solution sets of $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ are the same. This implies that if certain columns of A are independent then the same columns of the matrix B are independent. Conversely, if certain columns of A are dependent then the same columns of the matrix B are dependent. Therefore, since we have a basis for the span of the columns of B , we can simply take the same columns of the matrix A and conclude that those columns are a basis for the span of the columns of A .

Example 3.19 Consider the set of vectors:

$$S = \left\{ \begin{pmatrix} 0 \\ 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 8 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 8 \\ -2 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -6 \\ -1 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ 9 \\ 3 \\ 1 \end{pmatrix} \right\}$$

We will construct a basis for $\text{Span } S$. To that end, we first construct a matrix which has the vectors in S as columns:

$$A = \begin{pmatrix} 0 & 0 & 0 & 2 & -2 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 1 & 3 & 2 & -2 & 3 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix}$$

We determine an echelon form for the matrix A :

$$\begin{pmatrix} 0 & 0 & 0 & 2 & -2 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 1 & 3 & 2 & -2 & 3 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 2 & 8 & 8 & -6 & 4 \\ 2 & 3 & -2 & -1 & 9 \\ 0 & 0 & 0 & 2 & -2 \\ 2 & 5 & 2 & 3 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 2 & 4 & -2 & -2 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix}$$

where we first interchanged the first and fourth row and then subtracted 2 times the first row from the second, third and fifth row. Continuing:

$$\begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 2 & 4 & -2 & -2 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & -3 & -6 & 3 & 3 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & -1 & -2 & 7 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 6 & -6 \end{pmatrix}$$

where we first divided the second row by a factor 2 and then added 3 times the second to the third row and added the second row to the fifth row. Finally:

$$\begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 6 & -6 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & -6 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 2 & -2 & 3 \\ 0 & 1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where we first interchanged the third and fourth row as well as dividing the resulting third row by 2 and then subtracted 6 times the third row from the fifth row. We obtain an echelon form in which the first, second and fourth columns contain a pivot element and are hence independent. Therefore, also the first, second and fourth columns of the original matrix A are independent and we conclude that a basis for $\text{Span } S$ is given by:

$$\left\{ \begin{pmatrix} 0 \\ 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 8 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ -6 \\ -1 \\ -2 \\ 3 \end{pmatrix} \right\} \quad \square$$

One important thing to note when using elementary row operations to determine a basis is that elementary row operations change the span of the columns of a matrix. Let us illustrate this via a simple example:

Example 3.20 Consider the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Clearly the two columns are linearly independent. Next, we apply an elementary row operation:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$$

by adding the first row to the third row. Clearly, as the theory predicts the two columns remain independent. On the other hand, we clearly have that:

$$\text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\} \neq \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\} \quad \square$$

Therefore, we find in Example 3.19 that the first, second and fourth column of A are independent vectors but we cannot use the columns of the reduced echelon form to determine a basis for the span of the columns of the original matrix A .

Another method is to use elementary column operations. This technique will preserve the span and therefore we can often find a basis for the span which has a simpler structure.

Consider the set $\mathcal{A}$ given by (3.6) and again construct the matrix A whose columns are exactly $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. To find a basis for

$$V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

which is equal to $\text{Col } A$, we can use **elementary column operations**:

- Adding a multiple of one column to another column.
- Multiplying one column with a number unequal to zero.
- Interchanging two columns.

The main thing to remember is that elementary **row** operations on the coefficient matrix A do not affect the solution set of the associated linear system $A\mathbf{x} = \mathbf{0}$. Similarly, elementary **row** operations on the augmented matrix do not affect the solution set of the associated linear system $A\mathbf{x} = \mathbf{b}$. However, elementary **row** operations do change $\text{Col } A$.

On the other hand, elementary **column** operations do not affect $\text{Col } A$. However, elementary **column** operations applied to a coefficient or augmented matrix do affect the solution set of the associated linear system.

Hence if we are interested in the solution of a linear system then we apply elementary **row** operations to the augmented or coefficient matrix. We also use elementary **row** operations to compute $\text{Null } A$ which after all is nothing else than the solution set of $A\mathbf{x} = \mathbf{0}$. On the other hand, if we want to compute a basis for $\text{Col } A$ then we may use elementary **row** operations as well as elementary **column** operations, as is shown in the following example.

Example 3.21 Consider the set of vectors:

$$S = \left\{ \begin{pmatrix} 6 \\ 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ -3 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -3 \\ 4 \\ -2 \\ -5 \end{pmatrix} \right\}$$

We will construct a basis for $\text{Span } S$. To that end, we first construct a matrix which has the vectors in S as columns:

$$A = \begin{pmatrix} 6 & 1 & 3 & -3 \\ 3 & 2 & 1 & 4 \\ 2 & 2 & 1 & -2 \\ 1 & -3 & 1 & -5 \end{pmatrix}$$

If we compute an echelon form via **row** operations, we obtain:

$$\begin{pmatrix} 6 & 1 & 3 & -3 \\ 3 & 2 & 1 & 4 \\ 2 & 2 & 1 & -2 \\ 1 & -3 & 1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 1 & -5 \\ 0 & 11 & -2 & 19 \\ 0 & 0 & -5 & 64 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

We see pivot elements in the first three columns and therefore a basis for $\text{Span} S$ is given by the first three columns of the matrix A :

$$\left\{ \begin{pmatrix} 6 \\ 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ -3 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\} \quad (3.7)$$

An alternative approach is the use of column operations. In that case we simplify the matrix A by using elementary column operations. Let us first simplify the first row:

$$\begin{pmatrix} 6 & 1 & 3 & -3 \\ 3 & 2 & 1 & 4 \\ 2 & 2 & 1 & -2 \\ 1 & -3 & 1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 6 & 3 & -3 \\ 2 & 3 & 1 & 4 \\ 2 & 2 & 1 & -2 \\ -3 & 1 & 1 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & -9 & -5 & 10 \\ 2 & -10 & -5 & 4 \\ -3 & 19 & 10 & -14 \end{pmatrix}$$

Here we first interchanged the first and second column and in the second step we subtracted 6 times the first column from the second column, subtracted 3 times the first column from the third column and added 3 times the first column to the fourth column. Next,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & -9 & -5 & 10 \\ 2 & -10 & -5 & 4 \\ -3 & 19 & 10 & -14 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & -9 & 10 \\ 2 & 1 & -10 & 4 \\ -3 & -2 & 19 & -14 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -6 \\ 1 & -2 & 1 & 6 \end{pmatrix}$$

where we first divided the third column by -5 and then interchanged the second and third columns. In the second step we subtracted 2 times the second column from the first column, added 9 times the second column to the third column and subtracted 10 times the second column from the fourth column. Finally,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -6 \\ 1 & -2 & 1 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -6 \\ 1 & -2 & -1 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 \end{pmatrix} \quad (3.8)$$

where we first multiplied the third column by -1 and then subtracted the third column from the second column and added 6 times the third column to the fourth column.

Since we only used column operations, the span of the columns did not change during all these steps and hence we know that the first three columns of the resulting matrix in (3.8) (which are clearly independent) form a basis for $\text{Span } S$, i.e.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} \right\} \quad (3.9)$$

Both (3.7) and (3.9) yield a basis for $\text{Span } S$. So both methods obviously yield a correct result. However, (3.9) obtained through column operations yields a simpler structure compared to (3.7). $\square$

If we have a basis for a subspace V then we can express each vector in V in terms of the basis elements. This yields the coordinates of a vector $\mathbf{x}$ in terms of a basis for V .

Definition 3.22 Assume

$$\mathcal{A} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\} \quad (3.10)$$

is a basis for a subspace V in $\mathbb{R}^n$. In that case, for any $\mathbf{x} \in V$ there exists $\alpha_1, \dots, \alpha_p$ such that:

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_p \mathbf{v}_p$$

We call $\alpha_1, \dots, \alpha_p$ the **coordinates** of $\mathbf{x}$ with respect to the basis $\mathcal{A}$ and we use the notation:

$$[\mathbf{x}]_{\mathcal{A}} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_p \end{pmatrix}$$

Remark. Note that if we determine the coordinates of a vector with respect to a basis then the order of the elements of the basis is important. We describe a basis by $\{\cdot \cdot \cdot\}$ and these brackets, according to, for instance, the course Introduction to Mathematics, denote an *unordered* set. However, for a basis in Linear Algebra, such as (3.10), we define this as an ordered set. In other words, the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ is not the same as the basis $\{\mathbf{v}_2, \mathbf{v}_1\}$.

Example 3.23 Consider the set S defined in Example 3.19. We have:

$$\text{Span } S = \text{Span} \left\{ \begin{pmatrix} 0 \\ 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 8 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ -6 \\ -1 \\ -2 \\ 3 \end{pmatrix} \right\}$$

These three vectors form a basis for $\text{Span } S$. Consider

$$\mathbf{x} = \begin{pmatrix} 0 \\ 2 \\ 7 \\ 2 \\ 5 \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ -1 \\ 2 \end{pmatrix}$$

We would like to check whether $\mathbf{x}$ and/or $\mathbf{y}$ are in $\text{Span } S$ and, if so, find the coordinates with respect to our basis for $\text{Span } S$. For the vector $\mathbf{x}$ we need to solve the linear system:

$$\alpha_1 \begin{pmatrix} 0 \\ 2 \\ 2 \\ 1 \\ 2 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 8 \\ 3 \\ 3 \\ 5 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2 \\ -6 \\ -1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 7 \\ 2 \\ 5 \end{pmatrix}$$

or equivalently:

$$\begin{pmatrix} 0 & 0 & 2 \\ 2 & 8 & -6 \\ 2 & 3 & -1 \\ 1 & 3 & -2 \\ 2 & 5 & 3 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 7 \\ 2 \\ 5 \end{pmatrix}$$

This can be clearly solved using the techniques presented before. The associated augmented matrix has the following reduced echelon form:

$$\left(\begin{array}{ccc|c} 0 & 0 & 2 & 0 \\ 2 & 8 & -6 & 2 \\ 2 & 3 & -1 & 7 \\ 1 & 3 & -2 & 2 \\ 2 & 5 & 3 & 5 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Clearly, the associated linear system has a solution since the last column contains no pivot element. Actually the solution is unique since all first three columns contain a pivot element. It is then immediate that we get $\alpha_1 = 5$, $\alpha_2 = -1$ and $\alpha_3 = 0$.

Next, consider the vector $\mathbf{y}$ defined above. We again would like to check whether $\mathbf{y}$ is in $\text{Span } S$ and, if so, find the coordinates of $\mathbf{y}$ with respect to our basis for $\text{Span } S$. In this case, we need to solve the linear system:

$$\begin{pmatrix} 0 & 0 & 2 \\ 2 & 8 & -6 \\ 2 & 3 & -1 \\ 1 & 3 & -2 \\ 2 & 5 & 3 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ -1 \\ 2 \end{pmatrix}$$

The approach is the same as before. The associated augmented matrix has the following reduced echelon form:

$$\left(\begin{array}{ccc|c} 0 & 0 & 2 & 1 \\ 2 & 8 & -6 & 2 \\ 2 & 3 & -1 & 1 \\ 1 & 3 & -2 & -1 \\ 2 & 5 & 3 & 2 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Clearly, the associated linear system has **no** solution since the last column contains a pivot element and hence $\mathbf{y} \notin \text{Span } S$. $\square$

We also have the following, alternative, characterizations for the invertibility of a matrix. The proof of this theorem below will be addressed in Exercise 3.17.

Theorem 3.24 Consider a square matrix A .

- The matrix A is invertible if and only if

$$\text{Null } A = \{\mathbf{0}\}$$

- The matrix A is invertible if and only if

$$\text{Col } A = \mathbb{R}^n$$

Note that the condition $\text{Null } A = \{\mathbf{0}\}$ guarantees that for any $\mathbf{b}$, the linear system $A\mathbf{x} = \mathbf{b}$ has at most one solution. The condition $\text{Col } A = \mathbb{R}^n$ guarantees that the linear system $A\mathbf{x} = \mathbf{b}$ is consistent for any vector $\mathbf{b}$.

Combining the two conditions we find that A is invertible if and only if the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for any vector $\mathbf{b}$.

From the above we can conclude that

$$\text{Null } A = \{\mathbf{0}\}$$

if and only if

$$\text{Col } A = \mathbb{R}^n.$$

However, we should note that this is only true for **square** matrices.

Example 3.25 Consider

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

In that case

$$\text{Null } A = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right\}, \quad \text{Col } A = \mathbb{R}^2.$$

Verify this yourself! On the other hand for

$$B = \begin{pmatrix} 1 & 2 \\ 2 & -1 \\ 1 & 1 \end{pmatrix}$$

we get:

$$\text{Null } B = \{\mathbf{0}\}, \quad \text{Col } B = \text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}.$$

Verify this yourself!

□

3.5 Exercises

3.5.1 Self study exercises

3.1 The augmented matrix A and a solution vector $\mathbf{v}$ of a system of linear equations are given by:

$$A = \begin{pmatrix} 2 & -1 & 4 & -2 & 5 \\ -1 & 0 & -2 & 1 & -4 \\ 1 & -1 & 2 & -1 & 1 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 4 \\ 3 \\ -1 \\ -2 \end{pmatrix}$$

- (a) Show, *without* calculating the solution set, that $\mathbf{v}$ is indeed a solution of this system.
- (b) Determine the solution set of the system. Write the solution set in parametric vector form.
- (c) Let B be the coefficient matrix of this system. Determine, *without any further calculations* the solution set of the system $B\mathbf{x} = \mathbf{0}$. Motivate your answer.

3.2 Let $A \in \mathbb{R}^{3 \times 5}$. Suppose all columns of A are different and nonzero. Denote the column vectors of A by $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$.

- (a) Determine p and q such that $\text{Null } A \subseteq \mathbb{R}^p$ and $\text{Col } A \subseteq \mathbb{R}^q$.
- (b) How many vectors are in $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$?
- (c) How many vectors are in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$?
- (d) Show, using the definition of Span, that

$$\mathbf{a}_4 \in \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}.$$

3.3 Let D be a subspace in $\mathbb{R}^n$ which contains

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p.$$

Moreover, the vector $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. Show that $\mathbf{w} \in D$.

3.4 Consider Example 3.9. Show that:

$$\text{If } \mathbf{x} \in \begin{pmatrix} -58 \\ -101 \\ 9 \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -13 \\ -25 \\ 2 \\ 1 \end{pmatrix} \right\} \text{ then } \mathbf{x} \in \begin{pmatrix} -6 \\ -1 \\ 1 \\ -4 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 13 \\ 25 \\ -2 \\ -1 \end{pmatrix} \right\}$$

and that:

$$\text{If } \mathbf{x} \in \begin{pmatrix} -6 \\ -1 \\ 1 \\ -4 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 13 \\ 25 \\ -2 \\ -1 \end{pmatrix} \right\} \text{ then } \mathbf{x} \in \begin{pmatrix} -58 \\ -101 \\ 9 \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -13 \\ -25 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

3.5 Let S_1 and S_2 be given by

$$S_1 = \left\{ \begin{pmatrix} 1 \\ -3 \\ 4 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ -7 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix} \right\}; \quad S_2 = \left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -2 \\ 7 \end{pmatrix}, \begin{pmatrix} -5 \\ 1 \\ -6 \end{pmatrix} \right\}.$$

- (a) Examine if S_1 is a linearly independent set. If not, express one of the vectors in S_1 as a linear combination of the other vectors.
- (b) Repeat part (a) for the set S_2 .
- (c) Determine bases for $\text{Span } S_1$ and $\text{Span } S_2$.
- (d) What are the dimensions of $\text{Span } S_1$ and $\text{Span } S_2$?
- (e) Explain that S_1 is a basis for $\mathbb{R}^3$ and that S_2 is not.
- (f) Determine the coordinates of

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

with respect to the basis S_1 .

3.5.2 Tutorial exercises

3.6 Matrix A and vector $\mathbf{v}$ are given by:

$$A = \begin{pmatrix} 3 & -12 & 0 & 9 \\ 2 & -8 & 1 & 10 \\ -1 & 4 & 1 & 1 \end{pmatrix}; \quad \mathbf{v} = \begin{pmatrix} -5 \\ -2 \\ 4 \\ -1 \end{pmatrix}; \quad \mathbf{w} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

- (a) Verify, without calculating $\text{Null } A$, whether $\mathbf{v} \in \text{Null } A$ and $\mathbf{w} \in \text{Null } A$.
- (b) Determine $\text{Null } A$ and write it in parametric vector form.
- (c) Let

$$\mathbf{p} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} -6 \\ 1 \\ 7 \end{pmatrix}$$

verify that $\mathbf{p}$ is a solution of the system $A\mathbf{x} = \mathbf{b}$, and determine without any further calculations the solution set of this system in parametric vector form.

- (d) Determine if the vector $\mathbf{b}$ in part (c) is in $\text{Col } A$.

3.7 The vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{w} \in \mathbb{R}^3$ are given by:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 5 \\ 6 \\ \alpha \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ -6 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 1 \\ \beta \end{pmatrix},$$

where $\alpha, \beta \in \mathbb{R}$. Let $W = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

(a) Take $\alpha = 14$. Examine if $\mathbf{w}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in case $\beta = 0$ and in case $\beta = 2$.

(b) Determine all values of α and β for which $\mathbf{w} \in W$.

3.8 Consider Example 3.2. A plane p in $\mathbb{R}^n$ through the origin can be described as:

$$p = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \alpha\mathbf{q} + \beta\mathbf{r}\},$$

where $\mathbf{q} \neq \mathbf{0}$ and $\mathbf{r} \neq \mathbf{0}$ are arbitrarily chosen vectors on p . Show that p is a linear subspace of $\mathbb{R}^n$.

3.9 Consider Definition 3.11. Show that $\text{Col } A$ is the subspace spanned by the columns of A . Hint: use Definition 2.4 to show that $A\mathbf{x} = \mathbf{b}$ implies that $\mathbf{b}$ is a linear combination of the columns of A .

3.10 Consider Theorem 3.10. Suppose that $\mathbf{p}$ is one particular solution of $A\mathbf{x} = \mathbf{b}$.

(a) Show that each $\mathbf{y} \in \mathbf{p} + \text{Null } A$ is a solution of $A\mathbf{x} = \mathbf{b}$.

(b) Show that if $\mathbf{y}$ is a solution of $A\mathbf{x} = \mathbf{b}$, then $\mathbf{y} \in \mathbf{p} + \text{Null } A$ (hint: consider the vector $\mathbf{y} - \mathbf{p}$).

3.11 The matrix A and the vector $\mathbf{b}$ are given by

$$A = \begin{pmatrix} 1 & -3 & 2 & 2 \\ -2 & 6 & -4 & -3 \\ -3 & 9 & -6 & -5 \end{pmatrix}; \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

(a) Show that $\mathbf{b} \in \text{Col } A$.

(b) Determine a basis $\mathcal{B}$ for $\text{Col } A$ and determine $[\mathbf{b}]_{\mathcal{B}}$.

3.12 The matrix A is given by:

$$A = \begin{pmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{pmatrix}.$$

(a) Determine all values of $\alpha \in \mathbb{R}$, for which the columns of A form a basis for $\mathbb{R}^3$.

(b) Take $\alpha = 1$. Determine a basis $\mathcal{B}$ for $\text{Null } A$. What is the dimension of $\text{Null } A$?

- (c) Take $\alpha = 1$ and let

$$\mathbf{p} = \begin{pmatrix} 3 \\ -5 \\ 2 \end{pmatrix}.$$

Show that $\mathbf{p} \in \text{Null } A$ and determine $[\mathbf{p}]_{\mathcal{B}}$, where $\mathcal{B}$ denotes the basis of part (b).

3.13 Let $A \in \mathbb{R}^{3 \times 4}$ be a matrix. Denote the columns of A by $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4$.

- (a) Suppose the second column of A equals three times the first column minus four times the last column ($\mathbf{a}_2 = 3\mathbf{a}_1 - 4\mathbf{a}_4$). Give a nontrivial solution of the system $A\mathbf{x} = \mathbf{0}$.
- (b) Explain that the columns of A must be linearly dependent using part (a).
- (c) Prove that if $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ is a linear independent set, then so is $\{\mathbf{a}_1, \mathbf{a}_2\}$.

3.14 Let $A \in \mathbb{R}^{m \times n}$ be a matrix.

- (a) How many pivot positions must an echelon form of A have if the columns of A are linearly independent?
- (b) How many pivot positions must an echelon form of A have if the columns of A span $\mathbb{R}^m$?
- (c) If the columns of A are linearly independent, what can you say about the number of solutions of the system $A\mathbf{x} = \mathbf{b}$? ($\mathbf{b} \in \mathbb{R}^m$)
- (d) Assume that A^T is already in echelon form. Show that the nonzero columns of A are linearly independent.

3.15 Let $A \in \mathbb{R}^{k \times m}$ and $B \in \mathbb{R}^{m \times n}$ be matrices.

- (a) Assume that the last column of the product AB is equal to the zero-vector, but the last column of the matrix B is not. Prove that the columns of A are linearly dependent.
- (b) Prove that if the columns of B are linearly dependent, then so are the columns of AB .

3.16 Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\} \subseteq \mathbb{R}^n$, where $p \geq 2$.

- (a) Show that if $\mathbf{v}_i$ can be expressed as a linear combination of the other vectors in S , then each linear combination of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ can be expressed as a linear combination of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_p$.

- (b) Show that if $\mathbf{v}_i = \mathbf{0}$ for some $i \in \{1, \dots, p\}$, then S is linearly dependent.
- (c) Prove Theorem 3.14.
- (d) Prove the first part of Theorem 3.17: if S is a basis, then $p \leq n$.
Hint: show that S is linearly dependent if $p > n$.
- (e) Prove that if $p = n$ and S is linearly independent, then S is a basis for $\mathbb{R}^n$.

3.17 Prove Theorem 3.24.

3.18 Consider Definition 3.22. Prove that the coordinates of $\mathbf{x}$ with respect to a basis $\mathcal{A} = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ are unique, i.e, if $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p$ and $\mathbf{x} = \beta_1 \mathbf{v}_1 + \dots + \beta_p \mathbf{v}_p$, then $\alpha_i = \beta_i$ for all $1 \leq i \leq p$. Hint: use that the vectors in a basis must be linearly independent and apply Theorem 3.14 to the difference of the two expressions for $\mathbf{x}$ described above.

3.19 In this exercise we will prove that all bases for a given subspace V contain the same number of vectors (cf. remark after Theorem 3.17). Let $\mathcal{A} = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ and $\mathcal{B} = \{\mathbf{w}_1, \dots, \mathbf{w}_q\}$ be bases of a linear subspace $V \subseteq \mathbb{R}^n$. Let A be the matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_p$ and let B be the matrix with columns $\mathbf{w}_1, \dots, \mathbf{w}_q$.

- (a) Show that for each $j \in \{1, \dots, q\}$, the system $A\mathbf{x} = \mathbf{w}_j$ is consistent.

Let, for $1 \leq j \leq q$, $\mathbf{c}_j$ be a solution of the system $A\mathbf{x} = \mathbf{w}_j$, so $A\mathbf{c}_j = \mathbf{w}_j$. Let C be the matrix with columns $\mathbf{c}_1, \dots, \mathbf{c}_q$.

- (b) What is the size of C ?
- (c) Explain that if $q > p$, there must exist a nonzero vector $\mathbf{u}$ with $C\mathbf{u} = \mathbf{0}$.
- (d) Show that if $q > p$, we have $B\mathbf{u} = \mathbf{0}$ for the vector $\mathbf{u}$ in part (c).
- (e) Show that if $q > p$ then $\mathcal{B}$ cannot be a basis for V .
- (f) Prove that $p = q$, i.e, all bases of V must have the same number of vectors.

3.20 Consider two planes in $\mathbb{R}^3$:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

and

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$$

Find the intersection of the two planes.

- 3.21 In Chapter 1, we have seen that we can describe a line in $\mathbb{R}^3$ as the intersection of two planes. Given the line:

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \right\}$$

Find two planes (described by a linear equation) whose intersection is the given line.

- 3.22 Consider three planes in $\mathbb{R}^3$. The first plane is described by:

$$x_1 - x_2 + 2x_3 = 1$$

The second plane is given by:

$$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$$

and the third plane is given by:

$$\left\{ \mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \text{ for some } \lambda, \mu \in \mathbb{R} \right\}$$

Find the intersection of these three planes.

- 3.23 Given are the three vectors:

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Determine for each of the following statements whether they are true or not:

- (a) $\mathbf{v}_1 \in \{\mathbf{v}_2, \mathbf{v}_3\}$
- (b) $\mathbf{v}_2 \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_3\}$
- (c) $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} = \text{Span}\{\mathbf{v}_2, \mathbf{v}_3\}$
- (d) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a basis for $\mathbb{R}^3$.

Chapter 4

Determinants

We can associate with every square matrix A a real number, the so-called determinant. Determinants play a role in the computation of the area of a parallelogram and the volume of a parallelepiped, in the computation of multiple integrals. It can also be used to check whether a matrix is invertible or even the invertibility of functions of multiple variables.

4.1 Definition

We first introduce the notation A_{ij} .

Definition 4.1 Let A be an $n \times n$ matrix with $n \geq 2$. In that case A_{ij} is the $(n-1) \times (n-1)$ submatrix we obtain by deleting the i th row and the j th column.

Example 4.2 Consider

$$A = \begin{pmatrix} 4 & 1 & -3 & 1 \\ -3 & -2 & 0 & 2 \\ 2 & 1 & 0 & -1 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

Then we have:

$$A_{13} = \begin{pmatrix} -3 & -2 & 2 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix} \quad A_{32} = \begin{pmatrix} 4 & -3 & 1 \\ -3 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix} \quad \square$$

The determinant will now be defined recursively.

Definition 4.3 Let A be an $n \times n$ matrix. The **determinant** of A is given by:

- If $n = 1$ then $\det A = a_{11}$.
- If $n = 2$ then $\det A = a_{11}a_{22} - a_{12}a_{21}$.
- If $n > 2$ then

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{n+1} a_{1n} \det A_{1n}$$

We will sometimes refer to $\det A_{ij}$ as M_{ij} . M_{ij} is called the (i, j) **minor** of the matrix A .

We will also use the notation $|A|$ for $\det A$.

It should be noted that although the above definition defines the determinant, it is not a very efficient technique to compute the determinant.

Example 4.4 Consider

$$A = \begin{pmatrix} 4 & 1 & -3 & 1 \\ -3 & -2 & 0 & 2 \\ 2 & 1 & 0 & -1 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

We have:

$$\begin{aligned} \det A &= (4) \det A_{11} - (1) \det A_{12} + (-3) \det A_{13} - (1) \det A_{14} \\ &= 4M_{11} - M_{12} - 3M_{13} - M_{14} \end{aligned}$$

This reduces the computation of a determinant of a 4×4 -matrix to the computation of 4 determinants of 3×3 -matrix. We obtain:

$$\begin{aligned} M_{11} &= \begin{vmatrix} -2 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{vmatrix} = (-2) \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} - (0) \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} + (2) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \\ &= (-2)(1) - (0)(2) + (2)(1) = 0 \end{aligned}$$

$$\begin{aligned} M_{12} &= \begin{vmatrix} -3 & 0 & 2 \\ 2 & 0 & -1 \\ 1 & 1 & 2 \end{vmatrix} = (-3) \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} - (0) \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} + (2) \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} \\ &= (-3)(1) - (0)(5) + (2)(2) = 1 \end{aligned}$$

$$\begin{aligned}
 M_{13} &= \begin{vmatrix} -3 & -2 & 2 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{vmatrix} = (-3) \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} - (-2) \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} + (2) \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} \\
 &= (-3)(2) - (-2)(5) + (2)(-1) = 2
 \end{aligned}$$

$$\begin{aligned}
 M_{14} &= \begin{vmatrix} -3 & -2 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = (-3) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - (-2) \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} + (0) \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} \\
 &= (-3)(1) - (-2)(2) + (0)(-1) = 1
 \end{aligned}$$

and hence

$$\det A = 4(0) - (1) - 3(2) - (1) = -8.$$

For larger matrices this becomes unwieldy. A determinant of a 10×10 matrix can be expressed in 10 determinants of 9×9 matrices. Continuing, we find this expressed in terms of

$$10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 604,800$$

determinants of 3×3 matrices. This quickly becomes unmanageable. $\square$

Note that in Definition 4.3 we used a specific recursion to define the determinant. However, there are alternative ways such as Theorem 4.5 below, to compute the determinant which in some cases are much easier to use.

Theorem 4.5 *Let A be an $n \times n$ matrix. In that case:*

• *We have:*

$$\det A = \sum_{k=1}^n (-1)^{i+k} a_{ik} M_{ik}$$

(expansion of $\det A$ with respect to the i 'th row).

• *We have:*

$$\det A = \sum_{k=1}^n (-1)^{k+j} a_{kj} M_{kj}$$

(expansion of $\det A$ with respect to the j 'th column).

Here a_{ij} and M_{ij} are the (i, j) element and (i, j) minor of A respectively.

Using the terminology of the above theorem, Definition 4.3 used an expansion along the first row to define the determinant. However, we can use an expansion along any row or column and we generally prefer a row or a column which contains many zeros.

Example 4.6 We have:

$$\det A = \begin{vmatrix} 7 & 1 & 0 & 7 \\ -3 & -2 & 0 & 2 \\ 2 & 1 & 0 & -1 \\ 1 & 0 & 1 & 2 \end{vmatrix} = - \begin{vmatrix} 7 & 1 & 7 \\ -3 & -2 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

where we use an expansion along the third column which is easiest because it contains three zeros. Next we have:

$$- \begin{vmatrix} 7 & 1 & 7 \\ -3 & -2 & 2 \\ 2 & 1 & -1 \end{vmatrix} = - \begin{vmatrix} 5 & 1 & 8 \\ 1 & -2 & 0 \\ 0 & 1 & 0 \end{vmatrix}$$

where we subtracted twice the second column from the first column and added the second column to the third column. This does not affect the determinant. Finally we can expand along the third row or third column (both of which are good choices since they have two zero elements). We get:

$$- \begin{vmatrix} 5 & 1 & 8 \\ 1 & -2 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 5 & 8 \\ 1 & 0 \end{vmatrix} = -8$$

where we used the expansion along the third row. □

4.2 Properties of the determinant

We first state:

Lemma 4.7 *Given a square matrix A we have:*

$$\det A = \det A^T$$

For the proof of the above lemma, we refer to Exercise 4.8. We will next show the effect of elementary row operations on the determinant of a matrix. Because of the above lemma, we conclude that the effect of elementary column operations on the determinant of a matrix is similar. For a proof of the following lemma, we refer to Exercise 4.9.

Lemma 4.8 *Given a square matrix A we have:*

- *If $\tilde{A}$ is obtained from A by adding a multiple of one row to another row then $\det \tilde{A} = \det A$.*
- *If $\tilde{A}$ is obtained from A by multiplying one row with a number $\alpha \neq 0$ then $\det \tilde{A} = \alpha \det A$.*
- *If $\tilde{A}$ is obtained from A by interchanging two rows then $\det \tilde{A} = -\det A$.*

Corollary 4.9 *If a square matrix A has two identical rows or two identical columns then $\det A = 0$.*

If a square matrix A has a zero row or a zero column then $\det A = 0$.

Proof : Note that if we interchange two identical rows then the matrix does not change and hence the determinant should not change. But using Lemma 4.8 we find that the determinant gets multiplied by -1 . These two properties can only be simultaneously true if $\det A = 0$.

Note that if we multiply a zero row by 2 then the matrix does not change and hence the determinant should not change. But using Lemma 4.8 we find that the determinant gets multiplied by 2. These two properties can only be simultaneously true if $\det A = 0$.

The fact that these properties also hold with identical columns or zero columns is a direct consequence of Lemma 4.7. ■

Next, we note that for a certain class of matrices, the determinant can be easily computed:

Theorem 4.10 *If A is an $n \times n$ lower or upper triangular matrix then:*

$$\det A = a_{11}a_{22} \cdots a_{nn} \quad (4.1)$$

For the proof of the above theorem, we refer to Exercise 4.7. Corollary 4.9 and the above theorem in combination with Lemma 4.8 enable us to compute determinants more efficiently. We basically use elementary row or column operations to get either a zero row or column (in which case the determinant equals zero) or we get a triangular matrix in which case the determinant is given by (4.1).

Example 4.11 Consider the following matrix A :

$$A = \begin{pmatrix} 3 & 2 & -2 & 10 \\ 3 & 1 & 1 & 2 \\ -2 & 2 & 3 & 4 \\ 1 & 1 & 5 & 2 \end{pmatrix}$$

We get

$$\begin{pmatrix} 3 & 2 & -2 & 10 \\ 3 & 1 & 1 & 2 \\ -2 & 2 & 3 & 4 \\ 1 & 1 & 5 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 3 & 1 & 1 & 2 \\ -2 & 2 & 3 & 4 \\ 3 & 2 & -2 & 10 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & -2 & -14 & -4 \\ 0 & 4 & 13 & 8 \\ 0 & -1 & -17 & 4 \end{pmatrix}$$

Here we first interchange the first and fourth row. In the second step we subtract 3 times the first row from the second row, add 2 times the first row to the third row and subtract 3 times the first row from the fourth row. Next we turn our attention to the second column:

$$\begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & -2 & -14 & -4 \\ 0 & 4 & 13 & 8 \\ 0 & -1 & -17 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 4 & 13 & 8 \\ 0 & -1 & -17 & 4 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 0 & -15 & 0 \\ 0 & 0 & -10 & 6 \end{pmatrix}$$

Here we first divide the second row by -2 and then subtract 4 times the second row from the third row and add the second row to the fourth row. Next we put our effort in the third column:

$$\begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 0 & -15 & 0 \\ 0 & 0 & -10 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -10 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix}$$

Here we first divide the third row by -15 and then add 10 times the third row to the fourth row. We note that the final matrix is an upper triangular matrix. Hence by Theorem 4.10 we have:

$$\det \begin{pmatrix} 1 & 1 & 5 & 2 \\ 0 & 1 & 7 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix} = 6$$

Note that this is not equal to the determinant of A . We have applied elementary row operations which affected the determinant. Interchanging rows yielded a factor -1 , dividing a row by -2 yields a factor -2 and finally dividing a row by -15 yields a factor -15 . Hence:

$$\det A = (-1)(-2)(-15)6 = -180.$$

□

An interesting property of the determinant is the following.

Theorem 4.12 For $n \times n$ matrices A and B we have:

$$\det(AB) = (\det A)(\det B)$$

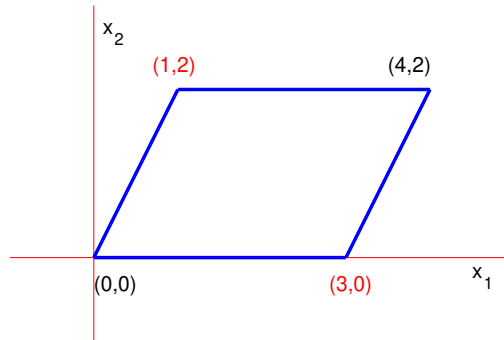
One of the first uses of the determinant is to check whether a matrix is invertible.

Theorem 4.13 Let A be an $n \times n$ matrix. The matrix A is invertible if and only if $\det A \neq 0$.

For the proof of the above theorem, we refer to Exercise 4.10. Note that the above two theorems for instance imply that AB is invertible if and only if A and B are both invertible.

Another usage of determinants is to compute the area of a parallelogram and the volume of a parallelepiped.

Example 4.14 Consider a parallelogram where one of the corners equals the origin:



The corners of the parallelogram are given by $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$ provided

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

□

We have the following result:

Theorem 4.15 Consider the parallelogram in $\mathbb{R}^2$ whose corners are given by the vectors $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$. Let A be the 2×2 matrix whose columns are $\mathbf{u}$ and $\mathbf{v}$.

In that case the surface area of the parallelogram is given by:

$$|\det A|$$

Example 4.16 Consider the parallelogram of Example 4.14. In that case:

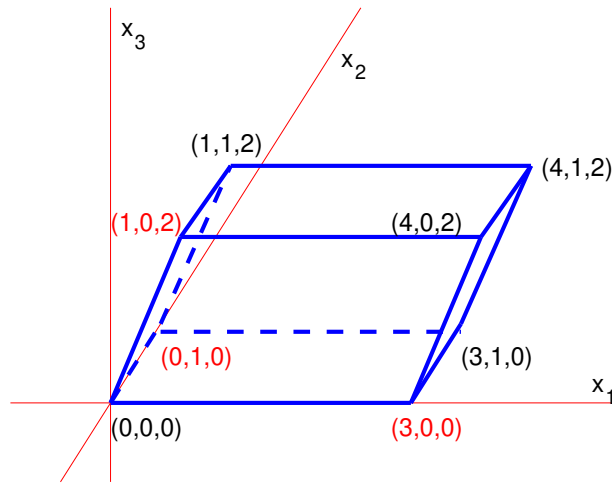
$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}$$

and $\det A = -6$. Therefore the area of the parallelogram is equal to

$$|\det A| = 6$$

□

Example 4.17 Next, consider a parallelepiped where one of the corners is the origin.



The corners of the parallelepiped are given by $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ as well as $\mathbf{u} + \mathbf{v}$, $\mathbf{u} + \mathbf{w}$, $\mathbf{v} + \mathbf{w}$ and $\mathbf{u} + \mathbf{v} + \mathbf{w}$ provided

$$\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

□

For a parallelepiped, we have the following result:

Theorem 4.18 Consider the parallelepiped in $\mathbb{R}^3$ whose corners are given by the vectors $\mathbf{0}$, $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ as well as $\mathbf{u} + \mathbf{v}$, $\mathbf{u} + \mathbf{w}$, $\mathbf{v} + \mathbf{w}$ and $\mathbf{u} + \mathbf{v} + \mathbf{w}$.

Let A be the 3×3 matrix whose columns are $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$.

In that case the volume of the parallelepiped is given by:

$$|\det A|$$

Example 4.19 Consider the parallelepiped of Example 4.17. In that case:

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix}$$

and $\det A = 6$. Therefore the volume of the parallelepiped is equal to

$$|\det A| = 6$$

Note that when we construct the matrix A we added the columns $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ in a rather arbitrary order. This does not affect the answer since interchanging columns only affects the sign of the determinant and hence the absolute value of the determinant is not affected. $\square$

4.3 Exercises

4.3.1 Self study exercises

4.1 The matrices A and B are given by:

$$A = \begin{pmatrix} 1 & -2 & -1 \\ -3 & 4 & 2 \\ 4 & -7 & -3 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 6 & -5 \\ 3 & -2 & 1 \\ 2 & 7 & -6 \end{pmatrix}.$$

(a) Determine $\det A$ and $\det B$ in three ways:

- (1) By expansion with respect to the first row (cf. Example 4.4).
- (2) By first transforming into a triangular matrix (cf. Example 4.11).
- (3) By first creating as many zero elements as possible in a certain row or column and then using expansion along that row or column (cf. Example 4.6).

(b) Determine if A and B are invertible.

4.2 Matrix A is given by:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 3 & 4 \end{pmatrix}$$

(a) Determine $\det A$.

(b) Use part (a) to determine

$$\det \begin{pmatrix} 3 & 2 & 2 \\ 5 & 3 & 2 \\ 4 & 3 & 2 \end{pmatrix} \quad \text{and} \quad \det \begin{pmatrix} p & 2p & 3p \\ 1+q & 3+2q & 5+3q \\ 1-r & 3-2r & 4-3r \end{pmatrix}$$

where $p, q, r \in \mathbb{R}$.

4.3 Let A and B be $n \times n$ -matrices, with $\det A = 5$ and $\det B = -3$.

Determine $\det(AB)$, $\det(B^3)$, $\det(3A)$, $\det(B^T B)$ and $\det(rB)$, ($r \in \mathbb{R}$).

4.3.2 Tutorial exercises

4.4 The matrix A is given by:

$$A = \begin{pmatrix} \alpha + 1 & -3 & 1 \\ -4 & \alpha & -1 \\ \alpha + 1 & -2 & 2 \end{pmatrix}, \text{ where } \alpha \in \mathbb{R}.$$

(a) Determine $\det A$.

(b) Use part (a) to determine all values of $\alpha \in \mathbb{R}$, for which A is invertible.

4.5 Let A and B be $n \times n$ -matrices.

(a) Suppose that $\det(A^3) = 0$. Prove that A is not invertible.

(b) Suppose that both A and B are invertible. Prove, using determinants, that also AB and $B^T A$ are invertible.

(c) Prove that if A is invertible, then $\det(A^{-1}) = \frac{1}{\det A}$.

(d) Prove that if B is invertible, then $\det(BAB^{-1}) = \det A$.

(e) Suppose that $A^T A = I_n$. Prove that $\det A = \pm 1$.

4.6 (a) Determine the area of the parallelogram in $\mathbb{R}^2$ with vertices

$$(0, 0), (-3, 1), (2, 2) \text{ and } (5, 1).$$

(b) Determine the area of the parallelogram in $\mathbb{R}^2$ with vertices

$$(-1, 4), (-2, 7), (-4, 9) \text{ and } (-3, 6).$$

Hint: first translate the parallelogram such that one vertex is in the origin.

(c) Determine the area of the triangle in $\mathbb{R}^2$ with vertices $(2, 1)$, $(5, 2)$ and $(8, 8)$.

(d) Determine the volume of the parallelepiped in $\mathbb{R}^3$ with vertices

$$(0, 0, 0), (-2, 3, 1), (-1, 2, 2), (1, -1, 1), (3, 4, -1), \\ (4, 3, 0), (5, 1, -2) \text{ and } (6, 0, -1).$$

(e) Consider the parallelepiped S in $\mathbb{R}^3$ spanned by the columns of the matrix A of Exercise 4.4. Determine the maximal volume of S in case $-4 \leq \alpha \leq 4$.

4.7 Prove Theorem 4.10 using mathematical induction to n (the size of matrix A).

(a) First note that Theorem 4.10 is correct for $n = 1$.

(b) Now let $k \geq 1$, and assume that Theorem 4.10 holds for all lower triangular $k \times k$ -matrices.

Use this induction hypothesis to show that then Theorem 4.10 also holds for all lower triangular $(k + 1) \times (k + 1)$ -matrices.

By part (a), (b) and the principle of mathematical induction, Theorem 4.10 holds for all lower triangular $n \times n$ -matrices.

(c) Deduce that Theorem 4.10 also holds for all upper triangular $n \times n$ -matrices.

4.8 Prove Lemma 4.7 using Theorem 4.5 and mathematical induction to n (the size of matrix A).

(a) First show that Lemma 4.7 is correct for $n = 1$.

(b) Now let $k \geq 1$, and assume that Lemma 4.7 holds for all $k \times k$ -matrices.

Use this induction hypothesis and Theorem 4.5 to show that then Lemma 4.7 also holds for all $(k + 1) \times (k + 1)$ -matrices.

Hint: determine $\det A^T$ using expansion with respect to the first column of A^T and determine $\det A$ using expansion with respect to the first row of A .

(c) By part (a), (b) and the principle of mathematical induction, we can conclude that Lemma 4.7 holds for all $n \times n$ -matrices.

4.9 Prove Lemma 4.8 using Theorem 4.5 and mathematical induction to n (the size of matrix A).

(a) First show that Lemma 4.8 is correct for $n = 1$ and $n = 2$.

(b) Now let $k \geq 2$, and assume that Lemma 4.8 holds for all $k \times k$ -matrices.

Use this induction hypothesis and Theorem 4.5 to show that then Lemma 4.8 also holds for all $(k + 1) \times (k + 1)$ -matrices.

Hint: Let B denote the matrix that arises from A after performing the elementary row operation considered. Choose a row of A that remains unchanged after this operation (why must such a row exist?). Then determine $\det B$ and $\det A$ using expansion with respect to this row.

- (c) By part (a), (b) and the principle of mathematical induction, we can conclude that Lemma 4.8 holds for all $n \times n$ -matrices.

4.10 Prove Theorem 4.13.

- (a) If A is invertible, then use Lemma 4.8 to show that $\det A \neq 0$.
(b) If $\det A \neq 0$, consider the reduced echelon form of A and show that A must be invertible.

4.11 Consider a matrix $A \in \mathbb{R}^{4 \times 4}$ for which the fourth column is the sum of the first two columns. What can you say about $\det A$ and why?

Chapter 5

Eigenvalues and eigenvectors

One of the most used concepts in linear algebra are eigenvalues and eigenvectors. Given the limited time available, we are not able to discuss the applications of these concepts extensively. We will briefly show in Section 5.3 how eigenvalues play a role in differential equations. The main issue is that eigenvalues allow us to split high dimensional problems into one-dimensional problems which results in a much better understanding of the structure of the problem.

5.1 Definition and computation

Let us start by giving a formal definition of these concepts of eigenvalues and eigenvectors:

Definition 5.1 Consider a square $n \times n$ matrix A . In that case $\lambda \in \mathbb{R}$ is called an **eigenvalue** of the matrix A if there exists a nonzero vector $\mathbf{x} \in \mathbb{R}^n$ such that

$$A\mathbf{x} = \lambda\mathbf{x}, \quad \mathbf{x} \neq \mathbf{0} \quad (5.1)$$

In that case $\mathbf{x}$ is called an **eigenvector** of A associated with the eigenvalue λ .

Note that the condition that $\mathbf{x} \neq \mathbf{0}$ is crucial since for any λ we have that $\mathbf{x} = \mathbf{0}$ satisfies (5.1). It is only the fact that (5.1) has a nonzero solution $\mathbf{x}$ that makes λ “special”.

Example 5.2 Consider the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

In this case, $\lambda = 1$ is an eigenvalue. After all we can check that in that case (5.1) is equivalent to

$$A\mathbf{x} = \mathbf{x} = I_2\mathbf{x}$$

or

$$(A - I_2)\mathbf{x} = \mathbf{0}$$

We have

$$A - I_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

and we find that

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

is an eigenvector associated with the eigenvalue 1. Clearly

$$\mathbf{x} = \begin{pmatrix} \alpha \\ 0 \end{pmatrix}$$

is an eigenvector for any $\alpha \neq 0$ and therefore the associated eigenvector is not unique. $\square$

Next, we formulate the set of all eigenvectors associated with an eigenvalue λ .

Definition 5.3 Let A be an $n \times n$ matrix and let λ be an eigenvalue of A . We define the **eigenspace** E_λ associated with the eigenvalue λ by:

$$E_\lambda = \{\mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \lambda\mathbf{v}\}$$

In the above example we found many eigenvectors for a given eigenvalue. This is a general property. It is easy to see for instance that if $\mathbf{x}$ is an eigenvector then also $2\mathbf{x}$ is an eigenvector associated with the same eigenvalue.

Lemma 5.4 Let A be an $n \times n$ matrix and let λ be an eigenvalue of A . In that case the eigenspace E_λ is a linear subspace of $\mathbb{R}^n$.

Proof : Note that (5.1) can be rewritten as:

$$\mathbf{x} \in \text{Null}(A - \lambda I_n)$$

We find:

$$E_\lambda = \text{Null}(A - \lambda I_n)$$

and since the null-space is a subspace of $\mathbb{R}^n$ (cf. Lemma 3.3 and Definition 3.4), we immediately find that the eigenspace E_λ is a subspace. ■

Note that finding eigenvectors for a given eigenvalue of a matrix amounts to computing a null space. This has already been studied before and we have good tools for this such as the reduced echelon form. However, we still need a tool to compute the eigenvalues. For this we rely on the determinant as discussed in Chapter 4. Let us first note the following:

Lemma 5.5 *λ is an eigenvalue of A if and only if $A - \lambda I$ is not invertible.*

Proof : We already noted that

$$E_\lambda = \text{Null}(A - \lambda I_n)$$

and hence an eigenvector $\mathbf{x} \in E_\lambda$ exists if and only if

$$\text{Null}(A - \lambda I_n) \neq \{\mathbf{0}\}$$

By Theorem 3.24, the latter is equivalent to the condition that $A - \lambda I$ is not invertible. ■

Note that in order to check whether a matrix is invertible, we can use the concept of determinant. We obtain the following result:

Theorem 5.6 *λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.*

Proof : This is an immediate consequence of Lemma 5.5 and Theorem 4.13. ■

Note that for an $n \times n$ matrix A we have that $\det(A - \lambda I)$ is always a polynomial of degree n , of the form:

$$p(\lambda) = (-1)^n \lambda^n + \alpha_1 \lambda^{n-1} + \dots + \alpha_{n-1} \lambda + \alpha_n$$

Therefore computation of the eigenvalues reduces to the computation of the zeros of a polynomial. This polynomial plays such a crucial role that it has a specific name:

Definition 5.7 Consider a square $n \times n$ matrix A . Then the polynomial p defined by

$$p(\lambda) = \det(A - \lambda I)$$

is called the **characteristic polynomial** of A .

Note that we know that a degree n polynomial has at most n distinct zeros and therefore we also have at most n eigenvalues.

Clearly, finding the zeros of a second-degree polynomial is quite easy. For degrees 3 and 4 there are still explicit formulas (although they are messy). For larger systems, we need to rely on numerical tools.

Example 5.8 Consider

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

We find that

$$A - \lambda I = \begin{pmatrix} 1 - \lambda & -1 & 0 \\ -1 & 2 - \lambda & -1 \\ 0 & -1 & 1 - \lambda \end{pmatrix}$$

If we expand the determinant along the first column, we get:

$$\begin{aligned} \begin{vmatrix} 1 - \lambda & -1 & 0 \\ -1 & 2 - \lambda & -1 \\ 0 & -1 & 1 - \lambda \end{vmatrix} &= (1 - \lambda) \begin{vmatrix} 2 - \lambda & -1 \\ -1 & 1 - \lambda \end{vmatrix} - (-1) \begin{vmatrix} -1 & 0 \\ -1 & 1 - \lambda \end{vmatrix} \\ &= (1 - \lambda) [(2 - \lambda)(1 - \lambda) - 1] + [\lambda - 1] \\ &= (1 - \lambda)(\lambda^2 - 3\lambda) \\ &= (1 - \lambda)(\lambda - 3)\lambda \end{aligned}$$

We immediately see the determinant is zero if and only if $\lambda = 1$, $\lambda = 3$ or $\lambda = 0$. These are therefore the eigenvalues of the matrix A .

Next, we can compute the associated eigenspaces. For $\lambda = 1$ we get:

$$A - I = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix} \sim \left(\begin{array}{ccc|ccc} \mathbf{1} & 0 & 1 & & & \\ 0 & \mathbf{1} & 0 & & & \\ 0 & 0 & 0 & & & \end{array} \right)$$

where we obtained the reduced echelon form through the techniques of Chapter 1. We find:

$$E_1 = \text{Null}(A - I) = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

For $\lambda = 3$ we get:

$$A - 3I = \begin{pmatrix} -2 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -2 \end{pmatrix} \sim \begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

From the reduced echelon form we find:

$$E_3 = \text{Null}(A - 3I) = \text{Span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

Finally for $\lambda = 0$ we get:

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \sim \begin{pmatrix} \mathbf{1} & 0 & -1 \\ 0 & \mathbf{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

From the reduced echelon form we find:

$$E_0 = \text{Null } A = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

□

Concerning the calculations of the eigenspaces above, it is very important to realize that, by Definitions 5.1 and 5.3, each eigenspace must contain more vectors than only the zero vector!

5.2 Diagonalization

We can often simplify matrices by choosing an appropriate basis. We can sometimes even make a matrix diagonal. This is directly related to eigenvalues and eigenvectors as will become obvious in this section.

Theorem 5.9 Let $\lambda_1, \lambda_2, \dots, \lambda_p$ be pairwise distinct eigenvalues of a matrix A with corresponding eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$. In that case, the set

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$$

is linearly independent.

The proof of the above theorem is addressed in Exercise 5.13.

Corollary 5.10 Assume A is an $n \times n$ matrix with n distinct eigenvalues. In that case $\mathbb{R}^n$ has a basis consisting of eigenvectors of A .

The proof of the above corollary is addressed in Exercise 5.14.

Example 5.11 In Example 5.8 we found that the matrix A has three distinct eigenvalues $\lambda_1 = 1$, $\lambda_2 = 3$ and $\lambda_3 = 0$. Associated eigenvectors are given by:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The above result claims these eigenvectors are independent. Indeed if we construct a matrix P whose columns are exactly the given vectors

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

then we see

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

where we first added the first row to the third row and then added the second row to the third row. We found an echelon form and it is immediately clear that the linear system $P\mathbf{x} = \mathbf{0}$ only has the trivial solution which implies that the columns of P are linearly independent. We do not need to continue with elementary row operations until we get a reduced echelon form to conclude this. Three independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$ which follows from Theorem 3.17. $\square$

The fact that we have a basis of eigenvectors turns out to be extremely useful:

Theorem 5.12 Let A be an $n \times n$ matrix and let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\mathbb{R}^n$ consisting of eigenvectors with associated eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Let

$$P = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n)$$

be the matrix with the $\mathbf{v}_i$'s as columns. In that case, we have:

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$$

Proof : Clearly P is invertible because its columns span $\mathbb{R}^n$ (see Theorem 3.24). Next we note that

$$P^{-1}AP = D$$

where

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{pmatrix}$$

is equivalent to:

$$AP = PD. \tag{5.2}$$

In order to verify the above equation, we look at each column. On the left, the i th column equals

$$A\mathbf{v}_i.$$

On the other hand, the right hand side of (5.2) yields

$$\lambda_i \mathbf{v}_i.$$

Clearly since $\mathbf{v}_i$ is an eigenvector associated with eigenvalue λ_i of the matrix A we have:

$$A\mathbf{v}_i = \lambda_i \mathbf{v}_i.$$

Since each column on the left and right of (5.2) is equal, we find the required result. ■

Note that the order of the eigenvectors in the columns of P corresponds to the order of the eigenvalues on the main diagonal of D ! The property obtained in the above theorem has a special name:

Definition 5.13 We call a square matrix A **diagonalizable** if there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix.

We have seen that the existence of a basis of eigenvectors guarantees that the matrix A is diagonalizable. This is the only case.

Theorem 5.14 A square matrix A in $\mathbb{R}^{n \times n}$ is diagonalizable if and only if there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of A .

Proof : The “if” part corresponds to Theorem 5.12. In order to prove the “only if” part, suppose that A is diagonalizable. We note that $P^{-1}AP = D$ implies $AP = PD$. If the columns of P are called $\mathbf{v}_1, \dots, \mathbf{v}_n$ and the diagonal elements of D are called $\lambda_1, \lambda_2, \dots, \lambda_n$ then looking at $AP = PD$ column by column, we find:

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i$$

and hence the columns of P (which by invertibility form a basis of $\mathbb{R}^n$) consist of eigenvectors of the matrix A . ■

There are two important cases when we know that the matrix A is diagonalizable:

Lemma 5.15 If a matrix A has n distinct eigenvalues, then the matrix is diagonalizable.

Lemma 5.16 If a matrix A is symmetric, i.e. $A = A^T$, then the matrix is diagonalizable.

The first lemma is a direct consequence of Corollary 5.10 and Theorem 5.14 but the second lemma requires more work (which we will not include in these notes).

However, the above two lemmas are not the only cases that a matrix is diagonalizable as the following example illustrates.

Example 5.17 Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & -1 \\ -2 & 4 & -2 \\ -2 & 2 & 0 \end{pmatrix}$$

To determine the eigenvalues, we will first determine the characteristic polynomial

$$\begin{aligned}\det(A - \lambda I_3) &= \begin{vmatrix} 1 - \lambda & 1 & -1 \\ -2 & 4 - \lambda & -2 \\ -2 & 2 & -\lambda \end{vmatrix} = \begin{vmatrix} 1 - \lambda & 1 & -1 \\ -2 & 4 - \lambda & -2 \\ 0 & \lambda - 2 & 2 - \lambda \end{vmatrix} \\ &= (\lambda - 2) \begin{vmatrix} 1 - \lambda & 1 & -1 \\ -2 & 4 - \lambda & -2 \\ 0 & 1 & -1 \end{vmatrix}\end{aligned}$$

where we subtracted the second row from the third row (which does not affect the determinant) and then extract the factor $\lambda - 2$ from the third row. We can then expand along the first column and get:

$$\begin{aligned}\begin{vmatrix} 1 - \lambda & 1 & -1 \\ -2 & 4 - \lambda & -2 \\ 0 & 1 & -1 \end{vmatrix} &= (1 - \lambda) \begin{vmatrix} 4 - \lambda & -2 \\ 1 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} \\ &= (1 - \lambda)(\lambda - 2) - (-2)(0) \\ &= (1 - \lambda)(\lambda - 2)\end{aligned}$$

Therefore, we get

$$\det(A - \lambda I_3) = (\lambda - 2)^2(1 - \lambda).$$

We note that A has eigenvalues 1 and 2. We see that 2 is a double zero of the characteristic polynomial. Next, let us determine the eigenvectors. For $\lambda = 1$ we get

$$A - I_3 = \begin{pmatrix} 0 & 1 & -1 \\ -2 & 3 & -2 \\ -2 & 2 & -1 \end{pmatrix} \sim \left(\begin{array}{ccc|ccc} \mathbf{1} & 0 & -\frac{1}{2} & & & \\ 0 & 1 & -1 & & & \\ 0 & 0 & 0 & & & \end{array} \right).$$

We find

$$E_1 = \text{Null}(A - I_3) = \text{Span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right\}$$

For $\lambda = 2$ we get

$$A - 2I_3 = \begin{pmatrix} -1 & 1 & -1 \\ -2 & 2 & -2 \\ -2 & 2 & -2 \end{pmatrix} \sim \left(\begin{array}{ccc|ccc} \mathbf{1} & -1 & 1 & & & \\ 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & & \end{array} \right).$$

We find

$$E_2 = \text{Null}(A - 2I_3) = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Even though we only have two distinct eigenvalues we can still find three independent eigenvectors:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

where the last two eigenvectors are related to the same eigenvalue 2. In other words, we could find two independent eigenvectors for this one eigenvalue and we therefore could obtain a basis of eigenvectors. This will not always be the case as we will see in Exercise 5.7. Since we have a basis of eigenvectors we find that A is indeed diagonalizable, i.e. A satisfies:

$$P^{-1}AP = D$$

where

$$P = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Note that the order of the eigenvectors in the columns of P corresponds to the order of the eigenvalues on the main diagonal of D ! $\square$

An interesting question is whether all square matrices have eigenvalues and eigenvectors. Consider the following example.

Example 5.18 We have:

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The characteristic polynomial is given by:

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = \lambda^2 + 1$$

The eigenvalues are determined by

$$\lambda^2 + 1 = 0.$$

But this has no real solution. Therefore there are no real eigenvalues. However, when we allow λ to be complex then the above equation clearly has a solution $\lambda = i$ or $\lambda = -i$. Note, however, that complex eigenvalues also require complex eigenvectors. In this case:

$$A \begin{pmatrix} 1 \\ i \end{pmatrix} = i \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad A \begin{pmatrix} 1 \\ -i \end{pmatrix} = -i \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad \square$$

In the above example we have seen that we can find complex eigenvalues in case the characteristic polynomial has complex roots. In the next section, we will motivate by example why one would consider complex eigenvalues and also illustrate the power of this technique.

In our original definition of the concept of vector space in Definition 2.1, we defined scalar multiplication by a constant $c \in \mathbb{R}$. However, we can also define complex vector spaces which have the same properties with the only difference that we use $c \in \mathbb{C}$.

Definition 5.19 A **complex vector space** V is a collection of elements, called vectors for which there are two operations: addition and scalar multiplication which satisfy the following ten axioms:

- $\mathbf{u}, \mathbf{v} \in V$ implies $\mathbf{u} + \mathbf{v} \in V$.
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.
- There exists a vector $\mathbf{0} \in V$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$.
- For every $\mathbf{u} \in V$ there exists a vector $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$.
- The scalar multiplication of $\mathbf{u} \in V$ with $c \in \mathbb{C}$, denoted by $c\mathbf{u}$, is in V .
- $c(d\mathbf{u}) = (cd)\mathbf{u}$ for all $\mathbf{u} \in V$ and $c, d \in \mathbb{C}$.
- $1\mathbf{u} = \mathbf{u}$ for all $\mathbf{u} \in V$.
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{C}$.
- $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ for all $\mathbf{u} \in V$ and $c, d \in \mathbb{C}$.

The space $\mathbb{C}^n$ consists of column vectors with exactly n elements in $\mathbb{C}$. This forms a complex vector space as defined above. Similarly $\mathbb{C}^{k \times n}$ are matrices whose elements are in $\mathbb{C}$. Matrix multiplication is defined the same as for real-valued matrices.

Clearly $\mathbb{R}^{k \times n}$ can be seen as a subset of $\mathbb{C}^{k \times n}$. Note that $\mathbb{R}^{k \times n}$ is **not** a subspace of $\mathbb{C}^{k \times n}$. After all,

$$\alpha A$$

with $A \in \mathbb{R}^{k \times n}$ is not in $\mathbb{R}^{k \times n}$ when α is complex-valued and $A \neq 0$.

Definition 5.20 Consider a square $n \times n$ matrix A . In that case $\lambda \in \mathbb{C}$ is called a (complex) **eigenvalue** of the matrix A if there exists a nonzero vector $\mathbf{x} \in \mathbb{C}^n$ such that

$$A\mathbf{x} = \lambda\mathbf{x} \tag{5.3}$$

In that case $\mathbf{x}$ is called an **eigenvector** of A associated with the eigenvalue λ .

Because $\mathbb{R}^{n \times n}$ is a subset of $\mathbb{C}^{n \times n}$ we can define complex eigenvalues for a real matrix. Let us recall that the complex conjugate of a complex number $a + bi$ is defined as $a - bi$. In the computation of eigenvalues and eigenvectors of a real matrix the following lemma is useful.

Lemma 5.21 Let A be a real, square matrix with a complex eigenvalue λ . In that case, the complex conjugate $\bar{\lambda}$ is also an eigenvalue of the matrix A . Moreover, if $\mathbf{x}$ is an eigenvector associated with eigenvalue λ then $\bar{\mathbf{x}}$ is an eigenvector associated with eigenvalue $\bar{\lambda}$.

The proof of the above theorem is addressed in Exercise 5.18bc. Note that in the above, $\bar{\mathbf{x}}$ is obtained from $\mathbf{x}$ by taking the complex conjugate of each component of the vector.

Example 5.22 Consider the matrix

$$A = \begin{pmatrix} 0 & -2 & -2 \\ 1 & 2 & -1 \\ -2 & 2 & 0 \end{pmatrix}$$

To determine the eigenvalues, we will first determine the characteristic poly-

nomial

$$\begin{aligned}
 \det(A - \lambda I_3) &= \begin{vmatrix} -\lambda & -2 & -2 \\ 1 & 2 - \lambda & -1 \\ -2 & 2 & -\lambda \end{vmatrix} = \begin{vmatrix} 0 & -2 + \lambda(2 - \lambda) & -\lambda - 2 \\ 1 & 2 - \lambda & -1 \\ 0 & 6 - 2\lambda & -2 - \lambda \end{vmatrix} \\
 &= (-1) \begin{vmatrix} -2 + \lambda(2 - \lambda) & -\lambda - 2 \\ 6 - 2\lambda & -2 - \lambda \end{vmatrix} \\
 &= -(-2 + \lambda(2 - \lambda))(-2 - \lambda) + (6 - 2\lambda)(-2 - \lambda) \\
 &= -(\lambda + 2)(\lambda^2 - 4\lambda + 8)
 \end{aligned}$$

where in the first step, we added λ times the second row to the first row and added 2 times the second row to the third row (which does not affect the determinant). In the second step we expanded along the first column. Therefore, we get

$$\det(A - \lambda I_3) = -(\lambda + 2)(\lambda^2 - 4\lambda + 8).$$

We note that A has real eigenvalue -2 . We also find two complex eigenvalues $2 + 2i$ and $2 - 2i$.

Next, let us first determine the eigenvector associated with the eigenvalue -2 . For $\lambda = -2$ we get

$$A + 2I_3 = \begin{pmatrix} 2 & -2 & -2 \\ 1 & 4 & -1 \\ -2 & 2 & 2 \end{pmatrix} \sim \left(\begin{array}{ccc|ccc} \mathbf{1} & 0 & -1 & & & \\ 0 & \mathbf{1} & 0 & & & \\ 0 & 0 & 0 & & & \end{array} \right).$$

From the reduced echelon form we find

$$E_{-2} = \text{Null}(A + 2I_3) = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Next we consider the complex eigenvalues. For $\lambda = 2 + 2i$ we get:

$$\begin{aligned}
 A - (2 + 2i)I_3 &= \begin{pmatrix} -2 - 2i & -2 & -2 \\ 1 & -2i & -1 \\ -2 & 2 & -2 - 2i \end{pmatrix} \sim \begin{pmatrix} 1 & -2i & -1 \\ -2 - 2i & -2 & -2 \\ -2 & 2 & -2 - 2i \end{pmatrix} \\
 &\sim \begin{pmatrix} 1 & -2i & -1 \\ 0 & 2 - 4i & -4 - 2i \\ 0 & 2 - 4i & -4 - 2i \end{pmatrix} \sim \begin{pmatrix} 1 & -2i & -1 \\ 0 & 2 - 4i & -4 - 2i \\ 0 & 0 & 0 \end{pmatrix} \\
 &\sim \begin{pmatrix} 1 & -2i & -1 \\ 0 & 1 & -i \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -i \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

In the first step, we interchanged the first and second rows. Next, we added 2 times the first row to the third row and added $2 + 2i$ times the first row to the second row. After this, we subtracted the second row from the third row and then we divided the second row by $2 - 4i$. In the final step we added $2i$ times the second row to the first row. We get

$$E_{2+2i} = \text{Null}(A - (2 + 2i)I_3) = \text{Span} \left\{ \begin{pmatrix} -1 \\ i \\ 1 \end{pmatrix} \right\}$$

Using Lemma 5.21 we immediately find

$$E_{2-2i} = \text{Null}(A - (2 - 2i)I_3) = \text{Span} \left\{ \begin{pmatrix} -1 \\ -i \\ 1 \end{pmatrix} \right\}$$

We therefore find three eigenvectors for the matrix A :

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -1 \\ i \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -1 \\ -i \\ 1 \end{pmatrix}. \quad \square$$

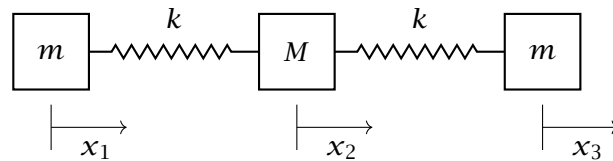
We end this section by stating (without proof) that Theorems 5.9, 5.12 and 5.14 can be generalized to complex eigenvalues and eigenvectors.

5.3 Differential equations

Eigenvalues and eigenvectors are used in many fields. Especially the concept of diagonalization is a very useful tool. In this section, we give two examples to illustrate the use of eigenvalues, eigenvectors and diagonalization: one mechanical and one electrical example. However, differential equations occur everywhere. We use them for modelling financial markets, for modelling the growth of a tumor and for modelling chemical reactions to name but a few examples.

5.3.1 Example: real eigenvalues

In this section we consider a mechanical example to illustrate the use of eigenvalues and eigenvectors. Consider three masses connected by identical springs with spring constant k , which can only move along the x -axis.



If we assume that the spring forces depend linearly on the extension of the spring (Hooke's law) then we get the following equations of motion (here $\ddot{x}_i$ denotes the second derivative of x_i with respect to time):

$$\begin{aligned} m\ddot{x}_1 &= -k(x_1 - x_2) \\ M\ddot{x}_2 &= -k(x_2 - x_1) - k(x_2 - x_3) \\ m\ddot{x}_3 &= -k(x_3 - x_2) \end{aligned} \quad (5.4)$$

If we introduce the vector $\mathbf{x}$ and the matrix A as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \frac{k}{m} \begin{pmatrix} -1 & 1 & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & 1 & -1 \end{pmatrix}$$

where

$$\alpha = \frac{m}{M}$$

this can be rewritten as (cf. Exercise 5.19a):

$$\ddot{\mathbf{x}} = A\mathbf{x}$$

The question is how to solve these three coupled differential equations. We can easily verify that (cf. Exercise 5.19b):

$$\det(A - \lambda I) = -\lambda \left(\lambda + \frac{k}{m} \right) \left(\lambda + 2\frac{k}{M} + \frac{k}{m} \right)$$

We get

$$\lambda_1 = 0, \quad \lambda_2 = -\frac{k}{m}, \quad \lambda_3 = -2\frac{k}{M} - \frac{k}{m}$$

Note that these eigenvalues are distinct, so by Corollary 5.10 and Theorem 5.14 the matrix A is diagonalizable. If we write

$$P^{-1}AP = D$$

then we can define:

$$\mathbf{p} = P^{-1}\mathbf{x}$$

and we get (cf. Exercise 5.19c):

$$\ddot{\mathbf{p}} = D\mathbf{p}.$$

If

$$D = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

then we find:

$$\begin{aligned}\ddot{p}_1 &= \lambda_1 p_1 \\ \ddot{p}_2 &= \lambda_2 p_2 \\ \ddot{p}_3 &= \lambda_3 p_3\end{aligned}$$

Since these differential equations are no longer coupled, these equations can be easily solved using the techniques from Calculus 1B (homogeneous second order linear differential equations with constant coefficients).

The first differential equation will then have a solution describing a linear translation over time:

$$p_1(t) = a_1 + a_2 t$$

The second equation is an oscillation where the middle mass stays put and the outer two masses move symmetrically with respect to each other.

$$p_2(t) = b_1 \cos\left(t\sqrt{\frac{k}{m}}\right) + b_2 \sin\left(t\sqrt{\frac{k}{m}}\right)$$

The final equation also yields an oscillation:

$$p_3(t) = c_1 \cos\left(t\sqrt{2\frac{k}{M} + \frac{k}{m}}\right) + c_2 \sin\left(t\sqrt{2\frac{k}{M} + \frac{k}{m}}\right)$$

However, this oscillation involves all three masses and is hence a bit more involved. The solution of the original differential equation is then given by:

$$\mathbf{x}(t) = P \begin{pmatrix} p_1(t) \\ p_2(t) \\ p_3(t) \end{pmatrix} = p_1(t)\mathbf{v}_1 + p_2(t)\mathbf{v}_2 + p_3(t)\mathbf{v}_3 \quad (5.5)$$

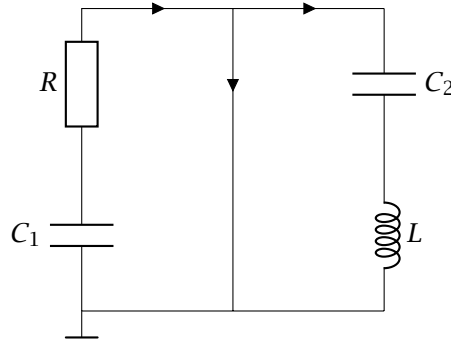
where $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are the eigenvectors associated with the eigenvalues $\lambda_1, \lambda_2, \lambda_3$. In this special case we get (cf. Exercise 5.19d):

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ -2\alpha \\ 1 \end{pmatrix}.$$

Eigenvalues help us to determine all possible solutions. Moreover, we see that the solution (5.5) is a combination of the translation and the two oscillations. By splitting the general solution into these three elementary movements we get a better insight into the type of movements that these masses can make.

5.3.2 Example: complex eigenvalues

In this section, we consider an electrical circuit to illustrate the use of eigenvalues and eigenvectors. Consider the following electrical circuit with two capacitors, one inductor and a resistor:



If we model this circuit using Kirchhoff's laws we get:

$$\begin{aligned} RC_1 \dot{x}_1 &= -x_1 \\ C_2 \dot{x}_2 &= x_3 \\ L \dot{x}_3 &= -x_2 \end{aligned} \tag{5.6}$$

where x_1 and x_2 are the voltages across the two capacitors and x_3 is the current through the inductor. For simplicity, we assume the capacitors are 1 farad, the resistor is 1 ohm and the inductor is 1 henry, i.e. $C_1 = C_2 = 1$, $L = 1$ and $R = 1$. Far from realistic values, but it gives a model which is easy to analyse and the essence of the result is the same. We get:

$$\dot{\mathbf{x}} = A\mathbf{x}$$

where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

Let us first compute the eigenvalues and associated eigenvectors of A . We have (cf. Exercise 5.20a):

$$\det(A - \lambda I) = -(\lambda + 1)(\lambda^2 + 1)$$

which yields three eigenvalues $\lambda_1 = -1$, $\lambda_2 = i$ and $\lambda_3 = -i$. For the associated eigenvectors, we find (cf. Exercise 5.20b):

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix}.$$

If we define:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & i & -i \end{pmatrix}, \quad D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{pmatrix}$$

then we have:

$$P^{-1}AP = D.$$

We define:

$$\mathbf{p} = P^{-1}\mathbf{x}$$

and we get:

$$\dot{\mathbf{p}} = D\mathbf{p}.$$

which can be written as:

$$\dot{p}_1 = \lambda_1 p_1$$

$$\dot{p}_2 = \lambda_2 p_2$$

$$\dot{p}_3 = \lambda_3 p_3$$

These equations are decoupled and therefore easy to solve. The question is of course whether this still has any connection with the solution of our electrical circuit since we are suddenly using complex numbers. But let us just formally solve these equations. We get using the techniques from Calculus 1(B):

$$p_1(t) = e^{-t} p_1(0), \quad p_2(t) = e^{it} p_2(0), \quad p_3(t) = e^{-it} p_3(0)$$

We then find:

$$\mathbf{x}(t) = P \begin{pmatrix} p_1(t) \\ p_2(t) \\ p_3(t) \end{pmatrix} = p_1(t)\mathbf{v}_1 + p_2(t)\mathbf{v}_2 + p_3(t)\mathbf{v}_3 \quad (5.7)$$

Furthermore:

$$\begin{pmatrix} p_1(0) \\ p_2(0) \\ p_3(0) \end{pmatrix} = P^{-1}\mathbf{x}(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{i}{2} \\ 0 & \frac{1}{2} & \frac{i}{2} \end{pmatrix} \begin{pmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \end{pmatrix}$$

and therefore

$$p_1(0) = x_1(0), \quad p_2(0) = \frac{1}{2}x_2(0) - \frac{i}{2}x_3(0), \quad p_3(0) = \frac{1}{2}x_2(0) + \frac{i}{2}x_3(0).$$

Using this in (5.7) we get:

$$\begin{aligned}
 \mathbf{x}(t) &= e^{-t} p_1(0) \mathbf{v}_1 + e^{it} p_2(0) \mathbf{v}_2 + e^{-it} p_3(0) \mathbf{v}_3 \\
 &= e^{-t} x_1(0) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + e^{it} \left(\frac{1}{2} x_2(0) - \frac{i}{2} x_3(0) \right) \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix} \\
 &\quad + e^{-it} \left(\frac{1}{2} x_2(0) + \frac{i}{2} x_3(0) \right) \begin{pmatrix} 0 \\ 1 \\ -i \end{pmatrix} \\
 &= x_1(0) \begin{pmatrix} e^{-t} \\ 0 \\ 0 \end{pmatrix} + x_2(0) \begin{pmatrix} 0 \\ \cos t \\ -\sin t \end{pmatrix} + x_3(0) \begin{pmatrix} 0 \\ \sin t \\ \cos t \end{pmatrix}
 \end{aligned}$$

(cf. Exercise 5.20d) where we use:

$$\cos t = \frac{e^{it} + e^{-it}}{2}, \quad \sin t = \frac{e^{it} - e^{-it}}{2i}.$$

Note that in the end we get an expression containing only real numbers for our solution of the differential equation even though we used complex numbers in our derivation. It is easy to verify that our solution satisfies the original differential equations (5.6). Therefore, complex eigenvalues enable us to solve these differential equations and obtain a real solution.

5.4 Exercises

5.4.1 Self study exercises

5.1 Let $A \in \mathbb{R}^{n \times n}$ be a matrix.

- (a) What is the easiest way to verify if a given vector $\mathbf{v} \in \mathbb{R}^n$ is an eigenvector of A ?
- (b) What is the easiest way to verify if a given number $\lambda \in \mathbb{R}$ is an eigenvalue of A ?
- (c) Let λ be an eigenvalue of A and E_λ the corresponding eigenspace. Is every vector in E_λ an eigenvector of A ?

5.2 The matrices A and the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are given by:

$$A = \begin{pmatrix} -4 & 3 & 3 \\ 2 & -3 & -2 \\ -1 & 0 & -2 \end{pmatrix}; \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}; \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix}.$$

- (a) Verify without computing the eigenvectors of A , if $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of A , and if so, determine the corresponding eigenvalue.
- (b) Verify, without computing the eigenvalues of A , if -5 and 2 are eigenvalues of A .

5.3 Show that (5.1) can be rewritten as $\mathbf{x} \in \text{Null}(A - \lambda I_n)$ (cf. proof of Lemma 5.4).

5.4 The matrices A and B are given by:

$$A = \begin{pmatrix} 3 & 2 \\ 3 & 8 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Determine the eigenvalues and corresponding eigenspaces of A and B .

5.5 Determine if the matrices in Exercise 5.4 are diagonalizable. If so, determine an invertible matrix P and a diagonal matrix D satisfying Definition 5.13.

5.6 Determine the eigenvalues and corresponding eigenspaces of the matrix

$$A = \begin{pmatrix} -1 & -2 \\ 5 & 1 \end{pmatrix}.$$

5.4.2 Tutorial exercises

5.7 The matrices A , B , C and E are given by:

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}; \quad B = \begin{pmatrix} 6 & 3 & -8 \\ 0 & -2 & 0 \\ 1 & 0 & -3 \end{pmatrix};$$

$$C = \begin{pmatrix} -4 & 1 & 1 \\ 2 & -3 & 2 \\ 3 & 3 & -2 \end{pmatrix}; \quad E = \begin{pmatrix} 4 & -4 & -5 \\ 1 & 0 & -3 \\ 0 & 0 & 2 \end{pmatrix}.$$

Determine for each of these matrices its eigenvalues and a basis for the corresponding eigenspaces.

5.8 Determine if the matrices in Exercise 5.7 are diagonalizable. If so, determine an invertible matrix P and a diagonal matrix D satisfying Definition 5.13.

5.9 The matrices A and B are given by:

$$A = \begin{pmatrix} 5 & 1 \\ -8 & 1 \end{pmatrix}; \quad B = \begin{pmatrix} -2 & 0 & 5 \\ 0 & 2 & 0 \\ -5 & 0 & 4 \end{pmatrix}.$$

- (a) Determine the eigenvalues and corresponding eigenspaces of A and B .
- (b) Determine if the matrices A and B in part (a) are diagonalizable, and if so, determine diagonal matrices D_1 and D_2 and invertible matrices P and Q such that $P^{-1}AP = D_1$ and $Q^{-1}BQ = D_2$.

5.10 We are given a matrix A :

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 4 & 2 - \alpha & 0 \\ 4 & 0 & \alpha \end{pmatrix}$$

where $\alpha \in \mathbb{R}$. Determine all $\alpha \in \mathbb{R}$ for which A is diagonalizable.

5.11 Let $\mathbf{x}_0$ be an eigenvector of A associated with eigenvalue λ . Define the vectors $\mathbf{x}_1, \mathbf{x}_2, \dots$ recursively by: $\mathbf{x}_k = A\mathbf{x}_{k-1}$ ($k = 1, 2, \dots$). Prove that $\mathbf{x}_k = \lambda^k \mathbf{x}_0$ for $k = 0, 1, 2, \dots$.

5.12 (a) Prove that if λ is an eigenvalue of A , then λ is an eigenvalue of A^T .

(b) Suppose that $A \in \mathbb{R}^{n \times n}$ is a matrix with the property that all row sums are equal to r (i.e., $\sum_{k=1}^n a_{ik} = r$, for all $i \in \{1, \dots, n\}$).

Determine an eigenvalue of A and a corresponding eigenvector.

(c) Determine an eigenvalue of A if all column sums are equal to r .

5.13 In this exercise we will construct a proof of Theorem 5.9. Let $\lambda_1, \dots, \lambda_p$ be the pairwise distinct eigenvalues of a matrix A with corresponding eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$. Suppose that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly dependent. We will deduce a contradiction.

(a) By Theorem 3.14, there exist $\alpha_1, \dots, \alpha_p \in \mathbb{R}$, not all zero, such that $\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$. Show that this implies that there must exist a vector $\mathbf{v}_k \in S$ which is a linear combination of its predecessors $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$.

(b) Let m be the least index in $\{1, \dots, p\}$ with the property that $\mathbf{v}_m$ is a linear combination of its predecessors $\mathbf{v}_1, \dots, \mathbf{v}_{m-1}$, say

$$\mathbf{v}_m = \beta_1 \mathbf{v}_1 + \dots + \beta_{m-1} \mathbf{v}_{m-1}. \quad (5.8)$$

Note that, by part (a), the fact that m is the least index implies that the set $\{\mathbf{v}_1, \dots, \mathbf{v}_{m-1}\}$ is linearly independent. Now multiply both sides in (5.8) on the left with A and deduce that

$$\lambda_m \mathbf{v}_m = \beta_1 \lambda_1 \mathbf{v}_1 + \dots + \beta_{m-1} \lambda_{m-1} \mathbf{v}_{m-1}. \quad (5.9)$$

- (c) Finally consider the equation (5.9) $-\lambda_m \cdot$ (5.8) and use Theorem 3.14 to deduce a contradiction.

5.14 Prove Corollary 5.10.

5.15 (a) Prove that the eigenvalues of a triangular matrix are the entries on the main diagonal.

- (b) Prove that if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of an $n \times n$ -matrix A , then

$$\det A = \lambda_1 \cdot \dots \cdot \lambda_n.$$

Hint: use that the characteristic polynomial $p(\lambda)$ of A has degree n and has zeros $\lambda_1, \dots, \lambda_n$. So $p(\lambda)$ can be factorized as:

$$p(\lambda) = c(\lambda_1 - \lambda) \cdot \dots \cdot (\lambda_n - \lambda).$$

Furthermore, the coefficient of λ^n in $p(\lambda)$ is $(-1)^n$, so $c = 1$.

- (c) Prove that if λ is an eigenvalue of A with corresponding eigenvector $\mathbf{x}$, then $A^k \mathbf{x} = \lambda^k \mathbf{x}$ for all $k \geq 1$.
- (d) Prove that if $A^k = 0$ for some $k \geq 1$, then $\lambda = 0$ is the only eigenvalue of A .
- (e) Prove that if A is invertible, and λ is an eigenvalue of A , then λ^{-1} is an eigenvalue of A^{-1} .

5.16 (a) Suppose that $B = P^{-1}AP$. Prove that A and B have the same eigenvalues.

Hint: show that $\det(P^{-1}AP - \lambda I)$ can be rewritten as $\det(P^{-1}(A - \lambda I)P)$ and then use Theorem 4.12.

- (b) Suppose that $D = P^{-1}AP$ for some invertible matrix P and diagonal matrix D . Prove that $A^k = PD^kP^{-1}$, for all $k \geq 1$.
- (c) Compute A^{100} for the matrix A in Exercise 5.4.

5.17 Let A be an $n \times n$ -matrix.

- (a) Prove that if A is invertible then A is diagonalizable, or give a counterexample.
- (b) Prove that if A is diagonalizable then A is invertible, or give a counterexample.

- (c) Prove that if A is both invertible and diagonalizable, then so is A^{-1} .

Hint: write $D = P^{-1}AP$, explain that D must be invertible, and use Exercise 2.16 to find an expression for D^{-1} . Finally explain that also D^{-1} is diagonal.

- 5.18 (a) Verify that $D = P^{-1}AP$ for the matrix A in Example 5.22. Here D is the diagonal matrix with the eigenvalues -2 , $2 + 2i$ and $2 - 2i$ of A on its main diagonal, and P the matrix whose columns are the corresponding eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.
- (b) Suppose $\lambda = a + bi \in \mathbb{C}$ is an eigenvalue of A with corresponding eigenvector $\mathbf{x} \in \mathbb{C}^n$. Write $\mathbf{x} = \mathbf{u} + i\mathbf{v}$, where $\mathbf{u} \in \mathbb{R}^n$ contains the real parts of $\mathbf{x}$ and $\mathbf{v} \in \mathbb{R}^n$ the imaginary parts. Assume that Lemma 2.5 also holds for $\alpha \in \mathbb{C}$. Show that

$$A\mathbf{x} = (a\mathbf{u} - b\mathbf{v}) + (b\mathbf{u} + a\mathbf{v})i.$$

- (c) Use part (b) to prove Lemma 5.21: show that if $A\mathbf{x} = \lambda\mathbf{x}$ then $A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$.

- 5.19 Consider the example in Subsection 5.3.1, where $\mathbf{x}$ denotes the positions of the three masses, $\ddot{\mathbf{x}}$ denotes the vector of second derivatives of $\mathbf{x}$ and A denotes the matrix containing the coefficients of the corresponding differential equations.

- (a) Verify that (5.4) can be rewritten as $\ddot{\mathbf{x}} = A\mathbf{x}$.
- (b) Show that $\det(A - \lambda I) = -\lambda(\lambda + \frac{k}{m})(\lambda + 2\frac{k}{M} + \frac{k}{m})$.
- (c) Show that $P^{-1}AP = D$ and $\mathbf{p} = P^{-1}\mathbf{x}$ implies $\ddot{\mathbf{p}} = D\mathbf{p}$.
- (d) Show that the vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ given at the end of this example are eigenvectors associated with the eigenvalues $\lambda_1 = 0$, $\lambda_2 = -\frac{k}{m}$ and $\lambda_3 = -2\frac{k}{M} - \frac{k}{m}$.

- 5.20 Consider the example in Subsection 5.3.2, where $\mathbf{x}$ denotes the vector with the voltages through the two capacitors and the current through the inductor, $\dot{\mathbf{x}}$ denotes the vector with the derivatives of $\mathbf{x}$ and A denotes the matrix containing the coefficients of the corresponding differential equations.

- (a) Show that $\det(A - \lambda I) = -(\lambda + 1)(\lambda^2 + 1)$.
- (b) Show that the vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ given in this example are eigenvectors associated with the eigenvalues $\lambda_1 = -1$, $\lambda_2 = i$ and $\lambda_3 = -i$ respectively.

(c) Show that

$$P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{i}{2} \\ 0 & \frac{1}{2} & \frac{i}{2} \end{pmatrix}.$$

(d) Verify the last equality in the derivation of

$$\mathbf{x}(t) = x_1(0) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2(0) \begin{pmatrix} 0 \\ \cos t \\ -\sin t \end{pmatrix} + x_3(0) \begin{pmatrix} 0 \\ \sin t \\ \cos t \end{pmatrix}.$$

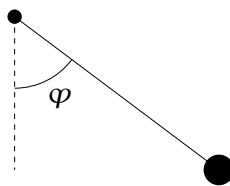
Chapter 6

Linear transformations

6.1 Functions from $\mathbb{R}^n$ to $\mathbb{R}^k$

In previous courses, we have seen many examples of functions from $\mathbb{R}$ to $\mathbb{R}$. We have also seen functions from $\mathbb{R}^2$ to $\mathbb{R}$ and investigated concepts such as partial derivatives. Even more generally, we can define functions from $\mathbb{R}^n$ to $\mathbb{R}^k$. We will write $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$ or sometimes $\mathbf{u} \mapsto F(\mathbf{u})$. We call $F(\mathbf{u})$, often also denoted by $F\mathbf{u}$, the *image* of $\mathbf{u}$ (under F). Conversely we call $\mathbf{u}$ the *original* of $F(\mathbf{u})$. Note that, by definition, for any vector $\mathbf{u} \in \mathbb{R}^n$ there exists a unique image $F\mathbf{u}$. However, given $\mathbf{y} \in \mathbb{R}^k$ there might exist multiple originals $\mathbf{u} \in \mathbb{R}^n$ for which $\mathbf{y} = F\mathbf{u}$ or there might not exist any original at all. Only in special cases, we have a unique original for every image.

Example 6.1 Consider a pendulum with initial position $\varphi(0)$ and initial velocity $\dot{\varphi}(0)$. Note that $\dot{\varphi}$ denotes the derivative of the function φ .



We define:

$$\mathbf{u} = \begin{pmatrix} \varphi(0) \\ \dot{\varphi}(0) \end{pmatrix}$$

The initial position and velocity determine the position and velocity at time 1. In other words we have a function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with

$$F\mathbf{u} = \begin{pmatrix} \varphi(1) \\ \dot{\varphi}(1) \end{pmatrix} \quad \square$$

Example 6.2 Consider the 2×3 matrix:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix}$$

The function $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is given by:

$$F(\mathbf{x}) = A\mathbf{x} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 + x_3 \\ x_1 - x_2 - x_3 \end{pmatrix}$$

In particular, the vector $\mathbf{x}$ is mapped to the vector $\mathbf{y}$ where:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 6 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \square$$

Instead of *function* F we will also use *mapping* or *transformation* F and we will sometimes denote transformations by other (capital) letters such as T, S, A instead of F .

6.2 Linear transformations

We will concentrate in this course on a special class of functions which “behave nicely” with respect to both of the vector operations (addition and scalar multiplication), the so-called linear transformations.

Definition 6.3 A function $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is called a **linear transformation** if

- $F(\mathbf{u} + \mathbf{v}) = F(\mathbf{u}) + F(\mathbf{v})$, and
- $F(\alpha\mathbf{u}) = \alpha F(\mathbf{u})$

for all vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ and all scalars α in $\mathbb{R}$.

It can be shown that Example 6.1 is not a linear transformation. However, Example 6.2 is a linear transformation:

Example 6.4 Consider Example 6.2. We have:

$$\begin{aligned}
 F(\mathbf{u} + \mathbf{v}) &= F \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{pmatrix} \\
 &= \begin{pmatrix} (u_1 + v_1) + (u_2 + v_2) + (u_3 + v_3) \\ (u_1 + v_1) - (u_2 + v_2) - (u_3 + v_3) \end{pmatrix} \\
 &= \begin{pmatrix} u_1 + u_2 + u_3 \\ u_1 - u_2 - u_3 \end{pmatrix} + \begin{pmatrix} v_1 + v_2 + v_3 \\ v_1 - v_2 - v_3 \end{pmatrix} \\
 &= F(\mathbf{u}) + F(\mathbf{v})
 \end{aligned}$$

Moreover,

$$\begin{aligned}
 F(\alpha \mathbf{u}) &= F \begin{pmatrix} \alpha u_1 \\ \alpha u_2 \\ \alpha u_3 \end{pmatrix} \\
 &= \begin{pmatrix} \alpha u_1 + \alpha u_2 + \alpha u_3 \\ \alpha u_1 - \alpha u_2 - \alpha u_3 \end{pmatrix} \\
 &= \alpha \begin{pmatrix} u_1 + u_2 + u_3 \\ u_1 - u_2 - u_3 \end{pmatrix} \\
 &= \alpha F(\mathbf{u})
 \end{aligned}$$

□

The above example is a linear transformation described by a matrix. The relationship between matrices and linear transformations is given by the following important theorem.

Theorem 6.5

- If A is a $k \times n$ matrix then the function $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$ defined by $F(\mathbf{u}) = A\mathbf{u}$ is a linear transformation.
- Every linear transformation $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$ can be represented by a **unique** $k \times n$ matrix A such that $F(\mathbf{u}) = A\mathbf{u}$.

Proof : For the first part, we assume $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$. In that case:

$$F(\mathbf{u} + \mathbf{v}) = A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = F(\mathbf{u}) + F(\mathbf{v})$$

where the second equality is a direct consequence of Lemma 2.5. Similarly for $\alpha \in \mathbb{R}$ and $\mathbf{u} \in \mathbb{R}^n$ we have:

$$F(\alpha \mathbf{u}) = A(\alpha \mathbf{u}) = \alpha A\mathbf{u} = \alpha F(\mathbf{u})$$

where the second equality is again a direct consequence of Lemma 2.5.

For the second part, we consider $\mathbf{e}_1, \dots, \mathbf{e}_n$, the columns of the identity matrix I_n , which are sometimes referred to as the standard basis. Define $\mathbf{a}_1, \dots, \mathbf{a}_n$ by:

$$\mathbf{a}_i = F(\mathbf{e}_i)$$

for $i = 1, \dots, n$. Next form a matrix A whose columns are given by $\mathbf{a}_1, \dots, \mathbf{a}_n$. Clearly,

$$F(\mathbf{e}_i) = \mathbf{a}_i = A\mathbf{e}_i$$

for $i = 1, \dots, n$. Next consider an arbitrary vector $\mathbf{u} \in \mathbb{R}^n$. We have:

$$\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2 + \dots + u_n\mathbf{e}_n$$

and therefore, since F is a linear transformation, we get:

$$\begin{aligned} F(\mathbf{u}) &= F(u_1\mathbf{e}_1 + \dots + u_n\mathbf{e}_n) \\ &= u_1F(\mathbf{e}_1) + \dots + u_nF(\mathbf{e}_n) \\ &= u_1\mathbf{a}_1 + \dots + u_n\mathbf{a}_n \\ &= A\mathbf{u} \end{aligned}$$

Regarding the uniqueness, we note that if there are two different matrices A_1 and A_2 such that

$$F(\mathbf{u}) = A_1\mathbf{u} \quad F(\mathbf{u}) = A_2\mathbf{u}$$

then we have $A_1\mathbf{u} = A_2\mathbf{u}$ for all $\mathbf{u}$. But $A_1\mathbf{e}_i = A_2\mathbf{e}_i$ implies that the i th column of A_1 is equal to the i th column of A_2 . Since this is true for $i = 1, \dots, n$ we find that each column of A_1 is equal to the corresponding column of A_2 which implies that $A_1 = A_2$. ■

Therefore, a linear transformation F uniquely determines a matrix A which will be denoted by $[F]$ and is called the *representation matrix* or *standard matrix* of the linear transformation F .

Example 6.6 Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be given by:

$$F \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} u_1 - u_2 \\ u_1 + u_2 \\ 2u_2 \end{pmatrix}$$

To determine $A = [F]$ we compute $F(\mathbf{e}_1)$ and $F(\mathbf{e}_2)$:

$$\mathbf{a}_1 = F(\mathbf{e}_1) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = F(\mathbf{e}_2) = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

and hence $[F]$ is the matrix with columns $\mathbf{a}_1$ and $\mathbf{a}_2$, i.e.

$$[F] = A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \square$$

For a matrix A we have defined two important subspaces $\text{Null } A$ and $\text{Col } A$ in Definition 3.4 and 3.11 respectively. The first space is related to the nonuniqueness of a solution of a linear system while the second was related to the solvability of linear systems. For a linear transformation we can define similar subspaces.

Definition 6.7 Consider a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$. We define the **kernel** and **image** (range) of the linear transformation T by

$$\ker T = \{ \mathbf{x} \in \mathbb{R}^n \mid T(\mathbf{x}) = \mathbf{0} \}$$

and

$$\text{im } T = \{ \mathbf{y} \in \mathbb{R}^k \mid \exists \mathbf{x} \text{ such that } T(\mathbf{x}) = \mathbf{y} \}$$

respectively.

Theorem 6.8 Consider a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$. Assume that A is the representation matrix of the linear transformation T . In that case:

$$\ker T = \text{Null } A$$

and

$$\text{im } T = \text{Col } A$$

The proof of the above theorem is addressed in Exercise 6.11a. Let us define:

Definition 6.9 A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is said to be **onto** if

$$\text{im } T = \mathbb{R}^k$$

A linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is said to be **one-to-one** if

$$T(\mathbf{u}) = T(\mathbf{v}) \text{ implies that } \mathbf{u} = \mathbf{v}$$

The following theorem gives an alternative characterization for the concept of one-to-one.

Theorem 6.10 Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a linear transformation. T is one-to-one if and only if $\ker T = \{\mathbf{0}\}$.

The proof of the above theorem is addressed in Exercise 6.10a. The next theorem connects the properties of one-to-one and onto for a linear transformation to corresponding properties for the associated representation matrix.

Theorem 6.11 Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a linear transformation with representation matrix A . We have

- (i) T is one-to-one if and only if $\text{Null } A = \{\mathbf{0}\}$.
- (ii) T is onto if and only if $\text{Col } A = \mathbb{R}^k$.

The proof of the above theorem is addressed in Exercise 6.10b.

6.3 Composition of linear transformations

Let two functions $F : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $G : \mathbb{R}^k \rightarrow \mathbb{R}^m$ be given. This situation gives us a natural way to define the composition of F and G , denoted by $G \circ F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, by

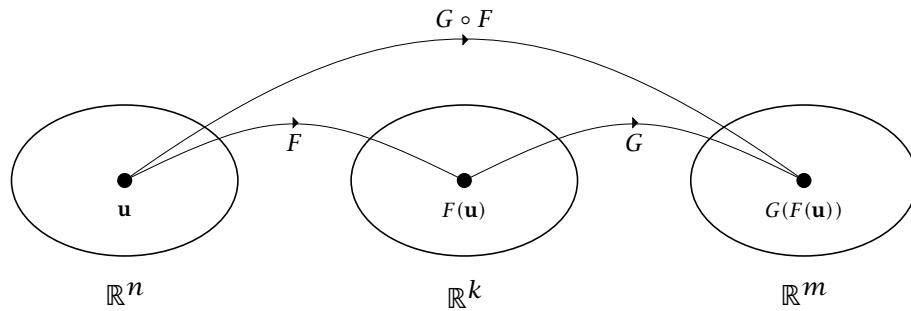
$$(G \circ F)(\mathbf{u}) = G(F(\mathbf{u}))$$

In other words, $G \circ F$ is the mapping $\mathbf{u} \mapsto G(F(\mathbf{u}))$ (see the picture below). We pronounce $G \circ F$ as “ G after F ”.

Note that if A is the representation matrix of F and B is the representation matrix of G , where A is a $k \times n$ matrix while B is an $m \times k$ matrix, then:

$$(G \circ F)(\mathbf{u}) = G(F(\mathbf{u})) = B(A\mathbf{u}) = (BA)\mathbf{u}.$$

Hence, BA is the representation matrix of the linear transformation $G \circ F$.



Example 6.12 Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $G : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear transformations given by:

$$F(\mathbf{u}) = \begin{pmatrix} u_1 + u_2 \\ u_1 - u_2 \end{pmatrix}$$

$$G(\mathbf{v}) = \begin{pmatrix} v_1 \\ v_1 + 2v_2 \\ 2v_2 \end{pmatrix}$$

The representation matrices of F and G are given by

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 2 \\ 0 & 2 \end{pmatrix},$$

respectively. The composition $G \circ F$ then has a representation matrix

$$BA = \begin{pmatrix} 1 & 0 \\ 1 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & -1 \\ 2 & -2 \end{pmatrix}$$

We have:

$$(G \circ F)(\mathbf{u}) = \begin{pmatrix} 1 & 1 \\ 3 & -1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} u_1 + u_2 \\ 3u_1 - u_2 \\ 2u_1 - 2u_2 \end{pmatrix} \quad \square$$

The function $\text{Id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $\text{Id}(\mathbf{u}) = \mathbf{u}$ is a linear transformation called the identity transformation. The associated representation matrix is the $n \times n$ identity matrix I_n .

Definition 6.13 A linear transformation $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called invertible if for every $\mathbf{w}$ there exists precisely one $\mathbf{u} \in \mathbb{R}^n$ such that $F(\mathbf{u}) = \mathbf{w}$.

If we write $\mathbf{u} = G(\mathbf{w})$ then we call G the inverse transformation of F and we write $G = F^{-1}$. It can be shown that F^{-1} is again linear. Moreover, by the construction of G we have:

$$(G \circ F)(\mathbf{u}) = G(F(\mathbf{u})) = \mathbf{u} \text{ and } (F \circ G)(\mathbf{w}) = F(G(\mathbf{w})) = \mathbf{w}$$

With the definition of the identity transformation we have:

$$G \circ F = \text{Id} \quad \text{and} \quad F \circ G = \text{Id}$$

We note, without proof, that we only need to check one of the above identities. The other identity then follows automatically.

Lemma 6.14 *Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be given with representation matrix A . The linear transformation F is invertible if and only if the matrix A is invertible.*

In that case, the transformation F^{-1} has representation matrix A^{-1} .

The proof of the above lemma is addressed in Exercise 6.11b. The above lemma enables us to check the existence of the inverse transformation since we already know how to check the existence of the inverse of a matrix (see Section 2.3). Moreover, we can use the inverse of a matrix to determine the inverse linear transformation.

Recall Theorem 3.24 which gives two equivalent conditions for a matrix A to be invertible. These two conditions can be equivalently expressed in terms of the kernel and image of a linear transformation.

Theorem 6.15 *Let a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be given. The following conditions are equivalent:*

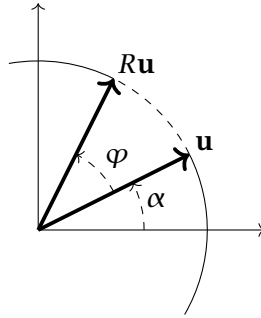
- T is invertible,
- T is onto,
- T is one-to-one.

The above theorem directly follows from Theorem 3.24 and Theorem 6.11. The proof of the above theorem is addressed in Exercise 6.11c. It is important to realize that a linear transformation can only be invertible if the image of a vector has the same size as the original vector. In other words, a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ can only be invertible if $k = n$.

6.4 Geometric examples of linear transformations

Consider linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$. We can visualize these transformations since each vector in $\mathbb{R}^2$ can be viewed as a coordinate in the plane. Let us consider some important examples, such as rotation, projection, reflection and scaling.

Example 6.16 Consider the linear transformation R which rotates a vector in $\mathbb{R}^2$ over an angle φ as pictured below:



This linear transformation is clearly invertible since rotating $R\mathbf{u}$ over an angle $-\varphi$ will get us back to $\mathbf{u}$. By Theorem 6.15 this implies this linear transformation is both onto and one-to-one.

If we describe the vector $\mathbf{u}$ in polar coordinates, i.e.

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} r \cos \alpha \\ r \sin \alpha \end{pmatrix},$$

where r denotes the distance from $\mathbf{u}$ to the origin, then

$$R\mathbf{u} = \begin{pmatrix} r \cos(\alpha + \varphi) \\ r \sin(\alpha + \varphi) \end{pmatrix}.$$

Using classical trigonometric equalities we get:

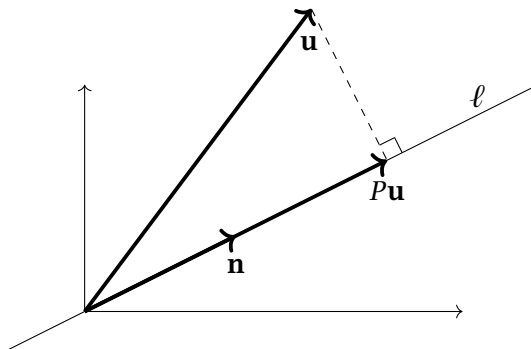
$$R\mathbf{u} = \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} r \cos \alpha \\ r \sin \alpha \end{pmatrix}$$

From the above description of R it is easily verified that this is a linear transformation. Moreover, the representation matrix of this linear transformation is

$$A = \begin{pmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{pmatrix}.$$

□

Example 6.17 Consider the linear transformation P which projects a vector in $\mathbb{R}^2$ (orthogonally) onto a line ℓ , as pictured below:



Note that this linear transformation is not onto since $P\mathbf{u}$ will always be on the line ℓ and cannot be equal to a point outside of this line. This linear transformation is not one-to-one since any point $\mathbf{v}$ on the line connecting $\mathbf{u}$ and $P\mathbf{u}$ will also be mapped onto $P\mathbf{u}$.

If we take $\mathbf{n}$ a unit vector (i.e., $\mathbf{n}$ has length 1) on the line ℓ , we have (Cf. Calculus 1B; Thomas Chapter 12)

$$P\mathbf{u} = (\mathbf{u} \cdot \mathbf{n})\mathbf{n}$$

where $\mathbf{u} \cdot \mathbf{n}$ denotes the dot product of $\mathbf{u}$ and $\mathbf{n}$. If

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \end{pmatrix}$$

then we get:

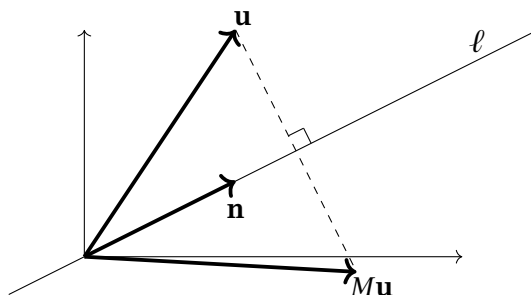
$$\begin{aligned} P\mathbf{u} = P \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} &= (u_1 n_1 + u_2 n_2) \begin{pmatrix} n_1 \\ n_2 \end{pmatrix} = \begin{pmatrix} u_1 n_1^2 + u_2 n_1 n_2 \\ u_1 n_1 n_2 + u_2 n_2^2 \end{pmatrix} \\ &= \begin{pmatrix} n_1^2 & n_1 n_2 \\ n_2 n_1 & n_2^2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \end{aligned}$$

From the above description of P it is easily verified that this is a linear transformation. Moreover, the representation matrix of this linear transformation is

$$A = \begin{pmatrix} n_1^2 & n_1 n_2 \\ n_2 n_1 & n_2^2 \end{pmatrix}.$$

□

Example 6.18 Consider the linear transformation M which mirrors (or reflects) a vector in $\mathbb{R}^2$ through a line as pictured below:



Note that this linear transformation is onto since any point $\mathbf{u}$ can be reached because $M(M\mathbf{u}) = \mathbf{u}$. This linear transformation is also one-to-one since $M\mathbf{u} = M\mathbf{v}$ implies

$$\mathbf{u} = M(M\mathbf{u}) = M(M\mathbf{v}) = \mathbf{v}.$$

Mirroring/reflection with respect to a line is closely related to projection. From the above picture, it can be easily seen that:

$$2P\mathbf{u} = M\mathbf{u} + \mathbf{u}$$

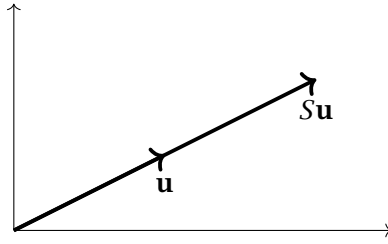
where P is the projection onto the line as defined in Example 6.17. This implies that:

$$M = 2P - \text{Id}$$

From the above description of M it is easily verified that this is a linear transformation. Moreover, the representation matrix of this linear transformation is

$$A = \begin{pmatrix} 2n_1^2 - 1 & 2n_1n_2 \\ 2n_2n_1 & 2n_2^2 - 1 \end{pmatrix}. \quad \square$$

Example 6.19 Consider the linear transformation S which scales a vector in $\mathbb{R}^2$ by a factor α as pictured below:



Scaling implies that $S = \alpha \text{Id}$. Note that this linear transformation is onto since any point $\mathbf{u}$ can be reached because $S(\alpha^{-1}\mathbf{u}) = \mathbf{u}$. This linear transformation is also one-to-one since $S\mathbf{u} = S\mathbf{v}$ implies $\alpha\mathbf{u} = \alpha\mathbf{v}$ which clearly yields $\mathbf{u} = \mathbf{v}$.

The representation matrix of this linear transformation is

$$A = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha \end{pmatrix}. \quad \square$$

Note that the above examples do not describe all possible linear transformations. Many other examples can be generated by the composition of the above examples. Consider for instance the linear transformation

$$Q\mathbf{u} = A\mathbf{u}$$

with

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}.$$

In other words, we have a scaling but we scale the entries of the vector by different factors. This can be actually written as:

$$Q = P + S$$

where P is a projection as in Example 6.17 onto the line $x_2 = 0$ and S is a scaling by a factor 2 as in Example 6.19.

Finally, note that in the above geometric examples we used explicit expressions for rotations as well as projection and mirroring with respect to a line. However, it is often easier just to evaluate the effect of the linear transformations on the standard basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$. This is illustrated in the following example.

Example 6.20 $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the linear transformation that reflects each point $(x_1, x_2) \in \mathbb{R}^2$ through the line $x_2 = -x_1$ and then rotates it about the origin through an angle of $\frac{\pi}{4}$ radians (counterclockwise).

Let us first consider $T(\mathbf{e}_1)$. The reflection maps the point $(1, 0)$ to the point $(0, -1)$. If we rotate this point counterclockwise over $\frac{\pi}{4}$ radians we get to the point $(\frac{1}{2}\sqrt{2}, -\frac{1}{2}\sqrt{2})$.

Let us next consider $T(\mathbf{e}_2)$. The reflection maps the point $(0, 1)$ to the point $(-1, 0)$. If we rotate this point counterclockwise over $\frac{\pi}{4}$ radians we get to the point $(-\frac{1}{2}\sqrt{2}, -\frac{1}{2}\sqrt{2})$.

We find

$$T(\mathbf{e}_1) = \begin{pmatrix} \frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} \end{pmatrix} \quad \text{and} \quad T(\mathbf{e}_2) = \begin{pmatrix} -\frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} \end{pmatrix}$$

These are the columns of the representation matrix A of our linear transformation T . In other words,

$$A = \begin{pmatrix} \frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \\ -\frac{1}{2}\sqrt{2} & -\frac{1}{2}\sqrt{2} \end{pmatrix}. \quad \square$$

6.5 Exercises

6.5.1 Self study exercises

6.1 The linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and the vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ are given by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3x_1 - 2x_2 \\ -x_1 + 4x_2 \\ x_1 - 3x_2 \end{pmatrix}; \quad \mathbf{v}_1 = \begin{pmatrix} 9 \\ 7 \\ -4 \end{pmatrix}; \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix}$$

(a) Determine the T -image of

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

(b) Determine the representation matrix A of T .

(c) Determine all $\mathbf{x} \in \mathbb{R}^2$ with $T(\mathbf{x}) = \mathbf{v}_1$.

(d) Determine if $\mathbf{v}_2 \in \text{im } T$.

(e) Is T one-to-one? Is T onto? Is T invertible?

6.2 Let $T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle of $\frac{3\pi}{4}$ radians (counterclockwise).

Let $T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that projects each point in $\mathbb{R}^2$ orthogonally onto the line $x_2 = -x_1$.

Let $T_3 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects each point in $\mathbb{R}^2$ through the line $x_2 = 0$.

Let A_1 , A_2 and A_3 be the representation matrices of T_1 , T_2 and T_3 respectively.

(a) Determine A_1 , A_2 and A_3 by examining the images of $\mathbf{e}_1$ and $\mathbf{e}_2$.

(b) Determine A_1 , A_2 and A_3 by using the matrices given in examples 6.16, 6.17 and 6.18 on pages 110-113.

(c) Determine $\ker T_2$,

(1) Using the definition of T_2 .

(2) Using the representation matrix A_2 .

(d) Determine if T_2 is one-to-one and if T_2 is onto,

(1) Using the definition of T_2 .

(2) Using the representation matrix A_2 .

6.3 (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = ax + b$.

Prove that f is a linear transformation if and only if $b = 0$.

(b) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the function given by

$$T(\mathbf{x}) = \begin{pmatrix} x_1 - x_2 + x_3 \\ 2x_1 - 4x_3 \\ x_1 + x_2 + 1 \end{pmatrix}$$

Show that T is not a linear transformation.

(c) Suppose that $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a linear transformation with a 4×5 representation matrix. Determine m and n .

6.5.2 Tutorial exercises

6.4 Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation with representation matrix A . This matrix A and the vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^3$ are given by

$$A = \begin{pmatrix} -2 & 4 & 4 & 2 \\ 3 & -2 & 6 & 5 \\ 1 & -3 & -5 & -3 \end{pmatrix}; \quad \mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}; \quad \mathbf{v}_2 = \begin{pmatrix} 6 \\ 7 \\ -7 \end{pmatrix}$$

- (a) Determine all $\mathbf{x} \in \mathbb{R}^4$ with $T(\mathbf{x}) = \mathbf{v}_1$.
- (b) Determine all $\mathbf{x} \in \mathbb{R}^4$ with $T(\mathbf{x}) = \mathbf{v}_2$.
- (c) Determine a basis for $\ker T$.
- (d) Determine a basis for $\operatorname{im} T$.
- (e) Is T one-to-one? Is T onto?

6.5 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation with representation matrix

$$A = \begin{pmatrix} 1 & 2 & -1 \\ -1 & -1 & 3 \\ 0 & 1 & 1 \end{pmatrix}$$

Show that T is invertible and determine

$$T^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

6.6 Let T_1 , T_2 , and T_3 be the linear transformations of Exercise 6.2. Let $F = T_2 \circ T_1$ and $G = T_1 \circ T_3$ and let B_1 and B_2 be the representation matrices of F and G respectively.

- (a) Determine B_1 and B_2 .
- (b) Determine the F -image of the vector

$$\begin{pmatrix} -4 \\ 1 \end{pmatrix}.$$

- (c) Determine all vectors whose F -image is

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix}.$$

(d) Determine if G is invertible and if so, determine

$$G^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

6.7 Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation that reflects each point in $\mathbb{R}^2$ through the line $x_2 = 3x_1$.

(a) Determine the representation matrix A of T .

(b) Determine the T -image of

$$\begin{pmatrix} -5 \\ 2 \end{pmatrix}.$$

6.8 $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear transformation that maps

$$\begin{pmatrix} 3 \\ -2 \end{pmatrix} \quad \text{onto} \quad \begin{pmatrix} -7 \\ 1 \end{pmatrix}$$

and maps

$$\begin{pmatrix} -5 \\ 4 \end{pmatrix} \quad \text{onto} \quad \begin{pmatrix} 13 \\ 3 \end{pmatrix}.$$

Use the linearity of T to determine

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

as well as the representation matrix of T .

6.9 Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a linear transformation.

(a) Prove that $T(\mathbf{0}) = \mathbf{0}$.

(b) Let $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^m$ and $\alpha_1, \dots, \alpha_p \in \mathbb{R}$. Prove that

$$T(\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p) = \alpha_1 T(\mathbf{v}_1) + \dots + \alpha_p T(\mathbf{v}_p).$$

The image of a linear combination of vectors is the linear combination of the images of these vectors, with the same weights.

(c) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a linearly dependent set in $\mathbb{R}^m$.

Prove that $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_p)\}$ is a linearly dependent set in $\mathbb{R}^n$.

(d) Let $\mathbf{p}, \mathbf{v} \in \mathbb{R}^m$ and let $\ell = \{\mathbf{p} + \alpha \mathbf{v} \mid \alpha \in \mathbb{R}\}$ be the line in $\mathbb{R}^m$ through $\mathbf{p}$ parallel to $\mathbf{v}$. Show that T maps ℓ onto a line in $\mathbb{R}^n$ or onto a single point.

- 6.10 (a) Prove Theorem 6.10.
(b) Prove Theorem 6.11.
(c) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be a linear transformation with representation matrix A . Prove that T is one-to-one if and only if the columns of A are linearly independent.
- 6.11 (a) Prove Theorem 6.8.
(b) Prove Lemma 6.14.
(c) Prove Theorem 6.15.
- 6.12 Given is a matrix $A \in \mathbb{R}^{2 \times 2}$. Define the mapping $T : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2}$ by:

$$T(X) = AX$$

that is, where $T(X)$ is a function of matrix X (transformation of X) which can be written as matrix A multiplied by matrix X .

- (a) Show that T is a linear transformation.
- (b) We choose

$$A = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$$

It can be shown that in this case the linear transformation T has eigenvalue 3. Determine an associated eigenvector.

Chapter 7

Answers to tutorial exercises

Chapter 1

$$1.5 \quad \begin{pmatrix} 1 & 0 & -14 & -7 \\ 0 & 1 & 9 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}; \quad \begin{pmatrix} 1 & 0 & 2 & 6 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$1.6 \quad \begin{cases} x_1 = \frac{1}{2} \\ x_2 = \frac{3}{2} \\ x_3 = \frac{9}{2} \end{cases}$$

1.7 No. Theorem 1.22: the last column of the (reduced) echelon form contains a pivot element

1.8 No, these planes do not have a common intersection point.

1.9 Yes: $f(x) = 2x^2 - 7x + 8$.

1.10 (a) No. If the system is consistent, then the reduced echelon form of the corresponding augmented matrix will always have a column (apart from the last column!) which does not have a pivot. The corresponding variable will then be a free variable in the parametric description of the solution set, meaning that the system has infinitely many solutions.

(b) Yes. For example (two equations, one variable):

$$\begin{cases} x_1 = 1 \\ 2x_1 = 2 \end{cases}$$

1.11 We have $f_1(x_1, x_2, x_3) = (x_1 - 1)^2 + x_2^2 + x_3^3$.

Clearly, f_1 is nonlinear, since $f_1(0, 0, 0) = 1$. And if f_1 would be linear then taking $(x_1, x_2, x_3) = (y_1, y_2, y_3) = (0, 0, 0)$ and $\alpha = 1$ in Definition 1.2, we must have $f_1(0, 0, 0) = f_1(0, 0, 0) + f_1(0, 0, 0)$, which is clearly not the case (a linear function always maps $\mathbf{0}$ onto $\mathbf{0}$).

The same argument applies to f_2 and f_3 .

1.12 We must show that $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i$. Well:

$$\begin{aligned}
 & a_{i1}x_1 + \cdots + a_{in}x_n \\
 &= a_{i1}(u_1 + \lambda(v_1 - u_1)) + \cdots + a_{in}(u_n + \lambda(v_n - u_n)) \\
 &= a_{i1}((1 - \lambda)u_1 + \lambda v_1) + \cdots + a_{in}((1 - \lambda)u_n + \lambda v_n) \\
 &= (1 - \lambda)(a_{i1}u_1 + \cdots + a_{in}u_n) + \lambda(a_{i1}v_1 + \cdots + a_{in}v_n) \\
 &= (1 - \lambda)b_i + \lambda b_i = b_i.
 \end{aligned}$$

- 1.13 (a) No solution: impossible;
 Exactly one solution: $\alpha \in \mathbb{R} - \{\frac{1}{2}\}$;
 Infinitely many solutions: $\alpha = \frac{1}{2}$.

$$\begin{cases} x_1 = -2 \\ x_2 = 0 \end{cases}$$

- (b) No solution: $\alpha = \frac{1}{2}$;
 Exactly one solution: $\alpha \in \mathbb{R} - \{\frac{1}{2}\}$;
 Infinitely many solutions: impossible.

$$\begin{cases} x_1 = \frac{7\alpha-4}{2-4\alpha} \\ x_2 = \frac{1}{2-4\alpha} \end{cases}$$

- (c) No solution: impossible;
 Exactly one solution: $\alpha \in \mathbb{R}$;
 Infinitely many solutions: impossible.

$$\begin{cases} x_1 = 4 - 7\alpha \\ x_2 = 2 - 4\alpha \end{cases}$$

- (d) No solution: $\beta = 3\alpha$; except the combinations
 $\alpha = 2, \beta = 6$ and $\alpha = -2, \beta = -6$;
 Exactly one solution: all $\alpha, \beta \in \mathbb{R}$ with $\beta \neq 3\alpha$;
 Infinitely many solutions: $\alpha = 2$ and $\beta = 6$ or
 $\alpha = -2$ and $\beta = -6$.

$$\begin{cases} x_1 = -\beta - 4\frac{\alpha\beta-12}{\beta-3\alpha} \\ x_2 = \frac{\alpha\beta-12}{\beta-3\alpha} \end{cases}$$

- 1.14 (a) The (reduced) echelon form of A does not have a pivot element in the last row. So there exists $\mathbf{b} \in \mathbb{R}^3$, for which $A\mathbf{x} = \mathbf{b}$ is not consistent.

- (b) The system $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^3$ satisfying $b_1 + b_2 - b_3 = 0$.

- 1.15 (a) The system is consistent if $\alpha \neq 14$ and if $\alpha = 14$ and $\beta = 2$.

(b)
$$\begin{cases} x_1 = 1 - 11x_3 \\ x_2 = 2x_3 \\ x_3 \text{ is free} \end{cases}$$

- (c) For $\alpha \neq 14$.

1.16 (a)
$$\begin{pmatrix} 1 & -2 & -3 & 1 & 3 \\ 0 & 0 & 1 & \alpha & -2 \\ 0 & 0 & 0 & \alpha+1 & \beta-4 \end{pmatrix}$$

- (b) The system is consistent if $\alpha \neq -1$ and if $\alpha = -1$ and $\beta = 4$.

(c)
$$\begin{cases} x_1 = -3 + 2x_2 + 2x_4 \\ x_2 \text{ is free} \\ x_3 = -2 + x_4 \\ x_4 \text{ is free} \end{cases}$$

- 1.17 The intersection is the point $(\frac{3}{5}, \frac{2}{5})$, i.e.

$$\begin{cases} x_1 = \frac{3}{5} \\ x_2 = \frac{2}{5} \end{cases}$$

- 1.18 The intersection is described by:

$$\begin{cases} x_1 = \frac{5}{3} - \frac{5}{3}x_3 \\ x_2 = \frac{1}{3} - \frac{1}{3}x_3 \\ x_3 \text{ is free} \end{cases}$$

which describes a line in $\mathbb{R}^3$.

1.19 The intersection is the point $(2, 0, -1)$, i.e.

$$\begin{cases} x_1 = 2 \\ x_2 = 0 \\ x_3 = -1 \end{cases}$$

Chapter 2

2.6 AB is not defined;

$$BA = \begin{pmatrix} 11 & 3 \end{pmatrix}; \quad AC = \begin{pmatrix} 12 & -6 & 8 \\ 16 & -4 & 8 \\ 8 & -11 & 10 \end{pmatrix}; \quad CA = \begin{pmatrix} 6 & 8 \\ -10 & 12 \end{pmatrix};$$

BC is not defined; CB is not defined.

$$2.7 \quad \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -7 \\ 8 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 5 \\ 12 \end{pmatrix}.$$

2.8 (a) A is a 5×4 -matrix.

(b) All elements in the last column of AB are zero, since $A\mathbf{0} = \mathbf{0}$ (cf. Definition 2.6)

(c) Solving the systems

$$A\mathbf{x} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \text{ and } A\mathbf{x} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

yields

$$B = \begin{pmatrix} -1 & 3 \\ 1 & -1 \end{pmatrix}.$$

$$2.9 \quad (a) \quad T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$(b) \quad T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

$$(c) \quad T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

$$(d) \quad T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (e) Let T be an elementary matrix. T is the matrix that results by performing the elementary row operation corresponding to T to the identity matrix I . This elementary row operation can be reversed, which is also an elementary row operation. Let S be the elementary matrix corresponding to this reverse. Then applying the reverse operation to T results in I , in other words: $ST = I$. So, by Lemma 2.13, T is invertible and $T^{-1} = S$.

2.10 Straightforward.

- 2.11 (a) $\mathbf{v}$ is a solution of $A\mathbf{x} = \mathbf{b}$ if $A\mathbf{v} = \mathbf{b}$. So we must show that $A(A^{-1}\mathbf{b}) = \mathbf{b}$.

This is straightforward using Lemma 2.8: $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I\mathbf{b} = \mathbf{b}$.

- (b) Suppose $A\mathbf{v} = \mathbf{b}$, then $A^{-1}(A\mathbf{v}) = A^{-1}\mathbf{b}$. So, by Lemma 2.8: $(A^{-1}A)\mathbf{v} = A^{-1}\mathbf{b}$, which implies $I\mathbf{v} = A^{-1}\mathbf{b}$. Hence $\mathbf{v} = A^{-1}\mathbf{b}$.

- 2.12 (a) Show that $AB = I$, and use Lemma 2.13.

$$(b) \quad B\mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}; \quad B\mathbf{b}_2 = \begin{pmatrix} -2 \\ -2 \\ 3 \end{pmatrix}; \quad B\mathbf{b}_3 = \begin{pmatrix} 23 \\ 7 \\ -10 \end{pmatrix}.$$

- 2.13 (a) Show that if $ad - bc \neq 0$ then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and use Lemma 2.13.

$$(b) \quad A^{-1} = \begin{pmatrix} -3/2 & -4 \\ -1/2 & -1 \end{pmatrix}.$$

$$2.14 \quad A \text{ is not invertible; } B^{-1} = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 1 & -3 \\ -1 & 0 & 2 \end{pmatrix}.$$

- 2.15 (a) Suppose $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^n$. Then the reduced echelon form of A must be I , otherwise the last row of the reduced echelon form is a zero-row and a vector $\mathbf{b}$ can be found for which $A\mathbf{x} = \mathbf{b}$ is inconsistent (add column $\mathbf{e}_n$ to the echelon form and reverse the elementary operations that transform A to its reduced echelon form: the last column of the final result can be taken for $\mathbf{b}$). Therefore, by Theorem 2.15, A must be invertible.
- (b) In this case the last row of the (reduced) echelon form must be a zero-row. Therefore (since A is square) the echelon form must have a non-pivot column, and therefore the solution set of $A\mathbf{x} = \mathbf{0}$ has a free variable. So the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
- (c) If A is singular, we have by Theorem 2.15 that its (reduced) echelon form contains a zero row.
Therefore there exists $\mathbf{b} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$ is inconsistent (cf. part (a)).
Moreover, if A is singular and $A\mathbf{x} = \mathbf{b}$ is consistent, the echelon form of A must have a non-pivot column (since A is square), and therefore the solution set of $A\mathbf{x} = \mathbf{b}$ has a free variable. So in that case the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.
- 2.16 (a) Use Lemma 2.13.
- (b) Use Lemma 2.13 and Lemma 2.20: $A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I$.
- (c) Use Lemma 2.13: show that $ABB^{-1}A^{-1} = I$ (cf. Lemma 2.8).
- (d) $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$.
- 2.17 (a) No. If $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^n$, then the (reduced) echelon form of A cannot have a zero-row (cf. Theorem 1.22). Therefore (since A is square), the (reduced) echelon form of A has a pivot in each column. Hence the system $A\mathbf{x} = \mathbf{c}$ does not have a free variable and the solution is unique.
- (b) If $A\mathbf{x} = \mathbf{0}$ has a unique solution, this system has no free variables. So the (reduced) echelon form of A cannot have a non-pivot column and therefore no zero-row (since A is square). Hence the reduced echelon form of A has a pivot in each row and the system $A\mathbf{x} = \mathbf{b}$ is consistent for each $\mathbf{b} \in \mathbb{R}^n$. Moreover (because the reduced echelon form of A does not have a non-pivot column) the system $A\mathbf{x} = \mathbf{b}$ does not have a free variable and therefore the solution is unique.
- 2.18 (a) AA^T is always defined because the number of columns of A equals the number of rows of A^T (n).

Also, $A^T A$ is always defined because the number of columns of A^T equals the number of rows of A (m).

AA is only defined if the number of columns of A equals the number of rows; in other words, if A is square.

- (b) If the product AB is defined, the number of columns of A equals the number of rows of B . But this does not guarantee that also the number of rows of A equals the number of columns of B , which is necessary for $A^T B^T$ to be defined (and even if this would be the case the equation does usually not hold).

(c) $(ABC)^T = ((AB)C)^T = C^T(AB)^T = C^T(B^T A^T) = C^T B^T A^T$.

- (d) Mathematical induction. The induction basis for $k = 0$ and $k = 1$ is trivial and the statement for $k = 2$ follows from Lemma 2.20: $(A^2)^T = (AA)^T = A^T A^T = (A^T)^2$. Let $r \geq 2$ and suppose the statement holds for r :

$$(A^r)^T = (A^T)^r \text{ (Induction Hypothesis).}$$

$$\text{We will show that: } (A^{r+1})^T = (A^T)^{r+1}.$$

Well: $(A^{r+1})^T = (AA^r)^T = (A^r)^T A^T = (A^T)^r A^T = (A^T)^{r+1}$, where the second equality follows from Lemma 2.20 and the third equality from the induction hypothesis.

Now the statement follows from the principle of mathematical induction.

- 2.19 (a) If $AB = AC$, then $A^{-1}AB = A^{-1}AC$. So $IB = IC$, thus $B = C$.
- (b) If $A = PBP^{-1}$, then $P^{-1}A = P^{-1}PBP^{-1} = IBP^{-1} = BP^{-1}$.
So $P^{-1}AP = BP^{-1}P = BI = B$. Hence $B = P^{-1}AP$.
- (c) Expand as $(AAX)^{-1} = (AB^{-1})^{-1}(ABA^{-1}ABA^{-1})$ and apply Exercise 2.16acd. Result: $X = A^{-1}(B^3)^{-1}$.
- (d) AB is invertible, so by Definition 2.11, there exist $C \in \mathbb{R}^{n \times n}$ with $(AB)C = I$ and $C(AB) = I$. Then, by Lemma 2.8, $A(BC) = I$ and $(CA)B = I$. And so, by Lemma 2.13, A and B are invertible (and $A^{-1} = BC$ and $B^{-1} = CA$).

2.20 Straightforward. Note however, that the zero-vector of V is in this case the zero-matrix!

Chapter 3

- 3.6 (a) $\mathbf{v} \in \text{Null } A$, because $A\mathbf{v} = \mathbf{0}$. $\mathbf{w} \notin \text{Null } A$, because $A\mathbf{w} \neq \mathbf{0}$.

(b)

$$\text{Null } A = \text{Span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ -4 \\ 1 \end{pmatrix} \right\}.$$

(c) Verify that $A\mathbf{p} = \mathbf{b}$. Solution set (cf. Theorem 3.10):

$$\begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ -4 \\ 1 \end{pmatrix} \right\}.$$

(d) Yes, $\mathbf{b} \in \text{Col } A$, since $A\mathbf{x} = \mathbf{b}$ is consistent (cf. part (c)).3.7 (a) $\beta = 0$: no. $\beta = 2$: yes.(b) $\mathbf{w} \in W$ if $\alpha \neq 14$ and if $\alpha = 14$ and $\beta = 2$.3.8 1. Obviously $\mathbf{0} \in p$: take $\alpha = \beta = 0$.2. If $\mathbf{u}, \mathbf{v} \in p$, then there exist $\alpha_1, \beta_1, \alpha_2, \beta_2 \in \mathbb{R}$ such that $\mathbf{u} = \alpha_1 \mathbf{q} + \beta_1 \mathbf{r}$ and $\mathbf{v} = \alpha_2 \mathbf{q} + \beta_2 \mathbf{r}$.Then $\mathbf{u} + \mathbf{v} = \alpha_1 \mathbf{q} + \beta_1 \mathbf{r} + \alpha_2 \mathbf{q} + \beta_2 \mathbf{r} = (\alpha_1 + \alpha_2) \mathbf{q} + (\beta_1 + \beta_2) \mathbf{r}$.Hence $\mathbf{u} + \mathbf{v} \in p$: take $\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$.3 If $\mathbf{v} \in p$ and $\lambda \in \mathbb{R}$, then there exists $\gamma, \delta \in \mathbb{R}$ with $\mathbf{v} = \gamma \mathbf{q} + \delta \mathbf{r}$.Then $\lambda \mathbf{v} = \lambda(\gamma \mathbf{q} + \delta \mathbf{r}) = \lambda\gamma \mathbf{q} + \lambda\delta \mathbf{r}$. Hence $\lambda \mathbf{v} \in p$: take $\alpha = \lambda\gamma$ and $\beta = \lambda\delta$.From 1, 2 and 3 it follows that p is a linear subspace of $\mathbb{R}^n$.3.9 Suppose $\mathbf{b} \in \text{Col } A$. Then by Definition 3.11, there exist $\mathbf{x}$ with $A\mathbf{x} = \mathbf{b}$.
Now by Definition 2.4,

$$\mathbf{b} = A\mathbf{x} = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix}.$$

So $\mathbf{b}$ is a linear combination of the columns of A .3.10 (a) Let $\mathbf{y} \in \mathbf{p} + \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_q\}$. Then $\mathbf{y} = \mathbf{p} + \alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q$ for certain $\alpha_1, \dots, \alpha_q \in \mathbb{R}$. Now applying Lemma 2.5, we obtain

$$A\mathbf{y} = A(\mathbf{p} + \alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q) = A\mathbf{p} + A(\alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q).$$

We have that $A\mathbf{p} = \mathbf{b}$, since $\mathbf{p}$ is a solution of $A\mathbf{x} = \mathbf{b}$.

Furthermore, $A(\alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q) = \mathbf{0}$, because $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_q\}$ is the solution set of $A\mathbf{x} = \mathbf{0}$. Therefore $A\mathbf{y} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, and so $\mathbf{y}$ is a solution of $A\mathbf{x} = \mathbf{b}$.

- (b) Suppose that if $\mathbf{y}$ is a solution of $A\mathbf{x} = \mathbf{b}$. Then $A\mathbf{y} = \mathbf{b}$. Since $\mathbf{p}$ is also a solution of $A\mathbf{x} = \mathbf{b}$ we have $A\mathbf{p} = \mathbf{b}$ as well. Now consider the vector $\mathbf{y} - \mathbf{p}$ and apply Lemma 2.5:

$$A(\mathbf{y} - \mathbf{p}) = A\mathbf{y} - A\mathbf{p} = \mathbf{b} - \mathbf{b} = \mathbf{0}.$$

So the vector $\mathbf{y} - \mathbf{p}$ is a solution of $A\mathbf{x} = \mathbf{0}$! Hence $\mathbf{y} - \mathbf{p} \in \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_q\}$, meaning that there exist $\alpha_1, \dots, \alpha_q \in \mathbb{R}$ with $\mathbf{y} - \mathbf{p} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q$. So $\mathbf{y} = \mathbf{p} + \alpha_1 \mathbf{v}_1 + \dots + \alpha_q \mathbf{v}_q$, and therefore $\mathbf{y} \in \mathbf{p} + \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_q\}$.

- 3.11 (a) Show, by calculating the echelon form of the augmented matrix, that the system $A\mathbf{x} = \mathbf{b}$ is consistent and use Definition 3.11.

(b) E.g. $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ -5 \end{pmatrix} \right\}$ and $[\mathbf{b}]_{\mathcal{B}} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}$.

- 3.12 (a) We must determine all values of $\alpha \in \mathbb{R}$, for which the columns of A are linearly independent, because then the columns of A form a basis for $\text{Col } A$, and therefore a basis for $\mathbb{R}^3$, by Theorem 3.17. By constructing an echelon form of A , we obtain that the columns of A are linearly independent for all $\alpha \in \mathbb{R}$ except for $\alpha = -2$ and $\alpha = 1$.

(b) E.g. $\mathcal{B} = \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$. The dimension of $\text{Null } A$ is 2.

- (c) Show that $A\mathbf{p} = \mathbf{0}$. $[\mathbf{p}]_{\mathcal{B}} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$, where $\mathcal{B}$ denotes the basis of part (b).

3.13 (a) From $3\mathbf{a}_1 - \mathbf{a}_2 - 4\mathbf{a}_4 = \mathbf{0}$ we derive that $A\mathbf{x} = \mathbf{0}$ for $\mathbf{x} = \begin{pmatrix} 3 \\ -1 \\ 0 \\ -4 \end{pmatrix}$.

- (b) This follows directly from Definition 3.13: the second column is a linear combination of the other columns.
- (c) If $\{\mathbf{a}_1, \mathbf{a}_2\}$ would be linear dependent, then, by Theorem 3.14, there exists $\alpha_1, \alpha_2 \in \mathbb{R}$, not both equal to zero, such that $\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 = \mathbf{0}$.

But then also $\alpha_1 \mathbf{a}_1 + \alpha_2 \mathbf{a}_2 + 0\mathbf{a}_3 = \mathbf{0}$, contradicting that $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ is linearly independent.

- 3.14 (a) n pivots, because then $A\mathbf{x} = \mathbf{0}$ has only the trivial solution (cf. Theorem 3.14).
- (b) m pivots because then $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$, and so $\text{Col } A = \mathbb{R}^m$.
- (c) The independence of the columns of A guarantees that $A\mathbf{x} = \mathbf{0}$ has only the trivial solution (cf. Theorem 3.14) so the system has no free variables. Hence the number of solutions of the system $A\mathbf{x} = \mathbf{b}$ is either 0 or 1.
- (d) Denote the nonzero columns of A by $\mathbf{v}_1, \dots, \mathbf{v}_p$, where $p \leq n$ and suppose $\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$ for certain $\alpha_1, \dots, \alpha_p \in \mathbb{R}$. Then $\alpha_1 = 0$, since the fact that A^T is on echelon form guarantees that there is an entry in $\mathbf{v}_1$ which is nonzero whereas this entry is zero in all other columns $\mathbf{v}_2, \dots, \mathbf{v}_p$. Furthermore $\alpha_2 = 0$, since there is an entry in $\mathbf{v}_2$ which is nonzero whereas this entry is zero in all columns $\mathbf{v}_3, \dots, \mathbf{v}_p$. Continuing this argument, we obtain that $\alpha_i = 0$ for all $i \in \{1, \dots, p\}$. Hence $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly independent by Theorem 3.14.
- 3.15 (a) By Definition 2.6, we have that the last column of AB equals $A\mathbf{b}_n$. So $A\mathbf{b}_n = \mathbf{0}$, whereas $\mathbf{b}_n \neq \mathbf{0}$. Note that $A\mathbf{b}_n$ is a nontrivial linear combination of the columns of A (since $\mathbf{b}_n$ is nonzero) that results in the zero vector. Hence, the columns of A are linearly dependent by Theorem 3.14.
- (b) Suppose that the columns of B are linearly dependent. Then, by Theorem 3.14, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, not all zero, such that $\alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n = \mathbf{0}$. But then $\alpha_1 (A\mathbf{b}_1) + \dots + \alpha_n (A\mathbf{b}_n) = A(\alpha_1 \mathbf{b}_1) + \dots + A(\alpha_n \mathbf{b}_n) = A(\alpha_1 \mathbf{b}_1 + \dots + \alpha_n \mathbf{b}_n) = A\mathbf{0} = \mathbf{0}$. So $\alpha_1 (A\mathbf{b}_1) + \dots + \alpha_n (A\mathbf{b}_n) = \mathbf{0}$ and the columns of AB are linearly dependent by Theorem 3.14.
- 3.16 (a) Suppose $\mathbf{v}_i$ is a linear combination of the other vectors in S , say $\mathbf{v}_i = \alpha_1 \mathbf{v}_1 + \dots + \alpha_{i-1} \mathbf{v}_{i-1} + \alpha_{i+1} \mathbf{v}_{i+1} + \dots + \alpha_p \mathbf{v}_p$, where $\alpha_1, \dots, \alpha_p \in \mathbb{R}$.

Now consider a linear combination $\mathbf{w}$ of $\mathbf{v}_1, \dots, \mathbf{v}_p$, say

$$\mathbf{w} = \beta_1 \mathbf{v}_1 + \dots + \beta_p \mathbf{v}_p.$$

Then $\mathbf{w}$ can be expressed as a linear combination of the vectors

$\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_p$ as follows:

$$\begin{aligned} \mathbf{w} &= \beta_1 \mathbf{v}_1 + \dots + \beta_{i-1} \mathbf{v}_{i-1} \\ &\quad + \beta_i (\alpha_1 \mathbf{v}_1 + \dots + \alpha_{i-1} \mathbf{v}_{i-1} + \alpha_{i+1} \mathbf{v}_{i+1} + \dots + \alpha_p \mathbf{v}_p) \\ &\quad + \beta_{i+1} \mathbf{v}_{i+1} + \dots + \beta_p \mathbf{v}_p \\ &= (\beta_1 + \alpha_1 \beta_i) \mathbf{v}_1 + \dots + (\beta_{i-1} + \alpha_{i-1} \beta_i) \mathbf{v}_{i-1} \\ &\quad + (\beta_{i+1} + \alpha_{i+1} \beta_i) \mathbf{v}_{i+1} + \dots + (\beta_p + \alpha_p \beta_i) \mathbf{v}_p. \end{aligned}$$

- (b) Let $j \in \{1, \dots, p\}$, $j \neq i$. Then $\mathbf{v}_i = 0\mathbf{v}_j$, so $\mathbf{v}_i$ is a linear combination of the other vectors in S , and therefore, S is linearly dependent.
- (c) (i) Suppose S is linearly independent and suppose that $\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$. If $\alpha_i \neq 0$ for some $i \in \{1, \dots, p\}$, then $\mathbf{v}_i$ can be expressed as a linear combination of the other vectors in S :

$$\mathbf{v}_i = -\frac{\alpha_1}{\alpha_i} \mathbf{v}_1 - \dots - \frac{\alpha_{i-1}}{\alpha_i} \mathbf{v}_{i-1} - \frac{\alpha_{i+1}}{\alpha_i} \mathbf{v}_{i+1} - \dots - \frac{\alpha_p}{\alpha_i} \mathbf{v}_p,$$

contradicting that S is linearly independent. Therefore $\alpha_i = 0$ for all $i \in \{1, \dots, p\}$.

- (ii) Suppose $\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$ implies $\alpha_1 = \dots = \alpha_p = 0$. If S would be linearly dependent, then, for some $i \in \{1, \dots, p\}$, $\mathbf{v}_i$ is a linear combination of the other vectors in S , say

$$\mathbf{v}_i = \beta_1 \mathbf{v}_1 + \dots + \beta_{i-1} \mathbf{v}_{i-1} + \beta_{i+1} \mathbf{v}_{i+1} + \dots + \beta_p \mathbf{v}_p.$$

But then

$$\beta_1 \mathbf{v}_1 + \dots + \beta_{i-1} \mathbf{v}_{i-1} - \mathbf{v}_i + \beta_{i+1} \mathbf{v}_{i+1} + \dots + \beta_p \mathbf{v}_p = \mathbf{0},$$

while the weight of $\mathbf{v}_i$ is nonzero. This contradiction proves that S must be linearly independent.

- (d) Let A be the $n \times p$ -matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_p$. If $p > n$, then $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\mathbf{y}$, since the echelon form of A will have a non-pivot column and therefore the system has a free variable. But then the linear combination of the columns of A with the weights of the vector $\mathbf{y}$ is equal to the zero vector, and so the columns of A are linearly dependent by Theorem 3.14. Hence S cannot be a basis if $p > n$.
- (e) Suppose that S is linearly independent. Then obviously, S is a basis for $\text{Span } S$, by Definition 3.16. Furthermore, if we consider the square matrix A , with columns $\mathbf{v}_1, \dots, \mathbf{v}_p$, then the independence of the columns guarantee that the system $A\mathbf{x} = \mathbf{0}$ has only the

trivial solution $\mathbf{x} = \mathbf{0}$ (cf. Theorem 3.14). Because A is square, this implies that the echelon form of A must have a pivot in every row, and so for each $\mathbf{b} \in \mathbb{R}^n$, the system $A\mathbf{x} = \mathbf{b}$ is consistent. So $\text{Span } S = \text{Col } A = \mathbb{R}^n$. So S is a basis for $\mathbb{R}^n$.

- 3.17 (i) If A is invertible then the reduced echelon form of A is equal to I_n , by Theorem 2.15. So the system $A\mathbf{x} = \mathbf{0}$ has no free variables, and therefore $\text{Null } A = \{\mathbf{0}\}$.

Conversely, if $\text{Null } A = \{\mathbf{0}\}$, then there are no free variables and each column of the echelon form of A must have a pivot. Then, since A is square, the reduced echelon form of A must be equal to I_n , and so A is invertible by Theorem 2.15.

- (ii) If A is invertible then the reduced echelon form of A is equal to I_n , by Theorem 2.15. Then the system $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^n$. So $\text{Col } A = \mathbb{R}^n$.

Conversely, if $\text{Col } A = \mathbb{R}^n$, then the system $A\mathbf{x} = \mathbf{b}$ is consistent for all $\mathbf{b} \in \mathbb{R}^n$. So each row of the echelon form of A must have a pivot. Hence, since A is square, the reduced echelon form of A is equal to I_n . So A is invertible by Theorem 2.15.

- 3.18 Suppose $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_p \mathbf{v}_p$ and $\mathbf{x} = \beta_1 \mathbf{v}_1 + \cdots + \beta_p \mathbf{v}_p$. We get:

$$(\alpha_1 - \beta_1) \mathbf{v}_1 + \cdots + (\alpha_p - \beta_p) \mathbf{v}_p = \mathbf{0}.$$

Now, by Theorem 3.14, the independence of $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ guarantees that $\alpha_i - \beta_i = 0$, for all $1 \leq i \leq p$. So $\alpha_i = \beta_i$ for all $1 \leq i \leq p$.

- 3.19 (a) The system $A\mathbf{x} = \mathbf{w}_j$ is consistent if $\mathbf{w}_j \in \text{Col } A$. This is clearly the case, since $\text{Col } A = V$ and $\mathcal{A}$ is a basis for V .
- (b) C is a $p \times q$ -matrix.
- (c) If $q > p$, then the echelon form of C will have a non-pivot column, so the system $C\mathbf{x} = \mathbf{0}$ will have a free variable. So there exists a nonzero vector $\mathbf{u}$ with $C\mathbf{u} = \mathbf{0}$.
- (d) We have $AC = (\mathbf{w}_1 \ \cdots \ \mathbf{w}_q) = B$.
So, by part (c), $B\mathbf{u} = (AC)\mathbf{u} = A(C\mathbf{u}) = A\mathbf{0} = \mathbf{0}$.
- (e) If $q > p$, then by part (d), there exists $\mathbf{u} \neq \mathbf{0}$ with $B\mathbf{u} = \mathbf{0}$. So the columns of B are linearly dependent by Theorem 3.14 and so, B cannot be a basis of V .
- (f) If $q > p$, B cannot be a basis of V (cf. part (e)). Similarly, if $p > q$, $\mathcal{A}$ cannot be a basis of V . Since both $\mathcal{A}$ and B are bases of V , we must have $p = q$.

3.20 We need to find λ, μ, ρ, η such that:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \rho \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \eta \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

We find:

$$\begin{cases} \lambda = -1 - 2\eta \\ \mu = -2\eta \\ \rho = -1 - 3\eta \\ \eta \text{ is free} \end{cases}$$

This yields that the intersection is given by:

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \right\}$$

3.21 There are many solutions to this problem. We note that

$$x_1 = 1 + \lambda, \quad x_2 = 2 + 3\lambda, \quad x_3 = 1 + \lambda$$

We see $x_1 = x_3$ so that is one plane. Next we see that $3x_1 - x_2 = 1$ and that is a second plane. We basically need to find two equations which no longer contain the variable λ and only x_1, x_2 and x_3 . It is easy to see that the intersection of these two planes is exactly the given line.

3.22 We first compute the intersection of the first and the third plane. We have according to the third plane $x_1 = 1 - \lambda$, $x_2 = 2 + 2\mu$ and $x_3 = \lambda + \mu$. Filling this into the equation for the first plane, we find $\lambda = 2$. So the intersection is given by:

$$\left\{ \mathbf{x} \in \mathbb{R}^3 \mid \mathbf{x} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \text{ for some } \lambda, \mu \in \mathbb{R} \right\}$$

We intersect this result with the second plane and we get:

$$\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \rho \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \eta \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

for some ρ, η . Solving this linear system of equations results in $\mu = 1$, $\rho = -2$ and $\eta = 3$. This results in an intersection given by a single point:

$$\begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix}.$$

3.23 It can be easily verified that the three vectors have a dependency:

$$\frac{1}{4}\mathbf{v}_1 + \frac{1}{4}\mathbf{v}_2 = \mathbf{v}_3$$

- (a) No. $\mathbf{v}_1$ is not equal to either $\mathbf{v}_2$ or $\mathbf{v}_3$.
- (b) Yes. $\mathbf{v}_2 = -\mathbf{v}_1 + 4\mathbf{v}_3$.
- (c) Yes. We can express both $\mathbf{v}_1$ and $\mathbf{v}_2$ in terms of $\mathbf{v}_2$ and $\mathbf{v}_3$ which implies:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} \subset \text{Span}\{\mathbf{v}_2, \mathbf{v}_3\}$$

Conversely, we can express both $\mathbf{v}_2$ and $\mathbf{v}_3$ in terms of $\mathbf{v}_1$ and $\mathbf{v}_2$ which implies:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\} \supset \text{Span}\{\mathbf{v}_2, \mathbf{v}_3\}$$

Combining these two results gives the required equality.

- No. After all the elements of a basis need to be independent which is not true.

Chapter 4

- 4.4 (a) $\det A = \alpha^2 + 2\alpha - 15$.
- (b) A is invertible if and only if $\det A \neq 0$. So if $\alpha \in \mathbb{R} - \{-5, 3\}$.
- 4.5 (a) $\det(A^3) = 0$, so $(\det A)^3 = 0$. Hence, $\det A = 0$ and so A is not invertible.
- (b) If A and B are invertible then $\det A \neq 0$ and $\det B \neq 0$.
Therefore $\det(AB) = \det A \det B \neq 0$. Hence, AB is invertible.
Furthermore, $\det(B^T A) = \det B^T \det A = \det B \det A \neq 0$. Hence, $B^T A$ is invertible.
- (c) Suppose A is invertible. Then $AA^{-1} = I_n$. So, $\det(AA^{-1}) = \det I_n = 1$, by Theorem 4.10. Also $\det(AA^{-1}) = \det A \det A^{-1}$.
Hence, $\det A \det A^{-1} = 1$ and so $\det(A^{-1}) = \frac{1}{\det A}$.

- (d) We have $\det(BAB^{-1}) = \det B \det A \det B^{-1}$. Because determinants are numbers, we can change the order of the factors in the product:

$$\det B \det A \det B^{-1} = \det B \det B^{-1} \det A = 1 \cdot \det A = \det A,$$

where the second equality follows from part (c). Hence,

$$\det(BAB^{-1}) = \det A.$$

- (e) We have

$$\det(A^T A) = \det I_n = 1,$$

by Theorem 4.10. On the other hand,

$$\det(A^T A) = \det A^T \det A = \det A \det A = (\det A)^2.$$

Hence, $(\det A)^2 = 1$ and so $\det A = \pm 1$.

- 4.6 (a) Note that the parallelogram is spanned by the vectors

$$\mathbf{u} = \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ and } \mathbf{v} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

The area equals 8.

- (b) The area equals 4.

- (c) First translate the triangle such that one vertex is in the origin. The area of the triangle equals half of the area of the parallelogram that is spanned by the corresponding vectors. The area of the triangle equals $\frac{15}{2}$.

- (d) Note that the parallelepiped is spanned by the vectors

$$\mathbf{u} = (-2, 3, 1), \mathbf{v} = (1, -1, 1) \text{ and } \mathbf{w} = (5, 1, -2).$$

The volume of the parallelepiped equals 25.

- (e) We have to maximize $f(\alpha) = |\alpha^2 + 2\alpha - 15|$ on the interval $[-4, 4]$. The maximum is obtained for $\alpha = -1$, so the maximal volume is $|f(-1)| = 16$.

- 4.7 (a) Theorem 4.10 is trivial for $n = 1$.

- (b) Now let $k \geq 1$, and assume that Theorem 4.10 holds for all lower triangular $k \times k$ -matrices.

Let A be a lower triangular $(k+1) \times (k+1)$ -matrix. Computing $\det A$ using expansion with respect to the first row, we get:

$$\det A = \sum_{j=1}^{k+1} (-1)^{1+j} a_{1j} \det A_{1j} = a_{11} \det A_{11},$$

where the last equality follows from the fact that $a_{1j} = 0$ for $j \geq 2$, since A is lower triangular. Now we obtain from the induction hypothesis:

$$\det A = a_{11} \det A_{11} = a_{11} \cdot (a_{22} \cdots a_{k+1,k+1}) = a_{11} \cdots a_{k+1,k+1},$$

and so the theorem holds for the lower triangular $(k+1) \times (k+1)$ matrix A .

- (c) From Lemma 4.7 we immediately obtain that Theorem 4.10 also holds for all upper triangular matrices, since a square upper triangular matrix is the transpose of a square lower triangular matrix with corresponding entries on the main diagonal.

- 4.8 (a) Trivial: if $n = 1$ then $A = A^T$.
 (b) Let $k \geq 1$, and assume that Lemma 4.7 holds for all $k \times k$ -matrices. Let A be a $(k+1) \times (k+1)$ -matrix. If we use expansion with respect to the first column to determine $\det A^T$, we obtain:

$$\det A^T = \sum_{i=1}^{k+1} (-1)^{i+1} a_{i1}^T \det(A^T)_{i1} = \sum_{i=1}^{k+1} (-1)^{1+i} a_{1i} \det(A_{1i})^T,$$

where the last equality follows from the observation that $a_{i1}^T = a_{1i}$ and $(A^T)_{i1} = (A_{1i})^T$. Now the induction hypothesis implies

$$\sum_{i=1}^{k+1} (-1)^{1+i} a_{1i} \det(A_{1i})^T = \sum_{i=1}^{k+1} (-1)^{1+i} a_{1i} \det A_{1i} = \det A,$$

where the last equality follows from computing $\det A$ by expansion with respect to the first row.

- 4.9 (a) If $n = 1$, then $A = (a_{11})$ and only the elementary row operation of type 2 can be performed. We have $\det(\alpha a_{11}) = \alpha a_{11} = \alpha \det A$. If $n = 2$ then verify that

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} + \alpha a_{11} & a_{22} + \alpha a_{12} \end{vmatrix} = \begin{vmatrix} a_{11} + \beta a_{21} & a_{12} + \beta a_{22} \\ a_{21} & a_{22} \end{vmatrix} \\ = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$\begin{vmatrix} \alpha a_{11} & \alpha a_{12} \\ a_{21} & a_{22} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{vmatrix} = \alpha \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$\begin{vmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{vmatrix} = - \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}.$$

- (b) Now let $k \geq 2$, and assume that Lemma 4.8 holds for all $k \times k$ -matrices.

Let A be a $(k + 1) \times (k + 1)$ -matrix. Since $k + 1 \geq 3$, and an elementary row operation concerns at most two rows, A has a row i which remains unchanged when performing the elementary row operation considered. Let B be the matrix that arises from A after performing the elementary row operation. If we compute $\det A$ and $\det B$ using expansion with respect to this row i , we have that B_{ij} arises from A_{ij} by the same type of row elementary row operation that transforms A into B . So, by the induction hypothesis, we obtain that $\det B_{ij} = \alpha \det A_{ij}$, where $\alpha = 1$ in case of an operation of type 1, $\alpha \in \mathbb{R}$ in case of an operation of type 2 and $\alpha = -1$ in case of an operation of type 3. Hence

$$\det B = \sum_{j=1}^{k+1} (-1)^{i+j} b_{ij} \det B_{ij} = \sum_{j=1}^{k+1} (-1)^{i+j} a_{ij} \alpha \det A_{ij} = \alpha \det A.$$

This proves that Lemma 4.8 also holds for the $(k + 1) \times (k + 1)$ -matrix A .

- 4.10 (a) If A is invertible, then the reduced echelon form of A is I_n . By Theorem 4.10, $\det I_n = 1 \neq 0$. Furthermore by Lemma 4.8 in each elementary row operation that transforms A into its reduced echelon, the determinant is multiplied with a number $\alpha \neq 0$. Therefore, if the reduced echelon form of A has a nonzero determinant, then so has A . Hence $\det A \neq 0$.
- (b) Suppose $\det A \neq 0$. Consider the reduced echelon form of A . Note that this reduced echelon form is an upper triangular matrix. If this reduced echelon form is not I_n , it must have a non-pivot column, and then the entry of this column that is on the main diagonal is zero. But then $\det A = 0$ by Theorem 4.10, yielding a contradiction. So the reduced echelon form of A must be I_n and therefore A must be invertible.

4.11 One option is to realize that we have:

$$A \begin{pmatrix} 1 \\ 1 \\ 0 \\ -1 \end{pmatrix} = \mathbf{0}$$

and hence A is not invertible which implies that $\det A = 0$. Another option is to realize that by subtracting the first and second column from the fourth column we do not change the determinant. But this creates

a matrix with a column identical to zero. This will result in a determinant of zero (for example compute the determinant by expanding along the fourth column).

Chapter 5

5.7 Eigenvalue of A : 3, with $E_3 = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$.

Eigenvalues of B : -2 and 5 .

Bases for the eigenspaces: $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ and $\left\{ \begin{pmatrix} 8 \\ 0 \\ 1 \end{pmatrix} \right\}$ respectively.

Eigenvalues of C : -5 and 1 .

Bases for the eigenspaces: $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$ and $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}$ respectively.

Eigenvalues of E : 2 .

Basis for the eigenspace: $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\}$.

5.8 A is not diagonalizable.

B is not diagonalizable.

C is diagonalizable with $P = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$ and $D = \begin{pmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

E is not diagonalizable.

5.9 (a) Eigenvalues A : $3 + 2i$ and $3 - 2i$,

with $E_{3+2i} = \text{Span} \left\{ \begin{pmatrix} -1-i \\ 4 \end{pmatrix} \right\}$ and $E_{3-2i} = \text{Span} \left\{ \begin{pmatrix} -1+i \\ 4 \end{pmatrix} \right\}$.

Eigenvalues B : 2 , $1 + 4i$ and $1 - 4i$,

with $E_2 = \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$; $E_{1+4i} = \text{Span} \left\{ \begin{pmatrix} 3-4i \\ 0 \\ 5 \end{pmatrix} \right\}$ and

$E_{1-4i} = \text{Span} \left\{ \begin{pmatrix} 3+4i \\ 0 \\ 5 \end{pmatrix} \right\}$.

(b) Both A and B are diagonalizable, with

$$D_1 = \begin{pmatrix} 3+2i & 0 \\ 0 & 3-2i \end{pmatrix}; \quad P = \begin{pmatrix} -1-i & -1+i \\ 4 & 4 \end{pmatrix};$$

$$D_2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1+4i & 0 \\ 0 & 0 & 1-4i \end{pmatrix}; \quad Q = \begin{pmatrix} 0 & 3-4i & 3+4i \\ 1 & 0 & 0 \\ 0 & 5 & 5 \end{pmatrix}.$$

5.10 We note that the given matrix has eigenvalues 2, $2 - \alpha$ and α . For $\alpha \neq 0, 1, 2$ we have three distinct eigenvalues and we know that this implies that the matrix is diagonalizable. Remains to verify these three cases.

For $\alpha = 0$ we have eigenvalues 2 and 0 and we can only find two independent eigenvectors so the matrix is not diagonalizable.

For $\alpha = 1$ we have eigenvalues 2 and 1 and we can find three independent eigenvectors (we have two independent eigenvectors for eigenvalue 1) so the matrix is diagonalizable.

For $\alpha = 2$ we have eigenvalues 2 and 0 and we can only find two independent eigenvectors so the matrix is not diagonalizable.

In conclusion, the matrix is diagonalizable for all $\alpha \in \mathbb{R}$ except for $\alpha = 0, 2$.

5.11 With mathematical induction to k .

The induction basis for $k = 0$ is obvious: $\mathbf{x}_0 = \lambda^0 \mathbf{x}_0$.

Now let $k \geq 0$, and suppose that $\mathbf{x}_k = \lambda^k \mathbf{x}_0$.

We will show that this induction hypothesis implies that

$$\mathbf{x}_{k+1} = \lambda^{k+1} \mathbf{x}_0.$$

Well,

$$\mathbf{x}_{k+1} = A\mathbf{x}_k = A(\lambda^k \mathbf{x}_0) = \lambda^k A\mathbf{x}_0 = \lambda^k \lambda \mathbf{x}_0 = \lambda^{k+1} \mathbf{x}_0,$$

where the second equality follows from the induction hypothesis and the fourth equality from the fact that $\mathbf{x}_0$ is an eigenvector of A associated with eigenvalue λ .

5.12 (a) Suppose λ is an eigenvalue of A . Note that $A^T - \lambda I = (A - \lambda I)^T$, so

$$\det(A^T - \lambda I) = \det(A - \lambda I)^T = \det(A - \lambda I) = 0,$$

where the second equality follows from Lemma 4.7 and the last equality from Theorem 5.6. So λ is also an eigenvalue of A^T .

- (b) Note that $A\mathbf{v} = r\mathbf{v}$, where $\mathbf{v} \in \mathbb{R}^n$ denotes the vector with all entries equal to 1. So $\mathbf{v}$ is an eigenvector of A corresponding to eigenvalue r .
- (c) From parts (a) and (b) we obtain that r is an eigenvalue of A .

- 5.13 (a) Consider the equation $\alpha_1\mathbf{v}_1 + \cdots + \alpha_p\mathbf{v}_p = \mathbf{0}$ and take the highest index $k \in \{1, \dots, p\}$ for which $\alpha_k \neq 0$. Then

$$\mathbf{v}_k = -\frac{\alpha_1}{\alpha_k}\mathbf{v}_1 - \cdots - \frac{\alpha_{k-1}}{\alpha_k}\mathbf{v}_{k-1}.$$

So, $\mathbf{v}_k$ is a linear combination of its predecessors $\mathbf{v}_1, \dots, \mathbf{v}_{k-1}$.

- (b) Multiplying both sides in (5.8) on the left with A , we get

$$A\mathbf{v}_m = A(\beta_1\mathbf{v}_1 + \cdots + \beta_{m-1}\mathbf{v}_{m-1}) = \beta_1 A\mathbf{v}_1 + \cdots + \beta_{m-1} A\mathbf{v}_{m-1}.$$

And so:

$$\lambda_m \mathbf{v}_m = \beta_1 \lambda_1 \mathbf{v}_1 + \cdots + \beta_{m-1} \lambda_{m-1} \mathbf{v}_{m-1}.$$

- (c) From (5.9) – $\lambda_m \cdot$ (5.8) we deduce

$$\mathbf{0} = \beta_1(\lambda_1 - \lambda_m)\mathbf{v}_1 + \cdots + \beta_{m-1}(\lambda_{m-1} - \lambda_m)\mathbf{v}_{m-1}.$$

Since all λ 's are distinct and the set $\{\mathbf{v}_1, \dots, \mathbf{v}_{m-1}\}$ is linearly independent, we obtain that $\beta_1 = \cdots = \beta_{m-1} = 0$. But then (5.8) implies that $\mathbf{v}_m = \mathbf{0}$, contradicting that $\mathbf{v}_m$ is an eigenvector.

- 5.14 Let $\lambda_1, \dots, \lambda_n$ be the distinct eigenvalues of A . Then, by Theorem 5.9, the corresponding eigenvectors $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ form a linearly independent set and thus a basis for $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ (cf. Definition 3.16). Now it follows from Theorem 3.17 that $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \mathbb{R}^n$ and so, $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for $\mathbb{R}^n$.

- 5.15 (a) By Theorem 4.10, we have that

$$\det(A - \lambda I) = (a_{11} - \lambda) \cdots (a_{nn} - \lambda).$$

Therefore $\det(A - \lambda I) = 0$ if and only if $\lambda = a_{11}$ or $\cdots$ or $\lambda = a_{nn}$.

So the eigenvalues of A are the entries on the main diagonal.

- (b) The factorized expression for $p(\lambda) = \det(A - \lambda I)$ holds for all $\lambda \in \mathbb{R}$.

Taking $\lambda = 0$ yields the required result.

- (c) With mathematical induction to k .

The induction basis for $k = 1$ is trivial: $A\mathbf{x} = \lambda\mathbf{x}$, by definition of λ and $\mathbf{x}$.

Now let $k \geq 1$ and suppose that $A^k\mathbf{x} = \lambda^k\mathbf{x}$. Then

$$A^{k+1}\mathbf{x} = A(A^k\mathbf{x}) = A(\lambda^k\mathbf{x}) = \lambda^k A\mathbf{x} = \lambda^k \lambda\mathbf{x} = \lambda^{k+1}\mathbf{x},$$

where the second equality follows from the induction hypothesis and the fourth equality from the fact that $\mathbf{x}$ is an eigenvector corresponding to eigenvalue λ .

- (d) Suppose $A^k = 0$ and λ is an eigenvalue of A . Then $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq \mathbf{0}$.

Now we deduce from part (c) that $\lambda^k\mathbf{x} = A^k\mathbf{x} = \mathbf{0}$.

Hence $\lambda^k = 0$ and therefore $\lambda = 0$.

- (e) Suppose A is invertible and λ is an eigenvalue of A . Then $\lambda \neq 0$ and $A\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq \mathbf{0}$ and hence

$$A^{-1}(A\mathbf{x}) = A^{-1}(\lambda\mathbf{x}).$$

Therefore,

$$(A^{-1}A)\mathbf{x} = \lambda A^{-1}\mathbf{x}.$$

We get $I\mathbf{x} = \lambda A^{-1}\mathbf{x}$, so, $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$. This implies that λ^{-1} is an eigenvalue of A^{-1} .

- 5.16 (a) Suppose λ is an eigenvalue of B . Then we have

$$\begin{aligned} \det(B - \lambda I) &= 0 \\ \Leftrightarrow \det(P^{-1}AP - \lambda I) &= 0 \Leftrightarrow \det(P^{-1}AP - \lambda P^{-1}P) = 0 \\ \Leftrightarrow \det(P^{-1}AP - P^{-1}\lambda P) &= 0 \Leftrightarrow \det(P^{-1}(AP - \lambda P)) = 0 \\ \Leftrightarrow \det(P^{-1}(AP - \lambda IP)) &= 0 \Leftrightarrow \det(P^{-1}(A - \lambda I)P) = 0 \\ \Leftrightarrow \det(P^{-1}) \det(A - \lambda I) \det P &= 0 \\ \Leftrightarrow \det(P^{-1}) \det P \det(A - \lambda I) &= 0 \\ \Leftrightarrow \det(P^{-1}P) \det(A - \lambda I) &= 0 \\ \Leftrightarrow \det I \det(A - \lambda I) &= 0 \Leftrightarrow \det(A - \lambda I) = 0. \end{aligned}$$

The last expression is equivalent to λ being an eigenvalue of A .

- (b) With mathematical induction to k .

The induction basis for $k = 1$ follows from multiplying both sides of the equation $D = P^{-1}AP$ on the left with P and on the right with P^{-1} , which yields $PDP^{-1} = A$.

Now let $k \geq 1$ and suppose that $A^k = PD^kP^{-1}$.

Then

$$\begin{aligned} A^{k+1} &= AA^k = (PDP^{-1})(PD^kP^{-1}) = PD(P^{-1}P)D^kP^{-1} \\ &= PDID^kP^{-1} = PDD^kP^{-1} = PD^{k+1}P^{-1}, \end{aligned}$$

where the second equality follows from the induction hypothesis.

(c)

$$\begin{aligned} \begin{pmatrix} 3 & 2 \\ 3 & 8 \end{pmatrix}^{100} &= \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 9 \end{pmatrix}^{100} \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} -2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2^{100} & 0 \\ 0 & 9^{100} \end{pmatrix} \cdot \frac{1}{7} \begin{pmatrix} -3 & 1 \\ 1 & 2 \end{pmatrix} \\ &= \frac{1}{7} \begin{pmatrix} 6 \cdot 2^{100} + 9^{100} & -2^{101} + 2 \cdot 9^{100} \\ -3 \cdot 2^{100} + 3 \cdot 9^{100} & 2^{100} + 6 \cdot 9^{100} \end{pmatrix}. \end{aligned}$$

5.17 (a) Counterexample: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(b) Counterexample: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

(c) Suppose A is both invertible and diagonalizable. Then A^{-1} is invertible by Definition 2.11. Furthermore, by Lemma 5.5, none of the eigenvalues of A (which are on the main diagonal of D) are zero, and so $\det D \neq 0$ by Theorem 4.10. Therefore D is invertible by Theorem 4.13. Now we obtain from $D = P^{-1}AP$ and Exercise 2.16:

$$D^{-1} = (P^{-1}AP)^{-1} = P^{-1}A^{-1}(P^{-1})^{-1} = P^{-1}A^{-1}P.$$

Hence A^{-1} is diagonalizable, since D^{-1} is the diagonal matrix with on its main diagonal entries the inverses of the corresponding entries on the main diagonal of D (prove this by verifying that $DD^{-1} = I$).

5.18 (a)

$$A = \begin{pmatrix} 0 & -2 & -2 \\ 1 & 2 & -1 \\ -2 & 2 & 0 \end{pmatrix}; P = \begin{pmatrix} 1 & -1 & -1 \\ 0 & i & -i \\ 1 & 1 & 1 \end{pmatrix}; D = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 2 + 2i & 0 \\ 0 & 0 & 2 - 2i \end{pmatrix}$$

Show that

$$AP = PD = \begin{pmatrix} -2 & -2 - 2i & -2 + 2i \\ 0 & -2 + 2i & 2 - 2i \\ -2 & -2 + 2i & 2 - 2i \end{pmatrix}$$

(b)

$$A\mathbf{x} = \lambda\mathbf{x} = (a + bi)(\mathbf{u} + i\mathbf{v}) = (a\mathbf{u} - b\mathbf{v}) + (b\mathbf{u} + a\mathbf{v})i.$$

(c) Suppose $A\mathbf{x} = \lambda\mathbf{x}$. We have $A\mathbf{x} = A(\mathbf{u} + i\mathbf{v}) = A\mathbf{u} + iA\mathbf{v}$.Also, by part (b), $A\mathbf{x} = (a\mathbf{u} - b\mathbf{v}) + (b\mathbf{u} + a\mathbf{v})i$.Comparing the real and imaginary parts, we get $A\mathbf{u} = a\mathbf{u} - b\mathbf{v}$ and $A\mathbf{v} = b\mathbf{u} + a\mathbf{v}$. Hence

$$A\bar{\mathbf{x}} = A(\mathbf{u} - i\mathbf{v}) = A\mathbf{u} - iA\mathbf{v} = (a\mathbf{u} - b\mathbf{v}) - (b\mathbf{u} + a\mathbf{v})i.$$

Also

$$\bar{\lambda}\bar{\mathbf{x}} = (a - bi)(\mathbf{u} - i\mathbf{v}) = (a\mathbf{u} - b\mathbf{v}) - (b\mathbf{u} + a\mathbf{v})i.$$

Therefore $A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$, and so $\bar{\mathbf{x}}$ is an eigenvector associated with eigenvalue $\bar{\lambda}$.

5.19 (a)

$$\begin{aligned} A\mathbf{x} &= \frac{k}{m} \begin{pmatrix} -1 & 1 & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{k}{m} \begin{pmatrix} -x_1 + x_2 \\ \alpha x_1 - 2\alpha x_2 + \alpha x_3 \\ x_2 - x_3 \end{pmatrix} \\ &= \begin{pmatrix} k(-x_1 + x_2)\frac{1}{m} \\ k(x_1 - 2x_2 + x_3)\frac{1}{M} \\ k(x_2 - x_3)\frac{1}{m} \end{pmatrix} = \begin{pmatrix} m\ddot{x}_1\frac{1}{m} \\ M\ddot{x}_2\frac{1}{M} \\ m\ddot{x}_3\frac{1}{m} \end{pmatrix} = \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix}. \end{aligned}$$

(b)

$$\det(A - \lambda I)$$

$$\begin{aligned} &= \left| \frac{k}{m} \begin{pmatrix} -1 & 1 & 0 \\ \alpha & -2\alpha & \alpha \\ 0 & 1 & -1 \end{pmatrix} - \lambda I \right| = \left| \begin{pmatrix} -\frac{k}{m} & \frac{k}{m} & 0 \\ \frac{k}{M} & -2\frac{k}{M} & \frac{k}{M} \\ 0 & \frac{k}{m} & -\frac{k}{m} \end{pmatrix} - \lambda I \right| \\ &= \begin{vmatrix} -\frac{k}{m} - \lambda & \frac{k}{m} & 0 \\ \frac{k}{M} & -2\frac{k}{M} - \lambda & \frac{k}{M} \\ 0 & \frac{k}{m} & -\frac{k}{m} - \lambda \end{vmatrix} \\ &= \begin{vmatrix} -\frac{k}{m} - \lambda & \frac{k}{m} & 0 \\ 0 & -2\frac{k}{M} - \lambda & \frac{k}{M} \\ \frac{k}{m} + \lambda & \frac{k}{m} & -\frac{k}{m} - \lambda \end{vmatrix} \end{aligned}$$

$$\begin{aligned}
&= \begin{vmatrix} 0 & 2\frac{k}{m} & -\frac{k}{m} - \lambda \\ 0 & -2\frac{k}{M} - \lambda & \frac{k}{M} \\ \frac{k}{m} + \lambda & \frac{k}{m} & -\frac{k}{m} - \lambda \end{vmatrix} \\
&= \left(\frac{k}{m} + \lambda\right) \begin{vmatrix} 2\frac{k}{m} & -\frac{k}{m} - \lambda \\ -2\frac{k}{M} - \lambda & \frac{k}{M} \end{vmatrix} \\
&= \left(\frac{k}{m} + \lambda\right) \left(\frac{2k^2}{mM} - \left(-\frac{2k}{M} - \lambda\right) \left(-\frac{k}{m} - \lambda\right)\right) \\
&= \left(\frac{k}{m} + \lambda\right) \left(\frac{2k^2}{mM} - \left(\frac{2k^2}{mM} + \frac{2k}{M}\lambda + \frac{k}{m}\lambda + \lambda^2\right)\right) \\
&= -\left(\frac{k}{m} + \lambda\right) \left(\frac{2k}{M}\lambda + \frac{k}{m}\lambda + \lambda^2\right) \\
&= -\lambda \left(\frac{k}{m} + \lambda\right) \left(\frac{2k}{M} + \frac{k}{m} + \lambda\right) \\
&= -\lambda \left(\lambda + \frac{k}{m}\right) \left(\lambda + 2\frac{k}{M} + \frac{k}{m}\right)
\end{aligned}$$

(c) From $\mathbf{p} = P^{-1}\mathbf{x}$, we derive

$$\ddot{\mathbf{p}} = P^{-1}\ddot{\mathbf{x}} = P^{-1}(A\mathbf{x}) = (P^{-1}A)\mathbf{x} = (DP^{-1})\mathbf{x} = D(P^{-1}\mathbf{x}) = D\mathbf{p},$$

where the fourth equality follows from $P^{-1}AP = D$.

(d) $A\mathbf{v}_1 = \mathbf{0} = 0\mathbf{v}_1 = \lambda_1\mathbf{v}_1$;

$$A\mathbf{v}_2 = \begin{pmatrix} \frac{k}{m} \\ 0 \\ -\frac{k}{m} \end{pmatrix} = -\frac{k}{m}\mathbf{v}_2 = \lambda_2\mathbf{v}_2;$$

$$A\mathbf{v}_3 = \begin{pmatrix} -\frac{k}{m} - 2\frac{k}{M} \\ 2\frac{k}{M} + 4\frac{km}{M^2} \\ -2\frac{k}{M} - \frac{k}{m} \end{pmatrix} = \left(-2\frac{k}{M} - \frac{k}{m}\right) \begin{pmatrix} 1 \\ -2\frac{m}{M} \\ 1 \end{pmatrix} = \lambda_3\mathbf{v}_3$$

5.20 (a) Straightforward.

(b) Show that $A\mathbf{v}_j = \lambda_j\mathbf{v}_j$, for $j = 1, 2, 3$.

(c) Show that $PP^{-1} = I_3$.

(d) Straightforward using the identities

$$\cos t = \frac{e^{it} + e^{-it}}{2} \quad \text{and} \quad \sin t = \frac{e^{it} - e^{-it}}{2i}.$$

Chapter 6

- 6.4 (a) Solution set: $\emptyset$. Recall that this denotes the empty set. In other words, there are no solutions.

(b) Solution set: $\begin{pmatrix} 5 \\ 4 \\ 0 \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -4 \\ -3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -2 \\ 0 \\ 1 \end{pmatrix} \right\}.$

- (c) $\ker T = \text{Null } A$ by Theorem 6.8.

A basis for $\ker T$ is (cf. part (b))

$$\mathcal{A} = \left\{ \begin{pmatrix} -4 \\ -3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -2 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Note: many other bases are possible.

- (d) $\text{im } T = \text{Col } A$ by Theorem 6.8.

Using elementary column operations we can determine the basis

$$\mathcal{B} = \left\{ \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ -1 \end{pmatrix} \right\}.$$

Note: many other bases are possible. For instance

$$\left\{ \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ -2 \\ -3 \end{pmatrix} \right\}.$$

- (e) No. In the reduced echelon form of A , not every column has a pivot, so $A\mathbf{x} = \mathbf{0}$ has nontrivial solutions. Hence $\text{Null } A \neq \{\mathbf{0}\}$. So T is not one-to-one by Theorem 6.11.

No, e.g., $\mathbf{v}_1$ is not in the range of T . Another way to see this is, is to use Theorem 6.11: the columns of T do not span $\mathbb{R}^3$ since in the reduced echelon form of A , not every row has a pivot.

- 6.5 By Theorem 2.15 we obtain that A is invertible (the reduced echelon form of A is I_3) and therefore T is invertible by Lemma 6.14. Moreover, we have by Theorem 2.15

$$A^{-1} = \begin{pmatrix} 4 & 3 & -5 \\ -1 & -1 & 2 \\ 1 & 1 & -1 \end{pmatrix},$$

and hence, $T^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4x_1 + 3x_2 - 5x_3 \\ -x_1 - x_2 + 2x_3 \\ x_1 + x_2 - x_3 \end{pmatrix}.$

6.6 (a)

$$B_1 = A_2 A_1 = \begin{pmatrix} -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 \end{pmatrix}; \quad B_2 = A_1 A_3 = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

(b)

$$F \begin{pmatrix} -4 \\ 1 \end{pmatrix} = B_1 \begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} 2\sqrt{2} \\ -2\sqrt{2} \end{pmatrix}.$$

(c) Solve $B_1 \mathbf{x} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$. Result: $\begin{pmatrix} -2\sqrt{2} \\ 0 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$

(d) Use Lemma 6.14. We have that B_2 is invertible and

$$B_2^{-1} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}.$$

(note that $B_2^{-1} = B_2$; can you explain this geometrically?). Hence, we have

$$G^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2}x_1 + \frac{\sqrt{2}}{2}x_2 \\ \frac{\sqrt{2}}{2}x_1 + \frac{\sqrt{2}}{2}x_2 \end{pmatrix}$$

6.7 (a) Following Example 6.18, we must reflect through the line with direction:

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

The length of this vector is $\sqrt{1^2 + 3^2} = \sqrt{10}$. So the corresponding unit vector is $\mathbf{n} = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Hence,

$$A = \begin{pmatrix} 2(\frac{1}{\sqrt{10}})^2 - 1 & 2\frac{1}{\sqrt{10}}\frac{3}{\sqrt{10}} \\ 2\frac{3}{\sqrt{10}}\frac{1}{\sqrt{10}} & 2(\frac{3}{\sqrt{10}})^2 - 1 \end{pmatrix} = \begin{pmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{pmatrix}$$

(b)

$$T \begin{pmatrix} -5 \\ 2 \end{pmatrix} = A \begin{pmatrix} -5 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{26}{5} \\ -\frac{7}{5} \end{pmatrix}.$$

6.8 Put $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Then $\begin{pmatrix} 3 \\ -2 \end{pmatrix} = 3\mathbf{e}_1 - 2\mathbf{e}_2$ and $\begin{pmatrix} -5 \\ 4 \end{pmatrix} = -5\mathbf{e}_1 + 4\mathbf{e}_2$.

Hence, by the linearity of T ,

$$T \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \end{pmatrix} \quad \text{and} \quad T \begin{pmatrix} -5 \\ 4 \end{pmatrix} = \begin{pmatrix} 13 \\ 3 \end{pmatrix}$$

which implies

$$3 \cdot T(\mathbf{e}_1) - 2 \cdot T(\mathbf{e}_2) = \begin{pmatrix} -7 \\ 1 \end{pmatrix} \tag{*}$$

$$-5 \cdot T(\mathbf{e}_1) + 4 \cdot T(\mathbf{e}_2) = \begin{pmatrix} 13 \\ 3 \end{pmatrix} \tag{**}$$

Now we obtain from $2 \cdot (*) + (**)$ that

$$T(\mathbf{e}_1) = 2 \cdot \begin{pmatrix} -7 \\ 1 \end{pmatrix} + \begin{pmatrix} 13 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}.$$

Hence, by $(*)$,

$$3 \cdot \begin{pmatrix} -1 \\ 5 \end{pmatrix} - 2 \cdot T(\mathbf{e}_2) = \begin{pmatrix} -7 \\ 1 \end{pmatrix}.$$

$$\text{So } T(\mathbf{e}_2) = \begin{pmatrix} 2 \\ 7 \end{pmatrix}.$$

Therefore

$$A = \begin{pmatrix} -1 & 2 \\ 5 & 7 \end{pmatrix} \quad \text{and so} \quad T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -x_1 + 2x_2 \\ 5x_1 + 7x_2 \end{pmatrix}.$$

6.9 Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a linear transformation.

(a) Taking $\alpha = 0$ in Definition 6.3, we get

$$T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}.$$

(b) Repeatedly applying the two properties of a linear transformation in Definition 6.3 yields:

$$\begin{aligned} T(\alpha_1 \mathbf{v}_1 + \cdots + \alpha_p \mathbf{v}_p) &= T(\alpha_1 \mathbf{v}_1) + \cdots + T(\alpha_p \mathbf{v}_p) \\ &= \alpha_1 T(\mathbf{v}_1) + \cdots + \alpha_p T(\mathbf{v}_p). \end{aligned}$$

- (c) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ be a linearly dependent set in $\mathbb{R}^m$.

Then there exist $\alpha_1, \dots, \alpha_p \in \mathbb{R}$, not all zero, such that $\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}$.

But then, by part (a),

$$T(\alpha_1 \mathbf{v}_1 + \dots + \alpha_p \mathbf{v}_p) = T(\mathbf{0}) = \mathbf{0},$$

and so, by part (b),

$$\alpha_1 T(\mathbf{v}_1) + \dots + \alpha_p T(\mathbf{v}_p) = \mathbf{0}.$$

Hence, $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_p)\}$ is a linearly dependent set in $\mathbb{R}^n$.

- (d) An arbitrary vector $\mathbf{p} + \alpha \mathbf{v}$ on ℓ is mapped onto

$$T(\mathbf{p} + \alpha \mathbf{v}) = T(\mathbf{p}) + \alpha T(\mathbf{v}).$$

So, if $T(\mathbf{v}) = \mathbf{0}$, then ℓ is mapped onto the single vector $T(\mathbf{p})$, otherwise ℓ is mapped onto the line in $\mathbb{R}^n$ through $T(\mathbf{p})$ parallel to $T(\mathbf{v})$.

- 6.10 (a) Suppose that T is one-to-one. By Exercise 6.9(a) we know that $T(\mathbf{0}) = \mathbf{0}$. Now Definition 6.9 implies that $\mathbf{x} = \mathbf{0}$ is the only solution of $T(\mathbf{x}) = \mathbf{0}$.

Conversely, suppose that $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$. Suppose that $T(\mathbf{u}) = T(\mathbf{v})$ for some $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$. We show that then necessarily $\mathbf{u} = \mathbf{v}$.

The linearity of T implies that $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v}) = \mathbf{0}$. Hence, $\mathbf{u} - \mathbf{v} = \mathbf{0}$, because the zero vector is the only vector that is mapped onto $\mathbf{0}$. Hence, $\mathbf{u} = \mathbf{v}$.

- (b) (i) This is a direct consequence of Theorems 6.8 and 6.10.
(ii) Suppose that T is onto. Then $\text{im } T = \mathbb{R}^k$. So $\text{Col } A = \mathbb{R}^k$, by Theorem 6.8.
Conversely, if $\text{Col } A = \mathbb{R}^k$, then $\text{im } A = \mathbb{R}^k$ by Theorem 6.8. Hence, T is onto.
- (c) Suppose that T is one-to-one. Let $\alpha_1 \mathbf{a}_1 + \dots + \alpha_n \mathbf{a}_n = \mathbf{0}$. This is equivalent to

$$A \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathbf{0} \quad \text{and so} \quad T \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathbf{0}$$

So, by part (a), $\alpha_1 = 0, \dots, \alpha_n = 0$ and therefore the columns of A are linearly independent.

Conversely, suppose that the columns of A are linearly independent. Suppose that $T(\mathbf{u}) = T(\mathbf{v})$ for some $\mathbf{u}, \mathbf{v} \in \mathbb{R}^k$. We show that then necessarily $\mathbf{u} = \mathbf{v}$. We have $T(\mathbf{u}) = T(\mathbf{v})$, so $A\mathbf{u} = A\mathbf{v}$. Hence, $A\mathbf{u} - A\mathbf{v} = \mathbf{0}$, and so, $A(\mathbf{u} - \mathbf{v}) = \mathbf{0}$. Thus

$$(u_1 - v_1)\mathbf{a}_1 + \cdots + (u_n - v_n)\mathbf{a}_n = \mathbf{0}.$$

Now the independence of the columns of A implies that

$$u_1 - v_1 = 0, \dots, u_n - v_n = 0.$$

Hence, $\mathbf{u} = \mathbf{v}$.

6.11 We have:

(a)

$$\mathbf{x} \in \ker T \Leftrightarrow T\mathbf{x} = \mathbf{0} \Leftrightarrow A\mathbf{x} = \mathbf{0} \Leftrightarrow \mathbf{x} \in \text{Null } A.$$

$$\mathbf{y} \in \text{im } T \Leftrightarrow \exists \mathbf{x} : T\mathbf{x} = \mathbf{y} \Leftrightarrow \exists \mathbf{x} : A\mathbf{x} = \mathbf{y} \Leftrightarrow \mathbf{y} \in \text{Col } A,$$

where the last equivalence follows from Definition 3.11.

(b) Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and let A be the representation matrix of F .

Suppose F is invertible.

Then, by Definition 6.13, we have that for each $\mathbf{w} \in \mathbb{R}^n$, there exists exactly one $\mathbf{u} \in \mathbb{R}^n$ such that $F(\mathbf{u}) = \mathbf{w}$.

So, for each $\mathbf{w} \in \mathbb{R}^n$, there exists exactly one $\mathbf{u} \in \mathbb{R}^n$ such that $A\mathbf{u} = \mathbf{w}$.

Therefore, the reduced echelon form of A must be equal to I_n , and so A is invertible by Theorem 2.15.

The converse part of the proof, that A invertible implies F invertible is obtained by reversing the steps above.

Furthermore, if B denotes the representation matrix of F^{-1} , then we have for all $\mathbf{u} \in \mathbb{R}^n$:

$F^{-1}(F(\mathbf{u})) = \mathbf{u}$, so $B(A\mathbf{u}) = \mathbf{u}$. Hence $(BA)\mathbf{u} = \mathbf{u}$, and so $BA = I_n$. Therefore $B = A^{-1}$ by Lemma 2.13.

(c) We prove T invertible $\Rightarrow T$ onto $\Rightarrow T$ one-to-one $\Rightarrow T$ invertible.

Let A be the representation matrix of T .

If T is invertible, then T is onto by Definitions 6.13, 6.7 and 6.9.

If T is onto, then, by Theorem 6.11, the columns of A span $\mathbb{R}^n$. Therefore the reduced echelon form of A has a pivot in every row. So, since A is square, the reduced echelon form of A has a pivot in every column. Hence $\text{Null } A = \{\mathbf{0}\}$, and so T is one-to-one by Theorem 6.11.

If T is one-to-one, then, by Theorem 6.11, $\text{Null } A = \{\mathbf{0}\}$, so the reduced echelon form of A has a pivot in every column. Therefore, since A is square, the reduced echelon form of A is equal to I_n , and so A is invertible by Theorem 2.15. Hence, T is invertible by Lemma 6.14.

- 6.12 (a) Assume we have matrices $X_1, X_2 \in \mathbb{R}^{2 \times 2}$ and constants $\alpha, \beta \in \mathbb{R}$ then we have:

$$T(\alpha X_1 + \beta X_2) = A(\alpha X_1 + \beta X_2) = \alpha A X_1 + \beta A X_2 = \alpha T(X_1) + \beta T(X_2)$$

where the second equality is a consequence of Lemma 2.8.

- (b) In order to find an eigenvector associated with the eigenvalue 3, we need to find a matrix $X \neq 0$ such that:

$$T(X) = 3X.$$

Given the definition of T we get:

$$AX = 3X$$

or

$$(A - 3I)X = 0.$$

We have:

$$A - 3I = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix}$$

and hence we can choose, for instance

$$X = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$$

Note that T is a mapping from $\mathbb{R}^{2 \times 2}$ to $\mathbb{R}^{2 \times 2}$ and hence an eigenvector needs to be a 2×2 matrix (even though it can be confusing that an eigenvector is a matrix).

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